Casual Document for Probabilistic Principal Component Analysis MLE Asymptotic Normality

Wen, Zehai

2026-08-25 17:36:32-04:00

Chapter 1

Low Dimensional Asymptotic Normality

1.1 Basic Setup

Given a "high dimensional" data set $\{\boldsymbol{x}_i\}_{i=1}^n$ where $\boldsymbol{x}_i \in \mathbb{R}^p$ for all $i = 1, \ldots, n$, the statistician fixes a positive integer $q < p$ and wishes to find the q principal components. Assuming that the data is centred (that is, with mean $\mathbf{0}$), the probabilistic principal component analysis model is a statistical model that treats $\boldsymbol{x}_i$'s as realizations of independent random variables $\boldsymbol{X}_i = \boldsymbol{W}\boldsymbol{Z}_i + \boldsymbol{\epsilon_i}$, where $\boldsymbol{W}$ is an unknown $p \times q$ loading matrix of full rank, $\boldsymbol{Z}_i \sim N_q(\mathbf{0}, \boldsymbol{I}_q)$ is the latent normal variable, and $\boldsymbol{\epsilon_i} \sim N_p(\mathbf{0}, \sigma^2 \boldsymbol{I}_p)$ is the Gaussian noise with unknown standard deviation $\sigma > 0$, and $(\boldsymbol{Z}_1, \ldots, \boldsymbol{Z}_n, \boldsymbol{\epsilon}_1, \ldots, \boldsymbol{\epsilon}_n)$ are all independent (for now). Put $\theta = (\boldsymbol{W}_\theta, \sigma_\theta^2)$ for the unknown parameter to estimate and Θ for all possible choices of θ. Θ is thus the collection of all tuples $(\boldsymbol{W}, \sigma^2)$ such that $\boldsymbol{W}$ is a real $p \times q$ matrix with $\operatorname{rank}(\boldsymbol{W}) = q$ and $\sigma^2 \in (0, \infty)$. Given that the collection of all real $p \times q$ matrix with rank q is an open subset of the Euclidean space $\mathbb{R}^{pq}$, we consider Θ as an open subset of the Euclidean space $\mathbb{R}^{pq+1}$ and our statistical model is

$$\mathcal{P} = \left\{ N_p\left(\mathbf{0}, \boldsymbol{W}_\theta \boldsymbol{W}_\theta^T + \sigma_\theta^2 \boldsymbol{I}_p\right) \right\}_{\theta \in \Theta}. \tag{1.1}$$

To resolve the non-identifiability of $\mathcal{P}$, we introduce the quotient manifold structure. For $(\boldsymbol{A}, s) \in \Theta$ and $(\boldsymbol{B}, t) \in \Theta$, we define $(\boldsymbol{A}, s) \sim (\boldsymbol{B}, t)$ to mean that they give rise to the same covariance matrix, namely when $\boldsymbol{A}\boldsymbol{A}^T + s\boldsymbol{I}_p = \boldsymbol{B}\boldsymbol{B}^T + t\boldsymbol{I}_p$. The equivalence classes are denoted by $[(\boldsymbol{A}, s)]$ or more briefly $[\boldsymbol{A}, s]$. Unless otherwise mentioned, $(\boldsymbol{A}, s) \in \Theta$ is the default representative we pick when we write an equivalence class as $[\boldsymbol{A}, s]$. Writing $O(q)$ for the collection of all $q \times q$ real orthogonal matrices, it was shown in Datta's thesis that $[\boldsymbol{A}, s] = \{(\boldsymbol{A}\boldsymbol{Q}, s) \mid \boldsymbol{Q} \in O(q)\}$. Therefore, $M := \Theta/\sim$ can be identified as the orbits of the right action of $O(q)$ on Θ and, given that this Lie group action is smooth, proper, and free, the quotient manifold theorem (see Lee, 2013, Theorem 21.10) shows that there is a unique smooth structure making M a manifold of dimension

$$d = pq + 1 - \frac{q(q-1)}{2} \tag{1.2}$$

and the map $\pi : \Theta \to M$ defined by

$$\pi((\boldsymbol{A}, s)) = [\boldsymbol{A}, s] \tag{1.3}$$

is a smooth submersion. Moreover, this action is vertical, transitive on fibers and is isometric with respect to the Euclidean metric. Thus, there exists a unique metric g on M making (M, g) a Riemannian manifold and π a Riemannian submersion (see Lee, 2018, Theorem 2.28). In more concrete terms, if g^Θ is the standard Euclidean metric given by

$$g^\Theta_{(\boldsymbol{A},s)}\left((\boldsymbol{V}, c), (\boldsymbol{U}, b)\right) = \operatorname{tr}\left(\boldsymbol{V}\boldsymbol{U}^T\right) + cb, \tag{1.4}$$

whenever $(\boldsymbol{A}, s) \in \Theta$, $(\boldsymbol{V}, c)$, $(\boldsymbol{U}, b) \in T_{(\boldsymbol{A},s)}\Theta \cong \mathbb{R}^{p\times q} \times \mathbb{R}$, then

$$g_{[\boldsymbol{A},s]}(\boldsymbol{v}, \boldsymbol{h}) = g_{[\boldsymbol{A},s]}\left(d\pi_{(A,s)}((\boldsymbol{V}, c)), d\pi_{(A,s)}((\boldsymbol{U}, b))\right) = g^\Theta_{(\boldsymbol{A},s)}\left((\boldsymbol{V}, c), (\boldsymbol{U}, b)\right), \tag{1.5}$$

whenever $\boldsymbol{v}$, $\boldsymbol{h} \in T_{[\boldsymbol{A},s]}M$ lifting to $(\boldsymbol{V}, c)$ and $(\boldsymbol{U}, b)$ respectively. For later convenience, let us define two functions $\Phi : \Theta \to \mathbb{R}^{p\times p}$ and $\boldsymbol{\Sigma} : M \to \mathbb{R}^{p\times p}$ by setting, for every $[\boldsymbol{A}, s] \in M$:

$$\boldsymbol{\Sigma}([\boldsymbol{A}, s]) := \Phi((\boldsymbol{A}, s)) := \boldsymbol{A}\boldsymbol{A}^T + s\boldsymbol{I}_p. \tag{1.6}$$

In other words, $\boldsymbol{\Sigma} \circ \pi = \Phi$ gives the covariance matrix. (1.1) is then reparametrized as

$$\mathcal{P} = \{N_p(\boldsymbol{0}, \boldsymbol{\Sigma}([\boldsymbol{A}, s]))\}_{[\boldsymbol{A},s]\in M} \tag{1.7}$$

and this is the identifiable model we shall work with.

Finally, thoughout the entire document, our default norm for a matrix $\boldsymbol{A}$ will be the spectral norm, denoted by $\|\boldsymbol{A}\|$. We will also make use of the Frobenius norm $\|\boldsymbol{A}\|_F$ as well. The eigenvalues and the singular values of a matrix will always be labelled in decreasing order. In symbols, if $\boldsymbol{A}$ is a matrix of size $p \times q$, then we denote its q singular values as

$$\|\boldsymbol{A}\| = \sigma_1(\boldsymbol{A}) \geqslant \sigma_2(\boldsymbol{A}) \geqslant \cdots \geqslant \sigma_q(\boldsymbol{A}) = \sigma_{\min}(\boldsymbol{A}) \geqslant 0$$

and, if $\boldsymbol{S}$ is a symmetric matrix of size $p \times p$, then we denote its p eigenvalues as

$$\lambda_1(\boldsymbol{S}) \geqslant \lambda_2(\boldsymbol{S}) \geqslant \cdots \geqslant \lambda_p(\boldsymbol{S}).$$

1.2 Basic Facts about (M, g)

Working with Riemannian submersion, we shall as usual start with computations regarding the horizontal and vertical tangent spaces.

Proposition 1.

1. The map $\boldsymbol{\Sigma}$ defined in (1.6) is a smooth injective immersion.
2. At every $(\boldsymbol{A}, s) \in \Theta$, the vertical tangent space

$$V_{(\boldsymbol{A},s)} := \ker(\mathrm{d}\pi_{(\boldsymbol{A},s)}) = \ker(\mathrm{d}\Phi_{(\boldsymbol{A},s)}) \tag{1.8}$$

   equals to the collection of all tangent vectors of the form $(\boldsymbol{A}\boldsymbol{\Omega}, 0) \in T_{(\boldsymbol{A},s)}\Theta$, with $\boldsymbol{\Omega}$ ranged over all $q \times q$ skew symmetric matrices.
3. At every $(\boldsymbol{A}, s) \in \Theta$, the horizontal tangent space $H_{(\boldsymbol{A},s)} := V^{\perp}_{(\boldsymbol{A},s)}$ equals to the collection of all tangent vectors of the form $(\boldsymbol{V}, c) \in T_{(\boldsymbol{A},s)}\Theta$ ranged over all $c \in \mathbb{R}$ and all $p \times q$ matrices $\boldsymbol{V}$ satisfying $\boldsymbol{V}^T\boldsymbol{A} = \boldsymbol{A}^T\boldsymbol{V}$.

Proof. Fix a point $(\boldsymbol{A}, s) \in \Theta$ throughout this proof. Before we start the proof, let us first study the set $\ker(\mathrm{d}\Phi_{(\boldsymbol{A},s)})$, where Φ is given in (1.6). For $(\boldsymbol{V}, c) \in T_{(\boldsymbol{A},s)}\Theta$, define a curve $\gamma : (-\epsilon, \epsilon) \to \Theta$ by setting $\gamma(t) = (\boldsymbol{A} + t\boldsymbol{V}, s + tc)$. Then:

$$\mathrm{d}\Phi_{(\boldsymbol{A},s)}((\boldsymbol{V}, c)) = \left.\frac{d}{dt}\right|_{t=0} \Phi(\gamma(t)) = \boldsymbol{A}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{A}^T + c\boldsymbol{I}_p.$$

If $(\boldsymbol{V}, c) \in \ker(d\Phi_{(\boldsymbol{A},s)})$, then

$$\boldsymbol{A}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{A}^T + c\boldsymbol{I}_p = \boldsymbol{0}. \tag{1.9}$$

If $\boldsymbol{u} \in \mathbb{R}^p$ is any nonzero vector orthogonal to all column vectors of $\boldsymbol{A}$, then $\boldsymbol{A}^T\boldsymbol{u} = \boldsymbol{0}$ and

$$0 = \boldsymbol{u}^T(\boldsymbol{A}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{A}^T + c\boldsymbol{I}_p)\boldsymbol{u} = c\,\boldsymbol{u}^T\boldsymbol{u}.$$

Given that $\boldsymbol{u} \neq \boldsymbol{0}$, we must have $c = 0$ and (1.9) reduces to

$$\boldsymbol{A}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{A}^T = \boldsymbol{0}. \tag{1.10}$$

Apply $\boldsymbol{u}$ to the right to get:

$$\boldsymbol{A}\boldsymbol{V}^T\boldsymbol{u} = \boldsymbol{0}.$$

Given that $\boldsymbol{A}$ has full rank, rank nullity theorem forces

$$\boldsymbol{V}^T\boldsymbol{u} = \boldsymbol{0}.$$

Therefore, we have proven that every vector $\boldsymbol{u}$ orthogonal to columns vectors of $\boldsymbol{A}$ will also be orthogonal to column vectors of $\boldsymbol{V}$. This means $\boldsymbol{V} = \boldsymbol{A}\boldsymbol{\Omega}$ for some $q \times q$ matrix $\boldsymbol{\Omega}$. Substitute this back to (1.10) gives

$$\boldsymbol{A}(\boldsymbol{\Omega}^T + \boldsymbol{\Omega})\boldsymbol{A}^T = \boldsymbol{0}.$$

Applying $(\boldsymbol{A}^T\boldsymbol{A})^{-1}\boldsymbol{A}^T$ to the left and $\boldsymbol{A}(\boldsymbol{A}^T\boldsymbol{A})^{-1}$ to the right yields $\boldsymbol{\Omega}^T + \boldsymbol{\Omega} = \boldsymbol{0}$. Therefore, $\boldsymbol{\Omega}$ is skew-symmetric. Conversely, (1.10) is satisfied by $\boldsymbol{V} = \boldsymbol{A}\boldsymbol{\Omega}$ with any skew-symmetric matrix $\boldsymbol{\Omega}$. We have proven that $\ker(\mathrm{d}\Phi_{(\boldsymbol{A},s)})$ equals to the collection of all tangent vectors of the form $(\boldsymbol{A}\boldsymbol{\Omega}, 0) \in T_{(\boldsymbol{A},s)}\Theta$, with $\boldsymbol{\Omega}$ ranged over all $q \times q$ skew symmetric matrices.

We now move back to prove 1 and 2 at the same time by showing (1.8). Clearly, $\boldsymbol{\Sigma}$ is well-defined and is injective. The smoothness of $\boldsymbol{\Sigma}$ is guaranteed by relation $\boldsymbol{\Sigma} \circ \pi = \Phi$ (see for example Lee, 2013, Theorem 4.29). By the chain rule, we have

$$\mathrm{d}\Phi_{(\boldsymbol{A},s)} = \mathrm{d}\boldsymbol{\Sigma}_{[\boldsymbol{A},s]} \circ \mathrm{d}\pi_{(\boldsymbol{A},s)}. \tag{1.11}$$

It follows immediately that $\ker(\mathrm{d}\pi_{(\boldsymbol{A},s)}) \subseteq \ker(\mathrm{d}\Phi_{(\boldsymbol{A},s)})$. The other direction will follow if we can prove that $\mathrm{d}\boldsymbol{\Sigma}_{[\boldsymbol{A},s]}$ is injective.

Suppose that $\boldsymbol{v} \in T_{[\boldsymbol{A},s]}M$ is such that $\mathrm{d}\boldsymbol{\Sigma}_{[\boldsymbol{A},s]}(\boldsymbol{v}) = 0$. We prove that $\boldsymbol{v} = \boldsymbol{0}$. Let $(\boldsymbol{V}, c) \in T_{(\boldsymbol{A},s)}\Theta$ be the horizontal lift of $\boldsymbol{v}$ at $(\boldsymbol{A}, s)$, meaning that $(\boldsymbol{V}, c)$ is the unique horizontal tangent vector at $(\boldsymbol{A}, s)$ such that $\boldsymbol{v} = \mathrm{d}\pi_{(\boldsymbol{A},s)}((\boldsymbol{V}, c))$. (1.11) then implies that $(\boldsymbol{V}, c) \in \ker(\mathrm{d}\Phi_{(\boldsymbol{A},s)})$ and the preceding discussion shows that $c = 0$ and $\boldsymbol{V} = \boldsymbol{A}\boldsymbol{\Omega}$ for some skew-symmetric $\boldsymbol{\Omega}$. To proceed, we look at the curve $\gamma : (-\epsilon, \epsilon) \to \Theta$ defined by $\gamma(t) = (\boldsymbol{A}e^{t\boldsymbol{\Omega}}, s)$. Clearly $\gamma(0) = (\boldsymbol{A}, s)$ and $\gamma'(0) = (\boldsymbol{A}\boldsymbol{\Omega}, 0)$. Moreover, the matrix $e^{t\boldsymbol{\Omega}}$ is always orthogonal for any t and any skew-symmetric $\boldsymbol{\Omega}$ so that $\pi(\gamma(t)) = [\boldsymbol{A}, s]$ for all t. We have

$$\boldsymbol{v} = \mathrm{d}\pi_{(\boldsymbol{A},s)}((\boldsymbol{A}\boldsymbol{\Omega}, 0)) = \mathrm{d}\pi_{(\boldsymbol{A},s)}(\gamma'(0)) = \left.\frac{d}{dt}\right|_{t=0} \pi(\gamma(t)) = \boldsymbol{0} \in T_{[\boldsymbol{A},s]}M.$$

Therefore, $\mathrm{d}\boldsymbol{\Sigma}_{[\boldsymbol{A},s]}$ is injective and the vertical tangent space is the same as $\ker(\mathrm{d}\Phi_{(\boldsymbol{A},s)})$. Since $[\boldsymbol{A}, s]$ is arbitrary, $\boldsymbol{\Sigma}$ is an immersion.

To compute the horizontal tangent space, any $(\boldsymbol{V}, c) \in V^{\perp}_{(\boldsymbol{A},s)}$ will satisfy

$$0 = g^{\Theta}_{(\boldsymbol{A},s)}\left((\boldsymbol{V}, c), (\boldsymbol{A}\boldsymbol{\Omega}, 0)\right) = \mathrm{tr}\big(\boldsymbol{A}^T\boldsymbol{V}\boldsymbol{\Omega}\big) \tag{1.12}$$

for all skew-symmetric $\boldsymbol{\Omega}$. Decomposing $\boldsymbol{A}^T\boldsymbol{V} = \boldsymbol{S} + \boldsymbol{K}$, where $\boldsymbol{S}$ is the symmetric part and $\boldsymbol{K}$ is the skew-symmetric part, we find out that

$$0 = \mathrm{tr}(\boldsymbol{S}\boldsymbol{\Omega}) + \mathrm{tr}(\boldsymbol{K}\boldsymbol{\Omega}) = 0 + \mathrm{tr}(\boldsymbol{K}\boldsymbol{\Omega}) = \mathrm{tr}(\boldsymbol{K}\boldsymbol{\Omega}).$$

Setting $\boldsymbol{\Omega} = \boldsymbol{K}^T$ will yield that $\boldsymbol{K} = \boldsymbol{0}$. Therefore, $\boldsymbol{A}^T\boldsymbol{V}$ is a symmetric matrix. Conversely, any $\boldsymbol{V}$ such that $\boldsymbol{A}^T\boldsymbol{V}$ is symmetric will satisfy (1.12). The proof is complete. $\square$

We now move to compute the exponential map and the logarithmic map. The following lemma is crucial for computing the orthogonal matrix based on given points in M.

Lemma 2. If $\boldsymbol{L}$ is an invertible $q \times q$ matrix and $\boldsymbol{Q} \in O(q)$ are such that $\boldsymbol{LQ}$ is symmetric, then there exists matrices $\boldsymbol{R} \in O(q)$, $\boldsymbol{P} \in O(q)$, a $q \times q$ diagonal matrix $\boldsymbol{D}$ whose diagonal entries are 1 or -1, and a $q \times q$ diagonal matrix $\boldsymbol{S}$ with diagonal entries $\sigma_1(\boldsymbol{L}) \geqslant \cdots \geqslant \sigma_q(\boldsymbol{L}) > 0$ such that $\boldsymbol{L} = \boldsymbol{RSP}^T$ and $\boldsymbol{Q} = \boldsymbol{PDR}^T$.

If we add furthermore the assumption that $\boldsymbol{L}$ is symmetric positive definite, then we can achieve the results mentioned above with $\boldsymbol{P} = \boldsymbol{R}$. In particular, this implies $\boldsymbol{Q} = \boldsymbol{Q}^T$ and $\|\boldsymbol{Q} - \boldsymbol{I}_q\| = \|\boldsymbol{D} - \boldsymbol{I}_q\|$ can take only 0 or 2 as values.

Proof. Let us start by applying the singular value decomposition of $\boldsymbol{L}$ to get matrices $\boldsymbol{R}' \in O(q)$, $\boldsymbol{P}' \in O(q)$ and a diagonal matrix $\boldsymbol{S}$ with diagonal entries $\sigma_1(\boldsymbol{L}) \geqslant \cdots \geqslant \sigma_q(\boldsymbol{L}) > 0$ such that $\boldsymbol{L} = \boldsymbol{R}'\boldsymbol{S}(\boldsymbol{P}')^T$. Note that all singular values are positive because $\boldsymbol{L}$ is invertible. Substituting the given condition that $\boldsymbol{LQ}$ is symmetric will yield that $\boldsymbol{SJ} = \boldsymbol{J}^T\boldsymbol{S}$, where $\boldsymbol{J} := (\boldsymbol{P}')^T\boldsymbol{Q}\boldsymbol{R}'$.

Clearly $\boldsymbol{J} \in O(q)$. The equation

$$\boldsymbol{S}^2 = \boldsymbol{SJJ}^T\boldsymbol{S} = (\boldsymbol{SJ})(\boldsymbol{SJ})^T = (\boldsymbol{J}^T\boldsymbol{S})(\boldsymbol{J}^T\boldsymbol{S})^T = \boldsymbol{J}^T\boldsymbol{S}^2\boldsymbol{J}$$

shows that

$$\boldsymbol{JS}^2 = \boldsymbol{S}^2\boldsymbol{J}.$$

Looking at each entry of both side, we end up with

$$\boldsymbol{J}_{ij}(\sigma_j(\boldsymbol{L})^2 - \sigma_i(\boldsymbol{L})^2) = 0$$

for $i, j = 1, \ldots q$. Given that all singular values are positive, we in fact have

$$\boldsymbol{J}_{ij}(\sigma_j(\boldsymbol{L}) - \sigma_i(\boldsymbol{L})) = 0 \tag{1.13}$$

for all i and all j so that $\boldsymbol{JS} = \boldsymbol{SJ} = \boldsymbol{J}^T\boldsymbol{S}$. Inverting $\boldsymbol{S}$ yields that $\boldsymbol{J}$ is actually symmetric. Being a symmetric orthogonal matrix, $\boldsymbol{J}^2 = \boldsymbol{I}_q$ and the eigenvalues of $\boldsymbol{J}$ must be 1 or -1. Another look at (1.13) shows that $\boldsymbol{J}$ is a block-diagonal matrix, diagonal if all singular values are distinct. Furthermore, each block of $\boldsymbol{J}$ must still be orthogonal and symmetric.

We now diagonalize $\boldsymbol{J}$ based on the above information. Suppose that $\boldsymbol{S} = \oplus_{i=1}^k \boldsymbol{s}_i \boldsymbol{I}_{m_i}$, where $\boldsymbol{s}_1 > \ldots > \boldsymbol{s}_k$ are the unique diagonal entries of $\boldsymbol{S}$ and each repeats for m_i times. Thus, $\boldsymbol{J}$ has k blocks. Write the ith block by $\boldsymbol{J}_i$ so that $\boldsymbol{J} = \oplus_{i=1}^k \boldsymbol{J}_i$. The spectral theorem then allows us to orthogonally diagonalize $\boldsymbol{J}_i = \boldsymbol{K}_i\boldsymbol{D}_i\boldsymbol{K}_i^T$, where $\boldsymbol{K}_i \in O(m_i)$ and $\boldsymbol{D}_i$ is an $m_i \times m_i$ diagonal matrix with all diagonal entries being 1 or -1. Put $\boldsymbol{K} = \oplus_{i=1}^k \boldsymbol{K}_i$ and $\boldsymbol{D} = \oplus_{i=1}^k \boldsymbol{D}_i$. Then $\boldsymbol{K} \in O(q)$, $\boldsymbol{D}$ is a $q \times q$ diagonal matrix with 1 or -1 diagonal entry, and $\boldsymbol{J} = \boldsymbol{KDK}^T$. This involved construction leads us to the outcome that $\boldsymbol{K}$ in fact commutes with $\boldsymbol{S}$:

$$\boldsymbol{KS} = \bigoplus_{i=1}^k \boldsymbol{s}_i\boldsymbol{K}_i = \boldsymbol{SK}.$$

Put $\boldsymbol{P} = \boldsymbol{P}'\boldsymbol{K}$, $\boldsymbol{R} = \boldsymbol{R}'\boldsymbol{K}$. We have

$$\boldsymbol{Q} = \boldsymbol{P}'\boldsymbol{J}(\boldsymbol{R}')^T = \boldsymbol{P}'\boldsymbol{KDK}^T(\boldsymbol{R}')^T = \boldsymbol{PDR}^T$$

and

$$\boldsymbol{RSP}^T = \boldsymbol{R}'\boldsymbol{KSK}^T(\boldsymbol{P}')^T = \boldsymbol{R}'\boldsymbol{SKK}^T(\boldsymbol{P}')^T = \boldsymbol{R}'\boldsymbol{S}(\boldsymbol{P}')^T = \boldsymbol{L}.$$

This completes the proof.

Finally, let us just note that, in case $\boldsymbol{L}$ is symmetric positive definite, the spectral theorem allows us to take $\boldsymbol{R}' = \boldsymbol{P}'$ from the start and it will yield in the end $\boldsymbol{L} = \boldsymbol{PSP}^T$ and $\boldsymbol{Q} = \boldsymbol{PDP}^T$. It then follows easily that $\boldsymbol{Q}^2 = \boldsymbol{I}_q$ and $\boldsymbol{Q}^T = \boldsymbol{Q}$. If $\boldsymbol{D} = \boldsymbol{I}_q$, then $\|\boldsymbol{Q} - \boldsymbol{I}_q\| = \|\boldsymbol{D} - \boldsymbol{I}_q\| = 0$. If any entry of $\boldsymbol{D}$ is -1, $\|\boldsymbol{D} - \boldsymbol{I}_q\| = 2$. $\square$

We now state and prove all the results about the exponential and the logarithmic map in one single proposition.

Proposition 3. Given $[\boldsymbol{A}, s] \in M$ and $\boldsymbol{v} \in T_{[\boldsymbol{A},s]}M$, geodesics in M starting at $[\boldsymbol{A}, s]$ with initial velocity $\boldsymbol{v}$ takes the format of curves $\gamma : I \to M$, where I is an open interval containing 0, defined by $\gamma(t) = [\boldsymbol{A} + t\boldsymbol{V}, s + tc]$, where $(\boldsymbol{V}, c) \in H_{(\boldsymbol{A},s)}$ is the horizontal lift of $\boldsymbol{v}$ at $(\boldsymbol{A}, s)$. Consequently, the exponential map is computed as

$$\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}) = [\boldsymbol{A} + \boldsymbol{V}, s + c]. \tag{1.14}$$

Let us use $\mathcal{E}_{[\boldsymbol{A},s]}$ to denote the open ball in $T_{[\boldsymbol{A},s]}M \cong \mathbb{R}^d$ centred at $\mathbf{0}$ of radius $\min(\sigma_{\min}(\boldsymbol{A}), s)$. Given that (1.14) is well-defined at least over all $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$, we from now on treat $\mathcal{E}_{[\boldsymbol{A},s]}$ as the domain of the exponential map. That is, $\mathrm{Exp}_{[\boldsymbol{A},s]} : \mathcal{E}_{[\boldsymbol{A},s]} \to U_{[\boldsymbol{A},s]}$, where $U_{[\boldsymbol{A},s]} := \mathrm{Exp}_{[\boldsymbol{A},s]}(\mathcal{E}_{[\boldsymbol{A},s]})$ is an open subset of M, is defined by (1.14). Note the following facts:

1. For every $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$, $\boldsymbol{A}^T(\boldsymbol{A} + \boldsymbol{V})$ is symmetric positive definite.
2. $\mathrm{Exp}_{[\boldsymbol{A},s]}$ is a diffeomorphism.
3. If $[\boldsymbol{B}, t] \in U_{[\boldsymbol{A},s]}$, then for every representative $(\boldsymbol{A}, s)$ of $[\boldsymbol{A}, s]$ and every representative $(\boldsymbol{B}, t)$ of $[\boldsymbol{B}, t]$, the matrix $\boldsymbol{A}^T\boldsymbol{B}$ will always be invertible. Given representative $(\boldsymbol{A}, s)$ of $[\boldsymbol{A}, s]$, *some* representative $(\boldsymbol{B}, t)$ of $[\boldsymbol{B}, t]$ will furthermore make $\boldsymbol{A}^T\boldsymbol{B}$ symmetric positive definite.
4. The logarithmic map $\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]} : U_{[\boldsymbol{A},s]} \to \mathcal{E}_{[\boldsymbol{A},s]}$ is computed as follows: for every $[\boldsymbol{B}, t] \in U_{[\boldsymbol{A},s]}$,
$$\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}([\boldsymbol{B}, t]) = \mathrm{d}\pi_{(\boldsymbol{A},s)}((\boldsymbol{B}\boldsymbol{Q}(\boldsymbol{A}, \boldsymbol{B}) - \boldsymbol{A}, t - s)), \tag{1.15}$$
where
$$\boldsymbol{Q}(\boldsymbol{A}, \boldsymbol{B}) = (\boldsymbol{B}^T\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{B})^{-\frac{1}{2}}\boldsymbol{B}^T\boldsymbol{A} \in O(q). \tag{1.16}$$
5. $U_{[\boldsymbol{A},s]}$ equals to the set of all $[\boldsymbol{B}, t] \in M$ such that $\boldsymbol{A}^T\boldsymbol{B}$ is invertible and
$$\|\boldsymbol{B}\boldsymbol{Q}(\boldsymbol{A}, \boldsymbol{B}) - \boldsymbol{A}\|_F^2 + (t - s)^2 < \min\left(\sigma_{\min}(\boldsymbol{A}), s\right)^2. \tag{1.17}$$

Proof. Before we start the proof, let us first check that all the statements are indeed well-defined. First of all, the curve γ is well defined. If $(\boldsymbol{A}\boldsymbol{Q}, s)$, where $\boldsymbol{Q} \in O(q)$, is another representative of $[\boldsymbol{A}, s]$, let $(\boldsymbol{V}', c')$ be the horizontal lift of $\boldsymbol{v}$ to $(\boldsymbol{A}\boldsymbol{Q}, s)$. With the function $R_{\boldsymbol{Q}} : \Theta \to \Theta$ defined by $R_{\boldsymbol{Q}}((\boldsymbol{A}, s)) = (\boldsymbol{A}\boldsymbol{Q}, s)$, we have

$$\boldsymbol{v} = \mathrm{d}\pi_{(\boldsymbol{A},s)}(\boldsymbol{V}, c) = \mathrm{d}(\pi \circ R_{\boldsymbol{Q}})_{(\boldsymbol{A},s)}(\boldsymbol{V}, c) = \mathrm{d}\pi_{(\boldsymbol{A}\boldsymbol{Q},s)}(dR_{\boldsymbol{Q}})_{(\boldsymbol{A},s)}(\boldsymbol{V}, c) = \mathrm{d}\pi_{(\boldsymbol{A}\boldsymbol{Q},s)}(\boldsymbol{V}\boldsymbol{Q}, c).$$

Thus, $\boldsymbol{V}' = \boldsymbol{V}\boldsymbol{Q}$ and $c' = c$. Substitute this into the definition of γ will yield the exact same curve. Secondly, if $\boldsymbol{R} \in O(q)$ and $\boldsymbol{P} \in O(q)$, then

$$\boldsymbol{Q}(\boldsymbol{A}\boldsymbol{P}, \boldsymbol{B}\boldsymbol{R}) = \boldsymbol{R}^T\boldsymbol{Q}(\boldsymbol{A}, \boldsymbol{B})\boldsymbol{P}. \tag{1.18}$$

Therefore, the same computation will show that (1.15) is indeed well-defined. Also, the contents of the inequality (1.17) is independent of representatives and the description of the set $U_{[\boldsymbol{A},s]}$ makes sense.

A well-known property of geodesics in case of Riemannian submersion (see for example O'Neill, 1967, Section 2 Corollary 2 or Valencia, 2021, Theorem 3.6) is that any geodesic c in Θ with initial velocity being horizontal will always have horizontal velocities and $\gamma = \pi \circ c$ will remain a geodesic in M. Therefore, (1.14) is valid for all vectors $\boldsymbol{v} \in T_{[\boldsymbol{A},s]}M$ of small enough norm making sure that $(\boldsymbol{A} + \boldsymbol{V}, s + c) \in \Theta$. To have a quantified upper bound of the norm, we use the classical result of Weyl's perturbation inequality (see for example Chapter 7.3 Problem 16 equation (7.3.15) of Horn and Johnson, 2013) to get

$$\sigma_{\min}(\boldsymbol{A} + \boldsymbol{V}) \geqslant \sigma_{\min}(\boldsymbol{A}) - \|\boldsymbol{V}\| \geqslant \sigma_{\min}(\boldsymbol{A}) - \|\boldsymbol{V}\|_F \geqslant \sigma_{\min}(\boldsymbol{A}) - |\boldsymbol{v}| > 0,$$

whenever $|\boldsymbol{v}| < \min(\sigma_{\min}(\boldsymbol{A}), s)$ or equivalently $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$. We have made use of the two facts that $\|\boldsymbol{V}\| \leqslant \|\boldsymbol{V}\|_F$ for any matrix $\boldsymbol{V}$ and that $|\boldsymbol{v}| = \sqrt{\|\boldsymbol{V}\|_F^2 + c^2}$ by isometry. Thus, for all $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$, $\operatorname{rank}(\boldsymbol{A}+\boldsymbol{V}) = q$ and the exponential map is well-defined.

We now start the proof of the five facts of the function $\operatorname{Exp}_{[\boldsymbol{A},s]} : \mathcal{E}_{[\boldsymbol{A},s]} \to U_{[\boldsymbol{A},s]}$.

The first fact can be quickly settled. Let $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$ with horizontal lift $(\boldsymbol{V}, c)$ to $(\boldsymbol{A}, s)$ be given. As proved in Proposition 1, $\boldsymbol{V}^T\boldsymbol{A} = \boldsymbol{A}^T\boldsymbol{V}$ and thus $\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V})$ is symmetric. If $\boldsymbol{x} \in \mathbb{R}^q$ is any unit vector, then

$$\boldsymbol{x}^T\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V})\boldsymbol{x} = |\boldsymbol{A}\boldsymbol{x}|^2 + (\boldsymbol{A}\boldsymbol{x})^T\boldsymbol{V}\boldsymbol{x} \geqslant |\boldsymbol{A}\boldsymbol{x}|(|\boldsymbol{A}\boldsymbol{x}| - |\boldsymbol{V}\boldsymbol{x}|) \geqslant \sigma_{\min}(\boldsymbol{A})(\sigma_{\min}(\boldsymbol{A}) - \|\boldsymbol{V}\|_F) > 0$$

because $\sigma_{\min}(\boldsymbol{A}) > |\boldsymbol{v}| \geqslant \|\boldsymbol{V}\|_F$.

The second fact has a longer proof. Being a well-defined exponential map, $\operatorname{Exp}_{[\boldsymbol{A},s]}$ is automatically a smooth map. To prove that it is actually a diffeomorphism in $\mathcal{E}_{[\boldsymbol{A},s]}$, we must first show that it is injective in $\mathcal{E}_{[\boldsymbol{A},s]}$ so that the inverse function, the logarithmic map $\operatorname{Exp}_{[\boldsymbol{A},s]}^{-1}$, exists. Once this is done, we prove that $U_{[\boldsymbol{A},s]}$ is open and that $\operatorname{Exp}_{[\boldsymbol{A},s]}^{-1}$ is smooth in one shot by proving that its differential $\mathrm{d}\left(\operatorname{Exp}_{[\boldsymbol{A},s]}\right)_{\boldsymbol{v}}$ is an isomorphism for every given $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$.

We thus move to show that $\operatorname{Exp}_{[\boldsymbol{A},s]}$ is injective in $\mathcal{E}_{[\boldsymbol{A},s]}$. Suppose that $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ are two vectors in $\mathcal{E}_{[\boldsymbol{A},s]}$ who lifts to $(\boldsymbol{V}_1, c_1)$ and $(\boldsymbol{V}_2, c_2)$ respectively at $(\boldsymbol{A}, s)$ such that $\operatorname{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}_1) = \operatorname{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}_2)$. We prove that $\boldsymbol{v}_1 = \boldsymbol{v}_2$. (1.14) immediately shows that $c_1 = c_2$ and

$$\boldsymbol{A}+\boldsymbol{V}_1 = (\boldsymbol{A}+\boldsymbol{V}_2)\boldsymbol{Q}$$

for some $\boldsymbol{Q} \in O(q)$. Lemma 2 with the symmetric positive definite matrix $\boldsymbol{L} = \boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V}_2)$ shows that $\|\boldsymbol{Q}-\boldsymbol{I}_q\|$ can only equal to 0 or 2. Given that

$$\sigma_{\min}(\boldsymbol{A})\|\boldsymbol{Q}-\boldsymbol{I}_q\| \leqslant \|\boldsymbol{A}\boldsymbol{Q}-\boldsymbol{A}\| = \|\boldsymbol{V}_1 - \boldsymbol{V}_2\boldsymbol{Q}\| \leqslant \|\boldsymbol{V}_1\| + \|\boldsymbol{V}_2\| < 2\sigma_{\min}(\boldsymbol{A}),$$

we have proved that $\|\boldsymbol{Q}-\boldsymbol{I}_q\| < 2$ and $\boldsymbol{Q} = \boldsymbol{I}_q$. Therefore, $\boldsymbol{V}_1 = \boldsymbol{V}_2$ and $\boldsymbol{v}_1 = \boldsymbol{v}_2$. The exponential map is indeed injective in $\mathcal{E}_{[\boldsymbol{A},s]}$.

Next, let $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$ be given. We prove that $\mathrm{d}\left(\operatorname{Exp}_{[\boldsymbol{A},s]}\right)_{\boldsymbol{v}}$ is an isomorphism because $\ker\left(\mathrm{d}\left(\operatorname{Exp}_{[\boldsymbol{A},s]}\right)_{\boldsymbol{v}}\right)$ consists only of the zero vector. Suppose that $\boldsymbol{v}$ lifts to $(\boldsymbol{V}, c)$ at $(\boldsymbol{A}, s)$. To make the task notationally feasible, we implicitly canonically identify the Euclidean spaces together. Let us write $l : T_{[\boldsymbol{A},s]}M \to H_{(\boldsymbol{A},s)}$ for the horizontal lift map: $l(\boldsymbol{v}) = (\boldsymbol{V}, c)$, which is a linear isometry. Define a map $F : l(\mathcal{E}_{[\boldsymbol{A},s]}) \to M$ by setting, for any $(\boldsymbol{U}, b) \in l(\mathcal{E}_{[\boldsymbol{A},s]}) \subseteq H_{(\boldsymbol{A},s)}$, that $F((\boldsymbol{U}, b)) = \pi((\boldsymbol{A}+\boldsymbol{U}, s+b))$. In this notation, $\operatorname{Exp}_{[\boldsymbol{A},s]} = F \circ l$ and the chain rule shows that

$$\mathrm{d}\left(\operatorname{Exp}_{[\boldsymbol{A},s]}\right)_{\boldsymbol{v}} = \mathrm{d}F_{(\boldsymbol{V},c)} \circ \mathrm{d}l_{\boldsymbol{v}}.$$

Therefore, the only thing left to show is that $\ker\left(\mathrm{d}F_{(\boldsymbol{V},c)}\right)$ contains only the zero vector. For any $(\boldsymbol{U}, b) \in l(\mathcal{E}_{[\boldsymbol{A},s]})$, we always have

$$\mathrm{d}F_{(\boldsymbol{V},c)}((\boldsymbol{U}, b)) = \left.\frac{d}{dt}\right|_{t=0} \pi((\boldsymbol{A}+\boldsymbol{V}+t\boldsymbol{U}, s+c+tb)) = \mathrm{d}\pi_{(\boldsymbol{A}+\boldsymbol{V},s+c)}((\boldsymbol{U}, b)).$$

Therefore, if $(\boldsymbol{U}, b) \in \ker\left(\mathrm{d}F_{(\boldsymbol{V},c)}\right)$, then $(\boldsymbol{U}, b) \in V_{(\boldsymbol{A}+\boldsymbol{V},s+c)}$ so that $b = 0$ and $\boldsymbol{U} = (\boldsymbol{A}+\boldsymbol{V})\boldsymbol{\Omega}$ for some $q \times q$ skew-symmetric matrix $\boldsymbol{\Omega}$. But $(\boldsymbol{U}, b) \in H_{(\boldsymbol{A},s)}$ as well. Thus, $\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V})\boldsymbol{\Omega} = \boldsymbol{\Omega}^T(\boldsymbol{A}+\boldsymbol{V})^T\boldsymbol{A}$. Given that $\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V})$ is also symmetric, we arrive at

$$\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V})\boldsymbol{\Omega} + \boldsymbol{\Omega}\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V}) = \boldsymbol{0}.$$

By the spectral theorem, choose $\boldsymbol{P} \in O(q)$ such that $\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V}) = \boldsymbol{P}\boldsymbol{S}\boldsymbol{P}^T$ where $\boldsymbol{S}$ is the $q \times q$ diagonal matrix with diagonal entries $\lambda_1(\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V})), \ldots, \lambda_q(\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V}))$. Substitute this yields

$$\boldsymbol{S}(\boldsymbol{P}^T\boldsymbol{\Omega}\boldsymbol{P}) + (\boldsymbol{P}^T\boldsymbol{\Omega}\boldsymbol{P})\boldsymbol{S} = \mathbf{0}.$$

Looking at each entry to get

$$(\lambda_i(\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V})) + \lambda_j(\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V})))(\boldsymbol{P}^T\boldsymbol{\Omega}\boldsymbol{P})_{ij} = 0,$$

for all i and all $j \in \{1,\dots,q\}$. Given that $\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V})$ is positive definite, this forces $\boldsymbol{P}^T\boldsymbol{\Omega}\boldsymbol{P} = \mathbf{0}$ and hence $\boldsymbol{\Omega} = \mathbf{0}$. We have proven that $\mathrm{Exp}_{[\boldsymbol{A},s]}$ is a diffeomorphism in $\mathcal{E}_{[\boldsymbol{A},s]}$ and that $U_{[\boldsymbol{A},s]}$ is an open subset of M.

Now that the logarithmic map $\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]} : U_{[\boldsymbol{A},s]} \to \mathcal{E}_{[\boldsymbol{A},s]}$ is well-defined and is smooth, we will compute it and show the three last facts in one shot. Let $[\boldsymbol{B},t] \in U_{[\boldsymbol{A},s]}$ be given. By (1.14), $[\boldsymbol{B},t] = [\boldsymbol{A}+\boldsymbol{V}, s+c]$, where $(\boldsymbol{V},c)$ is the horizontal lift of $\boldsymbol{v} = \mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}([\boldsymbol{B},t])$ to $(\boldsymbol{A},s)$. It immediately follows that $c = t-s$. The first fact shows that, with $\boldsymbol{B}_0 := \boldsymbol{A}+\boldsymbol{V}$, $\boldsymbol{A}^T\boldsymbol{B}_0$ is symmetric positive definite. Given that the value of $\left|\det\left(\boldsymbol{A}^T\boldsymbol{B}\right)\right|$ is independent of representatives, $\boldsymbol{A}^T\boldsymbol{B}$ is always invertible for any representative. The third fact is therefore established. Given that $\boldsymbol{Q}(\boldsymbol{A},\boldsymbol{B}_0) = \boldsymbol{I}_q$, we immediately see that (1.15) holds with $\boldsymbol{B}_0$ in place of $\boldsymbol{B}$. However, (1.15) is independent of representatives so it is valid for every representative $\boldsymbol{B}$. Finally, let us write C for the set of all points $[\boldsymbol{B},t] \in M$ such that $\boldsymbol{A}^T\boldsymbol{B}$ is invertible and satisfies (1.17). By using $\boldsymbol{B}_0$ in place of $\boldsymbol{B}$ in (1.17), it is clear that $U_{[\boldsymbol{A},s]} \subseteq C$. Conversely, let $[\boldsymbol{B},t] \in C$ be given. Put $\boldsymbol{B}_0 := \boldsymbol{B}\boldsymbol{Q}(\boldsymbol{A},\boldsymbol{B})$. Then $\boldsymbol{A}^T\boldsymbol{B}_0$ is again symmetric positive definite. In particular, $\boldsymbol{A}^T(\boldsymbol{B}_0-\boldsymbol{A})$ is symmetric and $(\boldsymbol{V},c) := (\boldsymbol{B}_0-\boldsymbol{A}, t-s)$ is a horizontal vector at $(\boldsymbol{A},s)$. If $\boldsymbol{v} := \mathrm{d}\pi_{(\boldsymbol{A},s)}((\boldsymbol{V},c))$, then (1.17) shows that $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$ and $[\boldsymbol{B},t] \in U_{[\boldsymbol{A},s]}$. We conclude that $U_{[\boldsymbol{A},s]} = C$. The proof is complete. □

Remark 4. In the above proof, it might seem that the proof of (1.15) is trivialized while the amazing matrix (1.16) seems to come from nowhere. This is not at all the case. The matrix $\boldsymbol{Q}(\boldsymbol{A},\boldsymbol{B})$ comes from considering the cases for a generic representative whose matrix $\boldsymbol{A}^T\boldsymbol{B}$ is only invertible, but not necessarily symmetric or positive definite. Given $[\boldsymbol{B},t] \in U_{[\boldsymbol{A},s]}$, (1.14) shows that

$$\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}([\boldsymbol{B},t]) = \mathrm{d}\pi_{(\boldsymbol{A},s)}((\boldsymbol{B}\boldsymbol{Q}-\boldsymbol{A}, t-s))$$

for some $\boldsymbol{Q} \in O(q)$. Given that $\boldsymbol{A}^T(\boldsymbol{B}\boldsymbol{Q}-\boldsymbol{A})$ is symmetric, it follows that $\boldsymbol{A}^T\boldsymbol{B}\boldsymbol{Q}$ is symmetric and Lemma 2 shows that there exists $\boldsymbol{R} \in O(q)$, $\boldsymbol{P} \in O(q)$ such that $\boldsymbol{A}^T\boldsymbol{B} = \boldsymbol{R}\boldsymbol{S}\boldsymbol{P}^T$ and $\boldsymbol{Q} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{R}^T$. Thus, by continuity of the logarithmic map, if $[\boldsymbol{B},t] \to [\boldsymbol{A},s]$ and for some representative $(\boldsymbol{B},t) \to (\boldsymbol{A},s)$, then it must be the case that $\boldsymbol{B}\boldsymbol{Q}-\boldsymbol{A} \to \mathbf{0}$ or equivalently $\boldsymbol{Q} \to \boldsymbol{I}_q$. Therefore, heurestically, the only reasonable choice of $\boldsymbol{D}$ is $\boldsymbol{I}_q$. This argument is not discarded: it is still in the proof of the injectivity of the exponential map but simplified by symmetric positive definiteness. So $\boldsymbol{Q} = \boldsymbol{P}\boldsymbol{R}^T$. This value is actually uniquely determined by $\boldsymbol{A}$ and $\boldsymbol{B}$ and gives exactly $\boldsymbol{Q} = \boldsymbol{Q}(\boldsymbol{A},\boldsymbol{B})$ with $\boldsymbol{Q}(\boldsymbol{A},\boldsymbol{B})$ as in (1.16), despite that neither $\boldsymbol{P}$ nor $\boldsymbol{R}$ is unique. This is shown in the following lemma:

Lemma 5. If $\boldsymbol{L}$ is an invertible $q\times q$ matrix with singular value decomposition $\boldsymbol{L} = \boldsymbol{R}\boldsymbol{S}\boldsymbol{P}^T$, then

$$\boldsymbol{P}\boldsymbol{R}^T = (\boldsymbol{L}^T\boldsymbol{L})^{-\frac{1}{2}}\boldsymbol{L}^T.$$

Proof. Given that

$$\boldsymbol{L}^T\boldsymbol{L} = \boldsymbol{P}\boldsymbol{S}^2\boldsymbol{P}^T,$$

we have

$$(\boldsymbol{L}^T\boldsymbol{L})^{\frac{1}{2}} = \boldsymbol{P}\boldsymbol{S}\boldsymbol{P}^T.$$

Therefore:

$$\boldsymbol{L} = \boldsymbol{R}\boldsymbol{P}^T\boldsymbol{P}\boldsymbol{S}\boldsymbol{P}^T = \boldsymbol{R}\boldsymbol{P}^T(\boldsymbol{L}^T\boldsymbol{L})^{\frac{1}{2}}.$$

The result follows from inverting $(\boldsymbol{L}^T\boldsymbol{L})^{\frac{1}{2}}$ and take transpose. □

We end this section with a result concerning the limit of a sequence.

Proposition 6. Given a sequence $\{\boldsymbol{A}_n\}_{n=1}^{\infty}$ of real $p \times q$ matrices with the property $\boldsymbol{A}_n\boldsymbol{A}_n^T \to \boldsymbol{A}\boldsymbol{A}^T$ as $n \to \infty$ for a $p \times q$ matrix $\boldsymbol{A}$ with $\operatorname{rank}(\boldsymbol{A}) = q$, then

$$\lim_{n\to\infty} \min_{\boldsymbol{Q}\in O(q)} \|\boldsymbol{A}_n\boldsymbol{Q} - \boldsymbol{A}\|_F = 0.$$

In particular, if $\{(\boldsymbol{A}_n, s_n)\}_{n=1}^{\infty}$ is a sequence in Θ such that, for some $(\boldsymbol{A}, s) \in \Theta$, we have both $s_n \to s$ and $\Phi((\boldsymbol{A}_n, s_n)) \to \Phi((\boldsymbol{A}, s))$ as $n \to \infty$, then $[\boldsymbol{A}_n, s_n] \to [\boldsymbol{A}, s]$ in M as $n \to \infty$.

Proof. Assume, for the sake of contradiction, that there exists an $\epsilon > 0$ and a subsequence $\{n_k\}_{k=1}^{\infty}$ such that, for all k,

$$\min_{\boldsymbol{Q}\in O(q)} \|\boldsymbol{A}_{n_k}\boldsymbol{Q} - \boldsymbol{A}\|_F \geqslant \epsilon.$$

Given that $\|\boldsymbol{A}_n\|_F^2 = \operatorname{tr}\left(\boldsymbol{A}_n\boldsymbol{A}_n^T\right)$ is convergent, in particular bounded, choose subsubsequence $\{n_{k_l}\}_{l=1}^{\infty}$ such that $\boldsymbol{A}_{n_{k_l}} \to \boldsymbol{B}$ as $l \to \infty$ for some $p \times q$ matrix $\boldsymbol{B}$. It follows that $\boldsymbol{B}\boldsymbol{B}^T = \boldsymbol{A}\boldsymbol{A}^T$. Therefore, $\operatorname{rank}(\boldsymbol{A}) = \operatorname{rank}(\boldsymbol{B}) = q$ and $\boldsymbol{B} = \boldsymbol{A}\boldsymbol{P}$ for some $\boldsymbol{P} \in O(q)$. We have

$$\min_{\boldsymbol{Q}\in O(q)} \|\boldsymbol{A}_{n_{k_l}}\boldsymbol{Q} - \boldsymbol{A}\|_F \leqslant \|\boldsymbol{A}_{n_{k_l}}\boldsymbol{P}^T - \boldsymbol{A}\|_F = \|\boldsymbol{A}_{n_{k_l}}\boldsymbol{P}^T - \boldsymbol{B}\boldsymbol{P}^T\|_F = \|\boldsymbol{A}_{n_{k_l}} - \boldsymbol{B}\|_F \to 0$$

as $l \to \infty$, a contradiction.

Now suppose that $\{(\boldsymbol{A}_n, s_n)\}_{n=1}^{\infty}$ is a sequence in Θ such that, for some $(\boldsymbol{A}, s) \in \Theta$, we have both $s_n \to s$ and $\Phi((\boldsymbol{A}_n, s_n)) \to \Phi((\boldsymbol{A}, s))$ as $n \to \infty$. It follows that $\boldsymbol{A}_n\boldsymbol{A}_n^T \to \boldsymbol{A}\boldsymbol{A}^T$ and the previous paragraph shows that there exists a sequence $\{\boldsymbol{Q}_n\}_{n=1}^{\infty}$ of $q \times q$ orthogonal matrices such that $\boldsymbol{A}_n\boldsymbol{Q}_n \to \boldsymbol{A}$ as $n \to \infty$. Thus,

$$\lim_{n\to\infty} [\boldsymbol{A}_n, s_n] = \lim_{n\to\infty} \pi((\boldsymbol{A}_n\boldsymbol{Q}_n, s_n)) = \pi((\boldsymbol{A}, s)) = [\boldsymbol{A}, s]\,.$$

The proof is complete. □

1.3 Properties of the Probabilistic Principal Component Analysis Model

We now show that the statistical model $\mathcal{P}$ given by (1.7) is *differentiable in quadratic mean* and has *positive definite Fisher information operator* in the sense developed in Sun et al. (2026), definition III.1 and III.2.

Let us introduce a few notations. For each $[\boldsymbol{W}, \sigma^2] \in M$, let

$$p(\boldsymbol{x}; [\boldsymbol{W}, \sigma^2]) = (2\pi)^{-\frac{p}{2}} \det{}^{-\frac{1}{2}}(\boldsymbol{\Sigma}([\boldsymbol{W}, \sigma^2])) \exp\left(-\frac{1}{2}\boldsymbol{x}^T(\boldsymbol{\Sigma}([\boldsymbol{W}, \sigma^2]))^{-1}\boldsymbol{x}\right) \tag{1.19}$$

be the density of $N_p(\boldsymbol{0}, \boldsymbol{\Sigma}([\boldsymbol{W}, \sigma^2]))$. For $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{W},\sigma^2]} \subset T_{[\boldsymbol{W},\sigma^2]}M \cong \mathbb{R}^d$, write $l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})$ for the log-likelihood function of the local experiment

$$l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) = \log\left(p(\boldsymbol{x}; \operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right). \tag{1.20}$$

Define the square root likelihood function $r_{[\boldsymbol{W},\sigma^2]} : \mathcal{E}_{[\boldsymbol{W},\sigma^2]} \to L^2(\mathbb{R}^p)$ by setting, for all $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{W},\sigma^2]}$, that

$$r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) = \sqrt{p(\boldsymbol{x}; \operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))}. \tag{1.21}$$

Theorem 7. *$r_{[\boldsymbol{W},\sigma^2]}$ is Fréchet differentiable at any $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{W},\sigma^2]}$. In particular, $\mathcal{P}$ is differentiable in quadratic mean with score*

$$s(\boldsymbol{x};[\boldsymbol{W},\sigma^2])(\boldsymbol{h}) = \frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-1}\left[\boldsymbol{x}\boldsymbol{x}^T - \left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)\right]\left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)\right\}, \tag{1.22}$$

where

$$D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2]) := \left.\frac{d}{dt}\right|_{t=0}\boldsymbol{\Sigma}(Exp_{[\boldsymbol{W},\sigma^2]}(t\boldsymbol{h})) = \boldsymbol{U}\boldsymbol{W}^T + \boldsymbol{W}\boldsymbol{U}^T + b\boldsymbol{I}_p, \tag{1.23}$$

if $\boldsymbol{h} \in T_{[\boldsymbol{W},\sigma^2]}M$ lifts to $(\boldsymbol{U}, b)$ at $(\boldsymbol{W},\sigma^2)$. Furthermore, $\mathcal{P}$ is a regular model in the sense that the Fisher information operator

$$G_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{h}_1,\boldsymbol{h}_2) = \frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-1}D_{\boldsymbol{h}_1}\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-1}D_{\boldsymbol{h}_2}\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right\} \tag{1.24}$$

is positive definite.

Proof. The proof for the Fréchet differentiability is long but routine. We compute the Gâteaux derivative of $r_{[\boldsymbol{W},\sigma^2]}$ at $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{W},\sigma^2]}$ in a given direction $\boldsymbol{h} \in T_{[\boldsymbol{W},\sigma^2]}M$ and show that the Gâteaux derivative is the Fréchet derivative by a direct analysis of the remainder term. At the very end, we specialize our computations to $\boldsymbol{v} = \boldsymbol{0}$ to prove (1.22) and (1.24).

Given $\boldsymbol{v}$ and $\boldsymbol{h}$, there exists an $\epsilon > 0$ such that $\boldsymbol{v} + t\boldsymbol{h} \in \mathcal{E}_{[\boldsymbol{W},\sigma^2]}$ whenever $t \in (-\epsilon,\epsilon)$. If they respectively lift to $(\boldsymbol{V}, c)$ and $(\boldsymbol{U}, b)$ at $(\boldsymbol{W},\sigma^2)$, then $\boldsymbol{v} + t\boldsymbol{h}$ lifts to $(\boldsymbol{V} + t\boldsymbol{U}, c + tb)$ at $(\boldsymbol{W},\sigma^2)$. First of all:

$$\begin{aligned}
D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})) &:= \left.\frac{d}{dt}\right|_{t=0}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})) \\
&= \left.\frac{d}{dt}\right|_{t=0}\left\{(\boldsymbol{W}+\boldsymbol{V}+t\boldsymbol{U})(\boldsymbol{W}+\boldsymbol{V}+t\boldsymbol{U})^T + (\sigma^2+c+tb)\boldsymbol{I}_p\right\} \\
&= \boldsymbol{U}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{U}^T + \boldsymbol{U}\boldsymbol{W}^T + \boldsymbol{W}\boldsymbol{U}^T + b\boldsymbol{I}_p,
\end{aligned} \tag{1.25}$$

proving (1.23) with special case $\boldsymbol{v} = \boldsymbol{0}$. Secondly, for all $\boldsymbol{x} \in \mathbb{R}^p$:

$$\begin{aligned}
&D_{\boldsymbol{h}}l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) := \left.\frac{d}{dt}\right|_{t=0} l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})(\boldsymbol{x}) \\
&= \left.\frac{d}{dt}\right|_{t=0}\left\{-\frac{p}{2}\log(2\pi) - \frac{1}{2}\log\det\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}))\right) - \frac{1}{2}\boldsymbol{x}^T\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}))\right)^{-1}\boldsymbol{x}\right\} \\
&= -\frac{1}{2}\operatorname{tr}\left[\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(\left.\frac{d}{dt}\right|_{t=0}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}))\right)\right] \\
&+ \frac{1}{2}\boldsymbol{x}^T\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(\left.\frac{d}{dt}\right|_{t=0}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h}))\right)\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\boldsymbol{x} \\
&= -\frac{1}{2}\operatorname{tr}\left[\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\right] \\
&+ \frac{1}{2}\boldsymbol{x}^T\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\boldsymbol{x} \\
&= \frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left[\boldsymbol{x}\boldsymbol{x}^T - \left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\right]\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\right\}.
\end{aligned} \tag{1.26}$$

Finally:

$$\begin{aligned}
D_{\boldsymbol{h}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) = D_{\boldsymbol{h}}\exp\left(\frac{1}{2}l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})\right) &= \frac{1}{2}\exp\left(\frac{1}{2}l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})\right)D_{\boldsymbol{h}}l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) \\
&= \frac{1}{2}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})D_{\boldsymbol{h}}l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}).
\end{aligned} \tag{1.27}$$

To prove that (1.27) indeed defines a valid Gâteaux derivative, we must prove that $D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}) \in L^2(\mathbb{R}^p)$. Luckily, the expectation and covariance of normal quadratic forms have already been very well studied in Gupta and Nagar (2000). For $\boldsymbol{X} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$, Theorem 7.7.1(i) in Gupta and Nagar (2000) states that

$$\mathbb{E}\left[\boldsymbol{X}^T \boldsymbol{Z} \boldsymbol{X}\right] = \operatorname{tr}\left(\boldsymbol{Z}\boldsymbol{\Sigma}\right)$$

for any $p \times p$ matrix $\boldsymbol{Z}$ and Theorem 7.7.2 in Gupta and Nagar (2000) states that

$$\operatorname{Cov}\left(\boldsymbol{X}^T \boldsymbol{Z} \boldsymbol{X}, \boldsymbol{X}^T \boldsymbol{Y} \boldsymbol{X}\right) = \operatorname{tr}\big(\boldsymbol{Z}\boldsymbol{\Sigma}\boldsymbol{Y}^T\boldsymbol{\Sigma}\big) + \operatorname{tr}\big(\boldsymbol{Z}^T\boldsymbol{\Sigma}\boldsymbol{Y}^T\boldsymbol{\Sigma}\big)$$

for any two $p \times p$ matrices $\boldsymbol{Y}$ and $\boldsymbol{Z}$. Therefore, we can confirm without effort that the expectation of derivative of log density is zero:

$$\begin{aligned}
&\int_{\mathbb{R}^p} D_{\boldsymbol{h}} l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) p(\boldsymbol{x}; \boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})))\mathrm{d}\boldsymbol{x} \\
&= -\frac{1}{2}\operatorname{tr}\left[\left(\boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\right] \\
&+ \frac{1}{2}\int_{\mathbb{R}^p}\left[\boldsymbol{x}^T\left(\boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\left(\boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\boldsymbol{x}\right] p(\boldsymbol{x}; \boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})))\mathrm{d}\boldsymbol{x} = 0
\end{aligned}$$

and, whenever $\boldsymbol{h}_1$ and $\boldsymbol{h}_2 \in T_{[\boldsymbol{W},\sigma^2]}M$:

$$\begin{aligned}
&\int_{\mathbb{R}^p} D_{\boldsymbol{h}_1} l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) D_{\boldsymbol{h}_2} l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) p(\boldsymbol{x}; \boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})))\mathrm{d}\boldsymbol{x} \\
&= \frac{1}{4}\operatorname{Cov}\left(\boldsymbol{X}^T\boldsymbol{Z}_1\boldsymbol{X},\ \boldsymbol{X}^T\boldsymbol{Z}_2\boldsymbol{X}\right) = \frac{1}{4}\cdot 2\operatorname{tr}\left(\boldsymbol{Z}_1\boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\boldsymbol{Z}_2\boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right) \\
&= \frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1} D_{\boldsymbol{h}_1}\boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\left(\boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1} D_{\boldsymbol{h}_2}\boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right\},
\end{aligned} \tag{1.28}$$

where $\boldsymbol{X} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})))$ and

$$\boldsymbol{Z}_j := \left(\boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}_j}\boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\left(\boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1} \tag{1.29}$$

is symmetric for $j = 1, 2$, so that the two terms in Theorem 7.7.2 of Gupta and Nagar (2000) coincide. Thus, in the particular case $\boldsymbol{h} = \boldsymbol{h}_1 = \boldsymbol{h}_2$:

$$\begin{aligned}
&\|D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|^2 = \int_{\mathbb{R}^p}\left[D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})\right]^2\mathrm{d}\boldsymbol{x} \\
&= \frac{1}{4}\int_{\mathbb{R}^p}\left(D_{\boldsymbol{h}} l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})\right)^2 p(\boldsymbol{x}; \boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})))\mathrm{d}\boldsymbol{x} = \frac{1}{8}\operatorname{tr}\left\{\left[\left(\boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1} D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\operatorname{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right]^2\right\}.
\end{aligned} \tag{1.30}$$

Therefore, $D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}) \in L^2(\mathbb{R}^p)$ and (1.27) is indeed a valid Gâteaux derivative at $\boldsymbol{v}$ in the direction $\boldsymbol{h}$.

To prove that this is indeed the Fréchet derivative, let us look at the remainder term:

$$R(\boldsymbol{v}, \boldsymbol{h})(\boldsymbol{x}) := r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v} + \boldsymbol{h})(\boldsymbol{x}) - r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x}) - D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})$$

and prove that

$$\lim_{\boldsymbol{h}\to\mathbf{0}} \frac{1}{|\boldsymbol{h}|^2}\|R(\boldsymbol{v}, \boldsymbol{h}); L^2(\mathbb{R}^p)\|^2 = 0.$$

To start, the fundamental theorem of calculus shows that

$$r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+\boldsymbol{h})(\boldsymbol{x})-r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})=\int_0^1\frac{d}{dt}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})(\boldsymbol{x})\mathrm{d}t$$
$$=\int_0^1\lim_{s\to0}\frac{r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+(t+s)\boldsymbol{h})(\boldsymbol{x})-r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})(\boldsymbol{x})}{s}\mathrm{d}t=\int_0^1 D_{\boldsymbol{h}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})(\boldsymbol{x})\mathrm{d}t.$$

Therefore, by Minkowski's integral inequality and the linearity of $D_{\boldsymbol{h}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})$ in $\boldsymbol{h}$, we have

$$\|R(\boldsymbol{v},\boldsymbol{h});L^2(\mathbb{R}^p)\|=\|\int_0^1\left(D_{\boldsymbol{h}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})(\boldsymbol{x})-D_{\boldsymbol{h}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})\right)\mathrm{d}t;L^2(\mathbb{R}^p)\|$$
$$\leqslant\int_0^1\|D_{\boldsymbol{h}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})-D_{\boldsymbol{h}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v});L^2(\mathbb{R}^p)\|\mathrm{d}t=|\boldsymbol{h}|\int_0^1\|D_{\frac{\boldsymbol{h}}{|\boldsymbol{h}|}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})-D_{\frac{\boldsymbol{h}}{|\boldsymbol{h}|}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v});L^2(\mathbb{R}^p)\|\mathrm{d}t$$

and

$$\frac{1}{|\boldsymbol{h}|^2}\|R(\boldsymbol{v},\boldsymbol{h});L^2(\mathbb{R}^p)\|^2\leqslant\left\{\int_0^1\|D_{\frac{\boldsymbol{h}}{|\boldsymbol{h}|}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})-D_{\frac{\boldsymbol{h}}{|\boldsymbol{h}|}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v});L^2(\mathbb{R}^p)\|\mathrm{d}t\right\}^2$$
$$\leqslant\left\{\int_0^1\sup_{|\boldsymbol{u}|=1}\|D_{\boldsymbol{u}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})-D_{\boldsymbol{u}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v});L^2(\mathbb{R}^p)\|\mathrm{d}t\right\}^2.$$

The proof will be complete once we can show that

$$\lim_{\boldsymbol{h}\to\boldsymbol{0}}\int_0^1\sup_{|\boldsymbol{u}|=1}\|D_{\boldsymbol{u}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})-D_{\boldsymbol{u}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v});L^2(\mathbb{R}^p)\|\mathrm{d}t=0.\tag{1.31}$$

We must therefore examine the continuity of $D_{\boldsymbol{h}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})$, which is easy to see if we revisit the formulas. (1.25) shows that $D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))$ is smooth in both $\boldsymbol{h}$ and $\boldsymbol{v}$. Thus, (1.27) shows that, for each fixed $\boldsymbol{x}\in\mathbb{R}^p$, $D_{\boldsymbol{h}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})$ is smooth in both $\boldsymbol{h}$ and $\boldsymbol{v}$ and (1.30) shows that $\|D_{\boldsymbol{h}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v});L^2(\mathbb{R}^p)\|$ is smooth in both $\boldsymbol{h}$ and $\boldsymbol{v}$. Scheffé's lemma shows that $D_{\boldsymbol{h}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})$, as a map to $L^2(\mathbb{R}^p)$, is continuous in $\boldsymbol{h}$ and in $\boldsymbol{v}$. The integrand inside (1.31) is thus bounded uniformly over all $t\in[0,1]$ and all $\boldsymbol{h}$ in a compact neighborhood of $\boldsymbol{0}$ and it suffices to show that, for every $t\in[0,1]$, that

$$\lim_{\boldsymbol{h}\to\boldsymbol{0}}\sup_{|\boldsymbol{u}|=1}\|D_{\boldsymbol{u}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})-D_{\boldsymbol{u}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v});L^2(\mathbb{R}^p)\|=0,\tag{1.32}$$

thanks to the bounded convergence theorem. To prove (1.32), let $\boldsymbol{e}_1,\dots,\boldsymbol{e}_d$ be a $g_{[\boldsymbol{W},\sigma^2]}$ orthonormal basis. Every unit vector $\boldsymbol{u}$ can be expressed as $\boldsymbol{u}=\sum_{i=1}^d u_i\boldsymbol{e}_i$ with $\sum_{i=1}^d u_i^2=1$. By linearity of $D_{\boldsymbol{h}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})$ in $\boldsymbol{h}$, we have

$$\|D_{\boldsymbol{u}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})-D_{\boldsymbol{u}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v});L^2(\mathbb{R}^p)\|\leqslant\sum_{i=1}^d|u_i|\|D_{\boldsymbol{e}_i}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})-D_{\boldsymbol{e}_i}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v});L^2(\mathbb{R}^p)\|$$
$$\leqslant\sqrt{d}\max_{i=1,\dots,d}\|D_{\boldsymbol{e}_i}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})-D_{\boldsymbol{e}_i}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v});L^2(\mathbb{R}^p)\|.$$

Taking supremum yields

$$\sup_{|\boldsymbol{u}|=1}\|D_{\boldsymbol{u}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})-D_{\boldsymbol{u}}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v});L^2(\mathbb{R}^p)\|\leqslant\sqrt{d}\max_{i=1,\dots,d}\|D_{\boldsymbol{e}_i}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})-D_{\boldsymbol{e}_i}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v});L^2(\mathbb{R}^p)\|.$$

Given that $D_{\boldsymbol{e}_i}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}+t\boldsymbol{h})\to D_{\boldsymbol{e}_i}r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})$ in $L^2(\mathbb{R}^p)$ as $\boldsymbol{h}\to\boldsymbol{0}$ for every i, every t, and every $\boldsymbol{v}$, the proof of Fréchet differentiability is complete.

Finally, the formulas (1.25), (1.26), and (1.27) simplifies when $\boldsymbol{v} = \boldsymbol{0}$:

$$D_{\boldsymbol{h}} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{0})(\boldsymbol{x}) = \frac{1}{2} r_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{0})(\boldsymbol{x}) D_{\boldsymbol{h}} l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{0})(\boldsymbol{x})$$
$$= \frac{1}{4}\sqrt{p(\boldsymbol{x};[\boldsymbol{W},\sigma^2])}\,\mathrm{tr}\left\{\left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-1}\left[\boldsymbol{x}\boldsymbol{x}^T - \left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)\right]\left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)\right\}.$$

The score $s(\boldsymbol{x};[\boldsymbol{W},\sigma^2])(\boldsymbol{h})$ therefore equals to $D_{\boldsymbol{h}} l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{0})(\boldsymbol{x})$ and is indeed given by (1.22). The Fisher information operator (1.24) is obtained by taking $\boldsymbol{v} = \boldsymbol{0}$ in (1.28). To show that $G_{[\boldsymbol{W},\sigma^2]}$ is positive definite, let $\boldsymbol{h} \in T_{[\boldsymbol{W},\sigma^2]}M$ be given. We show that $G_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{h},\boldsymbol{h}) \geqslant 0$ and equality holds only when $\boldsymbol{h} = \boldsymbol{0}$. The non-negative part is clear from (1.24):

$$G_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{h},\boldsymbol{h}) = \frac{1}{2}\,\mathrm{tr}\left\{\left[\left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-1} D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right]^2\right\} \geqslant 0.$$

Assume from now on that the equality holds. A slight annoyance here is that the product of two symmetric matrices is not necessarily symmetric and we cannot immediately conclude that $D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2]) = \boldsymbol{0}$. We easily overcome this difficulty by using the cyclic property of the trace to get a symmetric matrix

$$\boldsymbol{N} := \left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-\frac{1}{2}} D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\left(\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2])\right)^{-\frac{1}{2}}$$

such that

$$0 = G_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{h},\boldsymbol{h}) = \frac{1}{2}\,\mathrm{tr}\left(\boldsymbol{N}^2\right).$$

Therefore, $\boldsymbol{N} = \boldsymbol{0}$ and $D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W},\sigma^2]) = \boldsymbol{0}$. Now (1.23) reads the same as (1.9) in the proof of Proposition 1. We therefore get $(\boldsymbol{U},b) \in \ker(d\Phi_{(\boldsymbol{W},\sigma^2)}) = V_{(\boldsymbol{W},\sigma^2)}$. However, being lifted from $\boldsymbol{h}$, $(\boldsymbol{U},b)$ must also be a horizontal vector. Therefore, $\boldsymbol{h} = \boldsymbol{0}$. We have proven that $G_{[\boldsymbol{W},\sigma^2]}$ is positive definite. □

Proposition 8. Suppose that $[\boldsymbol{W},\sigma^2] \in M$ is the true parameter so that each $\boldsymbol{X}_i$ has density $p(\,\cdot\,;[\boldsymbol{W},\sigma^2])$ as in (1.19). We have the local asymptotic normality expansion

$$\sum_{i=1}^{n} \log\frac{p\left(\boldsymbol{X}_i;\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}\left(\frac{\boldsymbol{h}}{\sqrt{n}}\right)\right)}{p(\boldsymbol{X}_i;[\boldsymbol{W},\sigma^2])} = \sum_{i=1}^{n} s(\boldsymbol{X}_i;[\boldsymbol{W},\sigma^2])\left(\frac{\boldsymbol{h}}{\sqrt{n}}\right) - \frac{1}{2}G_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{h},\boldsymbol{h}) + o_{\mathbb{P}}(1) \tag{1.33}$$

whenever n large enough that $\frac{\boldsymbol{h}}{\sqrt{n}} \in \mathcal{E}_{[\boldsymbol{W},\sigma^2]}$. In particular,

$$\sum_{i=1}^{n} \log\frac{p\left(\boldsymbol{X}_i;\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}\left(\frac{\boldsymbol{h}}{\sqrt{n}}\right)\right)}{p(\boldsymbol{X}_i;[\boldsymbol{W},\sigma^2])} \to N\left(-\frac{1}{2}G_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{h},\boldsymbol{h}),\, G_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{h},\boldsymbol{h})\right) \tag{1.34}$$

in distribution as $n \to \infty$.

Proof. Given that $\mathcal{P}$ is differentiable in quadratic mean and has positive definite Fisher operator, this follows from Proposition III.1 in Sun et al. (2026). □

1.4 Properties of the Maximum Likelihood Estimator of $\mathcal{P}$

In this section, we assume that there is a true parameter $[\boldsymbol{W}_0,\sigma_0^2] \in M$ for the model $\mathcal{P}$ given in (1.7) and we investigate the consistency and asymptotic normality of the maximum likelihood estimator towards this true parameter.

More concretely, given data points $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^p$, we consider them as realizations of independent identically distributed random variables $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$ whose true density is $p(\boldsymbol{x}; [\boldsymbol{W}_0, \sigma_0^2])$, where p is given in (1.19). In what follows, we denote the true covariance matrix by

$$\boldsymbol{\Sigma}_0 := \boldsymbol{\Sigma}([\boldsymbol{W}_0, \sigma_0^2]) = \boldsymbol{W}_0 \boldsymbol{W}_0^T + \sigma_0^2 \boldsymbol{I}_p. \tag{1.35}$$

The eigenvalues of the true covariance matrix are

$$\lambda_j(\boldsymbol{\Sigma}_0) = \begin{cases} \sigma_j(\boldsymbol{W}_0)^2 + \sigma_0^2 & j = 1, \ldots, q \\ \sigma_0^2 & j = q+1, \ldots, p. \end{cases} \tag{1.36}$$

The sample covariance matrix is defined by

$$\boldsymbol{S}_n = \frac{1}{n} \sum_{i=1}^n \boldsymbol{x}_i \boldsymbol{x}_i^T. \tag{1.37}$$

Given that we will discuss $\boldsymbol{S}_n$ a lot, we shall write λ_j in place of $\lambda_j(\boldsymbol{S}_n)$. We need also to fix one choice of orthonormal basis of eigenvectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$ where $\boldsymbol{v}_j = \boldsymbol{v}_j(\boldsymbol{S}_n)$ corresponds to the eigenvalue λ_j. The facts we will use repeatedly is that, when p and q are both fixed, $\boldsymbol{S}_n \to \boldsymbol{\Sigma}_0$ almost surely as $n \to \infty$ and, due to the fact that eigenvalues depends continuously on the matrices, $\lambda_j \to \lambda_j(\boldsymbol{\Sigma}_0)$ almost surely as $n \to \infty$.

We now formulate the maximum likelihood estimator as computed in Tipping and Bishop (1999). Put $\boldsymbol{U} = [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_q]$, a $p \times q$ matrix, and Δ_q to be a $q \times q$ diagonal matrix with diagonal entries $\lambda_1, \ldots, \lambda_q$. Then, the collection of all choices of $(\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2)$ maximizing the likelihood is given by

$$\hat{\boldsymbol{W}}_n = \boldsymbol{U} \left(\Delta_q - \hat{\sigma}_n^2 \boldsymbol{I}_q\right)^{\frac{1}{2}} \boldsymbol{R} \tag{1.38}$$

and

$$\hat{\sigma}_n^2 = \frac{1}{p-q} \sum_{j=q+1}^p \lambda_j, \tag{1.39}$$

ranged over all rotation matrices $\boldsymbol{R} \in O(q)$.

It is immediately clear from (1.36) that, almost surely, for all sufficiently large n, $\lambda_j > 0$ for all j and $\hat{\sigma}_n^2 > 0$ so that $\hat{\sigma}_n^2$ is well-defined. Also, $\hat{\sigma}_n^2 \to \sigma_0^2$ almost surely as $n \to \infty$. Another look at (1.36) shows that, almost surely, for all sufficiently large n, $\lambda_j - \hat{\sigma}_n^2 > 0$ for all $j \in \{1, \ldots, q\}$, $\lambda_q > \lambda_{q+1}$, and $\operatorname{rank}(\hat{\boldsymbol{W}}_n) = q$ so that $\hat{\boldsymbol{W}}_n$ is well-defined. Thus, almost surely, for all sufficiently large n, the same model parametrized by M has a unique maximum likelihood estimator $\left[\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2\right]$.

First of all, I would like to argue that this quotient manifold parameter space allows us to settle consistency in an (arguably more straightforward) way that does not use Datta and Chakrabarty (2023) or Redner (1981).

Theorem 9. *The maximum likelihood estimator is consistent. That is,* $\left[\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2\right] \to [\boldsymbol{W}_0, \sigma_0^2]$ *almost surely as* $n \to \infty$.

Proof. Put

$$\hat{\boldsymbol{\Sigma}}_n := \boldsymbol{\Sigma}\left(\left[\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2\right]\right) = \hat{\boldsymbol{W}}_n \hat{\boldsymbol{W}}_n^T + \hat{\sigma}_n^2 \boldsymbol{I}_p. \tag{1.40}$$

In light of Proposition 6 and the fact that $\hat{\sigma}_n^2$ is consistent, it suffices to prove that $\hat{\boldsymbol{\Sigma}}_n \to \boldsymbol{\Sigma}_0$ almost surely as $n \to \infty$. The fact that $\sum_{j=1}^p \boldsymbol{v}_j \boldsymbol{v}_j^T = \boldsymbol{I}_p$ allows us to write

$$\hat{\boldsymbol{\Sigma}}_n = \boldsymbol{U}\left(\Delta_q - \hat{\sigma}_n^2 \boldsymbol{I}_q\right) \boldsymbol{U}^T + \hat{\sigma}_n^2 \boldsymbol{I}_p = \sum_{j=1}^q (\lambda_j - \hat{\sigma}_n^2) \boldsymbol{v}_j \boldsymbol{v}_j^T + \hat{\sigma}_n^2 \boldsymbol{I}_p = \sum_{j=1}^q \lambda_j \boldsymbol{v}_j \boldsymbol{v}_j^T + \hat{\sigma}_n^2 \sum_{j=q+1}^p \boldsymbol{v}_j \boldsymbol{v}_j^T$$

and

$$\boldsymbol{S}_n = \boldsymbol{S}_n \sum_{j=1}^{p} \boldsymbol{v}_j \boldsymbol{v}_j^T = \sum_{j=1}^{p} \lambda_j \boldsymbol{v}_j \boldsymbol{v}_j^T.$$

Thus,

$$\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{S}_n = \sum_{j=q+1}^{p} \left(\hat{\sigma}_n^2 - \lambda_j\right) \boldsymbol{v}_j \boldsymbol{v}_j^T$$

and

$$\|\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{S}_n\| \leqslant \sum_{j=q+1}^{p} \left|\hat{\sigma}_n^2 - \lambda_j\right| \|\boldsymbol{v}_j \boldsymbol{v}_j^T\| = \sum_{j=q+1}^{p} \left|\hat{\sigma}_n^2 - \lambda_j\right|.$$

Therefore, given that both $\hat{\sigma}_n^2$ and λ_j will converge almost surely to σ_0^2 when $j \in \{q+1, \ldots, p\}$, $\|\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{S}_n\| \to 0$ almost surely as $n \to \infty$. The proof is complete. □

We now move to prove that, locally at the true parameter, the maximum likelihood estimator is asymptotically normal in the tangent space of the true parameter by local linearity. Given that the manifold is locally Euclidean, this result can be attacked with the classical approach, namely by verifying conditions of Theorem 5.23 in van der Vaart (1998), without too much use of geometric analysis. We start with computational lemmas.

Lemma 10. Given $[\boldsymbol{W}, \sigma^2] \in M$ and any convex neighborhood U of $\mathbf{0}$ in $T_{[\boldsymbol{W},\sigma^2]}M$ such that $\overline{U} \subset \mathcal{E}_{[\boldsymbol{W},\sigma^2]}$, there exists a function $m : \mathbb{R}^p \to \mathbb{R}$ such that, whenever $\boldsymbol{v}_1$, $\boldsymbol{v}_2 \in U$ and $\boldsymbol{x} \in \mathbb{R}^p$, we always have

$$\left|l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}_1)(\boldsymbol{x}) - l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}_2)(\boldsymbol{x})\right| \leqslant m(\boldsymbol{x})|\boldsymbol{v}_1 - \boldsymbol{v}_2|.$$

This m has its second moment under any distribution μ on $\mathbb{R}^p$ that has the fourth moment. That is,

$$\int_{\mathbb{R}^p} (m(\boldsymbol{x}))^2 \mathrm{d}\mu(\boldsymbol{x}) < \infty$$

whenever

$$\int_{\mathbb{R}^p} |\boldsymbol{x}|^4 \mathrm{d}\mu(\boldsymbol{x}) < \infty.$$

Proof. Given $[\boldsymbol{W}, \sigma^2] \in M$ and a convex neighborhood U of $\mathbf{0}$ compactly contained in $\mathcal{E}_{[\boldsymbol{W},\sigma^2]}$, the result will follow by the mean value inequality if we can find a function m such that

$$\left|D_{\boldsymbol{h}} l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})\right| \leqslant m(\boldsymbol{x})$$

for all unit vector $\boldsymbol{h} \in T_{[\boldsymbol{W},\sigma^2]}M$, all $\boldsymbol{v} \in \overline{U}$, and all $\boldsymbol{x} \in \mathbb{R}^p$. Given that we have proved in Proposition 7 that $\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}$ is continuous in $\boldsymbol{v}$ and that $D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))$ is continuous in both $\boldsymbol{h}$ and $\boldsymbol{v}$, we choose constants $C_1 \in (0, \infty)$ and $C_2 \in (0, \infty)$ such that

$$\frac{1}{2}\left\|\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\right\|_F \leqslant C_1$$

and

$$\frac{1}{2}\left\|\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\right\|_F \leqslant C_2$$

for all all unit vector $\boldsymbol{h} \in T_{[\boldsymbol{W},\sigma^2]}M$ and all $\boldsymbol{v} \in \overline{U}$, thanks to the fact that $\overline{U}$ is compact. from Proposition 3. By (1.26), we have

$$\left|D_{\boldsymbol{h}} l_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})(\boldsymbol{x})\right| \leqslant \frac{1}{2}\|\boldsymbol{x}\boldsymbol{x}^T\|_F \left\|\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\right\|_F + \frac{1}{2}\left\|\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v}))\right)\right\|_F \leqslant C_1\boldsymbol{x}^T\boldsymbol{x} + C_2$$

Setting $m(\boldsymbol{x}) = C_1\boldsymbol{x}^T\boldsymbol{x} + C_2$ will finish the proof. □

Lemma 11. Define function $L : \mathcal{E}_{[\boldsymbol{W}_0,\sigma_0^2]} \to \mathbb{R}$ by setting, for all $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{W}_0,\sigma_0^2]}$,

$$L(\boldsymbol{v}) = \int_{\mathbb{R}^p} l_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{v})(\boldsymbol{x})p(\boldsymbol{x};\boldsymbol{\Sigma}_0)\mathrm{d}\boldsymbol{x}.$$

Then $\mathbf{0}$ is the unique maximizer of L and, for all $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{W}_0,\sigma_0^2]}$:

$$L(\boldsymbol{v}) = L(\mathbf{0}) - \frac{1}{2}G_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{v},\boldsymbol{v}) + R(\boldsymbol{v})$$

with the remainder of the second order Taylor expansion satisfying

$$\lim_{|\boldsymbol{v}|\to 0}\frac{R(\boldsymbol{v})}{|\boldsymbol{v}|^2} = 0.$$

Proof. One should immediately recognize that $L(\mathbf{0}) - L(\boldsymbol{v})$ equals to the Kullback-Leibler divergence of the true model $N_p(\mathbf{0},\boldsymbol{\Sigma}_0)$ with respect to the model $N_p\left(\mathbf{0},\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{v})\right)\right)$. Therefore, $L(\boldsymbol{v})$ is maximized at all $\boldsymbol{v}$ with $\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{v})\right) = \boldsymbol{\Sigma}_0$. Given that $\boldsymbol{\Sigma}$ is injective and that we have proven in Proposition 3 that $\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}$ is injective in $\mathcal{E}_{[\boldsymbol{W}_0,\sigma_0^2]}$, this only happens when $\boldsymbol{v} = \mathbf{0}$ and $\mathbf{0}$ is the unique maximizer of L.

To obtain the Taylor expansion, we basically have to repeat the computation part in the proof of Proposition 7 on a slightly different problem with the second derivative at $\mathbf{0}$ being computed as well. The first order derivative at $\mathbf{0}$ is automatically 0 because $\mathbf{0}$ is the maximizer.

Let $\boldsymbol{h} \in T_{[\boldsymbol{W}_0,\sigma_0^2]}M$ be given. We can easily compute that

$$L(t\boldsymbol{h}) = -\frac{p}{2}\log(2\pi) - \frac{1}{2}\log\left(\det\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\right) - \frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\boldsymbol{\Sigma}_0\right\}.$$

Suppose that $\boldsymbol{h}$ lifts to $(\boldsymbol{U}, b)$ at $(\boldsymbol{W}_0, \sigma_0^2)$. (1.25) immediately shows that

$$\left.\frac{d^2}{dt^2}\right|_{t=0}\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right) = 2\boldsymbol{U}\boldsymbol{U}^T.$$

We now compute the second derivative L term by term. For the log determinant term, we have

$$\frac{d}{dt}\left(-\frac{1}{2}\log\left(\det\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\right)\right) = -\frac{1}{2}\operatorname{tr}\left[\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\frac{d}{dt}\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\right],$$

$$\frac{d^2}{dt^2}\left(-\frac{1}{2}\log\left(\det\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\right)\right) = -\frac{1}{2}\frac{d}{dt}\operatorname{tr}\left[\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\frac{d}{dt}\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\right]$$
$$= -\frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\frac{d^2}{dt^2}\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\right\}$$
$$+\frac{1}{2}\operatorname{tr}\left\{\left[\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\frac{d}{dt}\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\right]^2\right\},$$

and therefore

$$\frac{d^2}{dt^2}\bigg|_{t=0}\left(-\frac{1}{2}\log\left(\det\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\right)\right)=-\operatorname{tr}\left\{\boldsymbol{\Sigma}_0^{-1}\boldsymbol{U}\boldsymbol{U}^T\right\}+\frac{1}{2}\operatorname{tr}\left\{\left[\boldsymbol{\Sigma}_0^{-1}D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W}_0,\sigma_0^2])\right]^2\right\}.$$

Next, for the trace term, we have

$$\begin{aligned}
&\frac{d}{dt}\left(-\frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\boldsymbol{\Sigma}_0\right\}\right)\\
&=\frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\frac{d}{dt}\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\boldsymbol{\Sigma}_0\right\},\\
&\frac{d^2}{dt^2}\left(-\frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\boldsymbol{\Sigma}_0\right\}\right)\\
&=\frac{1}{2}\operatorname{tr}\bigg\{-2\bigg\{\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\frac{d}{dt}\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\\
&\frac{d}{dt}\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\boldsymbol{\Sigma}_0\bigg\}\bigg\}\\
&+\frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\frac{d^2}{dt^2}\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\boldsymbol{\Sigma}_0\right\},
\end{aligned}$$

and therefore

$$\frac{d^2}{dt^2}\bigg|_{t=0}\left(-\frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}\left(\mathrm{Exp}_{[\boldsymbol{W}_0,\sigma_0^2]}(t\boldsymbol{h})\right)\right)^{-1}\boldsymbol{\Sigma}_0\right\}\right)=-\operatorname{tr}\left\{\left[\boldsymbol{\Sigma}_0^{-1}D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W}_0,\sigma_0^2])\right]^2\right\}+\operatorname{tr}\left(\boldsymbol{\Sigma}_0^{-1}\boldsymbol{U}\boldsymbol{U}^T\right).$$

We end up the negative of the Fisher information operator, as computed in (1.24).

$$\frac{d^2}{dt^2}\bigg|_{t=0}L(t\boldsymbol{h})=-\frac{1}{2}\operatorname{tr}\left\{\left[\boldsymbol{\Sigma}_0^{-1}D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{W}_0,\sigma_0^2])\right]^2\right\}=-G_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{h},\boldsymbol{h}).$$

This completes the proof. □

Definition 12. We define the **inverse Fisher information operator** $G^{-1}_{[\boldsymbol{W},\sigma^2]}$ as the inverse of the flat version of the Fisher information bilinear form. More concretely, instead of treating $G_{[\boldsymbol{W},\sigma^2]}$ as a bilinear form on $T_{[\boldsymbol{W},\sigma^2]}M$, we treat it as a map $G_{[\boldsymbol{W},\sigma^2]}:T_{[\boldsymbol{W},\sigma^2]}M\to T^*_{[\boldsymbol{W},\sigma^2]}M$ by setting $G_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v})=G_{[\boldsymbol{W},\sigma^2]}(\boldsymbol{v},\cdot)$ and we define $G^{-1}_{[\boldsymbol{W},\sigma^2]}:T^*_{[\boldsymbol{W},\sigma^2]}M\to T_{[\boldsymbol{W},\sigma^2]}M$ to be the inverse.

Proposition 13. The maximum likelihood estimator is asymptotically linear at $[\boldsymbol{W}_0,\sigma_0^2]$ with the influence function $\mathrm{IF}:\mathbb{R}^p\to T_{[\boldsymbol{W}_0,\sigma_0^2]}M$ given by

$$\mathrm{IF}(\boldsymbol{x})=G^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}(s(\boldsymbol{x};[\boldsymbol{W}_0,\sigma_0^2])). \tag{1.41}$$

As described in Sun et al. (2026) Lemma III.3, this means

$$\sqrt{n}\mathrm{Exp}^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}\left(\left[\hat{\boldsymbol{W}}_n,\hat{\sigma}_n^2\right]\right)-\frac{1}{\sqrt{n}}\sum_{i=1}^n\mathrm{IF}(\boldsymbol{X}_i)\to 0 \tag{1.42}$$

in probability as $n\to\infty$. Also, $\mathrm{IF}(\boldsymbol{X})$ has mean $\boldsymbol{0}$ and is square integrable if $\boldsymbol{X}$ follows the true distribution $N_p(\boldsymbol{0},\boldsymbol{\Sigma}_0)$.

In particular, once we fix a $g_{[\boldsymbol{W}_0,\sigma_0^2]}$ orthonormal basis $\boldsymbol{e}_1,\ldots,\boldsymbol{e}_d$ and we identify $T_{[\boldsymbol{W}_0,\sigma_0^2]}M$ canonically as $\mathbb{R}^d$ (in concrete terms, identify $\sum_{j=1}^d a_j\boldsymbol{e}_j$ to the point $(a_1,\ldots,a_d)\in\mathbb{R}^d$), then $\sqrt{n}\mathrm{Exp}^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}\left(\left[\hat{\boldsymbol{W}}_n,\hat{\sigma}_n^2\right]\right)$ converges in distribution to $N_d(\boldsymbol{0},\boldsymbol{G}^{-1})$ as $n\to\infty$, where $\boldsymbol{G}$ is the $d\times d$ matrix defined by $\boldsymbol{G}_{ij}=G_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{e}_i,\boldsymbol{e}_j)$.

Proof. Fix a $g_{[\boldsymbol{W}_0,\sigma_0^2]}$ orthonormal basis $\boldsymbol{e}_1,\ldots,\boldsymbol{e}_d$ of $T_{[\boldsymbol{W}_0,\sigma_0^2]}M$. Define the matrix $\boldsymbol{G}\in\mathbb{R}^{d\times d}$ as in the statement of lemma and the vector $\nabla l(\boldsymbol{x};[\boldsymbol{W}_0,\sigma_0^2])\in\mathbb{R}^d$ by setting $(\nabla l(\boldsymbol{x};[\boldsymbol{W}_0,\sigma_0^2]))_j = s(\boldsymbol{x};[\boldsymbol{W}_0,\sigma_0^2])(\boldsymbol{e}_j)$, for all $j=1,\ldots,d$. Identifying $\sqrt{n}\mathrm{Exp}^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}\left(\left[\hat{\boldsymbol{W}}_n,\hat{\sigma}_n^2\right]\right)\in T_{[\boldsymbol{W}_0,\sigma_0^2]}M$ to a point in $\mathbb{R}^d$ through this basis, we immediately have, by combining the two previous lemmas and Theorem 5.23 in van der Vaart (1998), that

$$\sqrt{n}\mathrm{Exp}^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}\left(\left[\hat{\boldsymbol{W}}_n,\hat{\sigma}_n^2\right]\right)-\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\boldsymbol{G}^{-1}\nabla l(\boldsymbol{X}_i;[\boldsymbol{W}_0,\sigma_0^2])\to 0 \tag{1.43}$$

in probability as $n\to\infty$. In particular, $\sqrt{n}\mathrm{Exp}^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}\left(\left[\hat{\boldsymbol{W}}_n,\hat{\sigma}_n^2\right]\right)$ converges in distribution to $N_d(\boldsymbol{0},\boldsymbol{G}^{-1})$ as $n\to\infty$. Also, if $\boldsymbol{X}$ follows the true distribution $N_p(\boldsymbol{0},\boldsymbol{\Sigma}_0)$, then $\mathbb{E}[\nabla l(\boldsymbol{X};[\boldsymbol{W}_0,\sigma_0^2])]=\boldsymbol{0}$ and thus $\mathbb{E}[\boldsymbol{G}^{-1}\nabla l(\boldsymbol{X};[\boldsymbol{W}_0,\sigma_0^2])]=\boldsymbol{0}$. Given that $\nabla l(\boldsymbol{X};[\boldsymbol{W}_0,\sigma_0^2])$ has its second moment, so does $\boldsymbol{G}^{-1}\nabla l(\boldsymbol{X};[\boldsymbol{W}_0,\sigma_0^2])$.

We remark that (1.42) is simply (1.43) rephrased in a way using only vectors in $T_{[\boldsymbol{W}_0,\sigma_0^2]}M$ defined without $\boldsymbol{e}_1,\ldots,\boldsymbol{e}_d$ and hence without the need of identifying $T_{[\boldsymbol{W}_0,\sigma_0^2]}M$ with $\mathbb{R}^d$. Let us for simplicity define $\boldsymbol{y}:=\boldsymbol{G}^{-1}\nabla l(\boldsymbol{x};[\boldsymbol{W}_0,\sigma_0^2])\in\mathbb{R}^d$. Put $\boldsymbol{v}\in T_{[\boldsymbol{W}_0,\sigma_0^2]}M$ by setting $\boldsymbol{v}=\sum_{j=1}^d \boldsymbol{y}_j\boldsymbol{e}_j\in T_{[\boldsymbol{W}_0,\sigma_0^2]}M$. We claim that $\boldsymbol{v}$ is the unique vector such that, for all $\boldsymbol{z}\in T_{[\boldsymbol{W}_0,\sigma_0^2]}M$:

$$G_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{v},\boldsymbol{z})=s(\boldsymbol{x};[\boldsymbol{W}_0,\sigma_0^2])(\boldsymbol{z}). \tag{1.44}$$

The uniqueness of $\boldsymbol{v}$ is immediate by the positive definiteness of $G_{[\boldsymbol{W}_0,\sigma_0^2]}$. Given that $(\boldsymbol{G}\boldsymbol{y})_j=(\nabla l(\boldsymbol{x};[\boldsymbol{W}_0,\sigma_0^2]))_j$ for all $j=1,\ldots,d$, we have, for all $j=1,\ldots,d$:

$$G_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{v},\boldsymbol{e}_j)=\sum_{i=1}^{d}\boldsymbol{y}_i G_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{e}_i,\boldsymbol{e}_j)=\sum_{i=1}^{d}\boldsymbol{y}_i\boldsymbol{G}_{ij}=s(\boldsymbol{x};[\boldsymbol{W}_0,\sigma_0^2])(\boldsymbol{e}_j).$$

The claim is therefore proved. By looking at Definition 12, (1.44) reads $G_{[\boldsymbol{W}_0,\sigma_0^2]}(\boldsymbol{v})=s(\boldsymbol{x};[\boldsymbol{W}_0,\sigma_0^2])$ and hence $\boldsymbol{v}=G^{-1}_{[\boldsymbol{W}_0,\sigma_0^2]}(s(\boldsymbol{x};[\boldsymbol{W}_0,\sigma_0^2]))$, giving (1.42). The results about the integral of $\mathrm{IF}(\boldsymbol{X})$ translates to here as well. This completes the proof. $\square$

Chapter 2

Draft

The first chapter produced three things: the quotient Riemannian manifold (M, g) together with its exponential and logarithmic maps; the verification that the probabilistic principal component analysis model is differentiable in quadratic mean with positive definite Fisher information operator, hence locally asymptotically normal; and, for the maximum likelihood estimator, consistency (Theorem 9) together with asymptotic linearity and asymptotic normality in the tangent space at the true parameter (Proposition 13).

That last result is a statement made under a single law and about a single estimator: it says nothing about what happens when the truth is perturbed. Section 2.1 computes the parallel transport of (M, g) and uses it to supply the missing statement, namely that the asymptotics do not degrade when the truth is moved at the scale $n^{-\frac{1}{2}}$, which is *local regularity*. Section 2.2 draws the inferential consequences, confidence sets with locally uniform coverage and the limit law of the fitted covariance matrix. Section 2.3 computes the Fisher information operator and its inverse in closed form, which is what makes any of the preceding usable in practice. Nothing from the first chapter is restated below; it is cited, and the complements it does not contain are proved where they are first needed.

Notation

Throughout this chapter we abbreviate

$$\theta_0 := [\boldsymbol{W}_0, \sigma_0^2] \in M, \qquad \hat{\theta}_n := [\hat{\boldsymbol{W}}_n, \hat{\sigma}_n^2] \in M, \qquad G := G_{\theta_0}, \tag{2.1}$$

and we keep $d = pq + 1 - \frac{q(q-1)}{2}$ of (1.2) for the dimension of M, together with $\boldsymbol{\Sigma}_0$ of (1.35), $\boldsymbol{S}_n$ of (1.37), its eigenvalues $\lambda_1 \geqslant \cdots \geqslant \lambda_p$ and eigenvectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$, the estimators (1.38) and (1.39) of Tipping and Bishop (1999), and the fitted covariance matrix $\hat{\boldsymbol{\Sigma}}_n$ of (1.40), all as in Section 1.4. The letter θ, which in Section 1.1 denoted a point of Θ, was retired at the beginning of Section 1.4 when the coordinates on Θ were renamed $(\boldsymbol{W}, \sigma^2)$; from here on θ always denotes a point of M. We write $\mathbb{P}_\theta^n$ for the law of $(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n)$ when the true parameter is θ, that is the n-fold product of $p(\,\cdot\,; \theta)\, \mathrm{d}\boldsymbol{x}$ with p as in (1.19), and $\rightsquigarrow$ for convergence in distribution; unless another law is named, $o_{\mathbb{P}}(1)$ and $O_{\mathbb{P}}(1)$ refer to $\mathbb{P}_{\theta_0}^n$. Finally $\mathrm{Sym}(m)$ denotes the space of real symmetric $m \times m$ matrices and $\preceq$ the Loewner order on it.

The horizontal lift was used repeatedly in the first chapter but never named; we name it now. For $(\boldsymbol{A}, s) \in \Theta$ let

$$\iota_{(\boldsymbol{A},s)} : T_{[\boldsymbol{A},s]}M \longrightarrow H_{(\boldsymbol{A},s)} \tag{2.2}$$

be the inverse of the isomorphism $\mathrm{d}\pi_{(\boldsymbol{A},s)}\big|_{H_{(\boldsymbol{A},s)}}$, so that $\iota_{(\boldsymbol{A},s)}(\boldsymbol{v})$ is the unique horizontal vector at $(\boldsymbol{A}, s)$ projecting to $\boldsymbol{v}$; by the very definition (1.5) of g, it is a linear isometry onto $H_{(\boldsymbol{A},s)}$. We abbreviate

$$\iota := \iota_{(\boldsymbol{W}_0, \sigma_0^2)}, \qquad \iota(\boldsymbol{h}) =: (\boldsymbol{U}_{\boldsymbol{h}},\, b_{\boldsymbol{h}}) \quad \text{for } \boldsymbol{h} \in T_{\theta_0}M, \tag{2.3}$$

so that, by part 3 of Proposition 1 and by (1.4),

$$\boldsymbol{U}_{\boldsymbol{h}}^T \boldsymbol{W}_0 = \boldsymbol{W}_0^T \boldsymbol{U}_{\boldsymbol{h}} \qquad \text{and} \qquad |\boldsymbol{h}|^2 = \|\boldsymbol{U}_{\boldsymbol{h}}\|_F^2 + b_{\boldsymbol{h}}^2. \tag{2.4}$$

Finally, put

$$\rho := \min\left(\sigma_{\min}(\boldsymbol{W}_0),\ \sigma_0^2\right) > 0, \qquad B_\epsilon := \{\boldsymbol{u} \in T_{\theta_0}M : |\boldsymbol{u}| < \epsilon\} \quad \text{for } 0 < \epsilon \leqslant \rho, \tag{2.5}$$

so that $B_\rho = \mathcal{E}_{\theta_0}$ is the domain of the exponential map supplied by Proposition 3, and put $\mathcal{U} := \operatorname{Exp}_{\theta_0}(B_\rho)$. That proposition makes $\operatorname{Exp}_{\theta_0} : B_\rho \to \mathcal{U}$ a diffeomorphism onto an open subset of M, and (1.14) makes it *affine* in the horizontal lift: $\operatorname{Exp}_{\theta_0}(\boldsymbol{u}) = [\boldsymbol{W}_0 + \boldsymbol{U}_{\boldsymbol{u}},\ \sigma_0^2 + b_{\boldsymbol{u}}]$, in the notation (2.3).

2.1 Parallel Transport and Local Regularity

Local regularity is the assertion that the conclusion of Proposition 13 survives a perturbation of the true parameter at the estimation scale $n^{-\frac{1}{2}}$. Stating it requires comparing the estimation error measured at θ_0 with the error measured at a nearby point $\theta_{n,\boldsymbol{h}}$, and these two errors live in *different* tangent spaces. Something is therefore needed to identify $T_{\theta_{n,\boldsymbol{h}}}M$ with $T_{\theta_0}M$, and the canonical choice is the parallel transport of (M, g) along the minimizing geodesic joining the two points, which we compute first.

The minimizing geodesics of M and their parallel transport

The metric g was constructed in Section 1.1 so that π is a Riemannian submersion from the *flat* manifold $\Theta \subseteq \mathbb{R}^{pq+1}$, and Proposition 3 exploited this to make the exponential map affine in the horizontal lift. The same structure turns the parallel transport into an explicit linear ordinary differential equation.

Parallel transport is an operation attached to a *curve*, whereas the object we need — the map written $\Pi_{x_1}^{x_2}$ in Section II of Sun et al. (2026) — is attached to a *pair of points*. The two are reconciled in the way that paper prescribes: between two points close enough to each other, transport is taken along the unique distance minimizing geodesic joining them. The distance in question is the Riemannian distance of (M, g),

$$d_M(x, y) := \inf\left\{\int_a^b |\dot{c}(\tau)|\,\mathrm{d}\tau \ :\ c : [a, b] \to M \text{ piecewise smooth},\ c(a) = x,\ c(b) = y\right\}, \tag{2.6}$$

the infimum of the g-lengths of the curves joining x to y (Lee, 2018, Chapter 2). That this infimum is taken over a nonempty set, so that d_M is finite on all of $M \times M$, is a global property of M; it is proved in Lemma 34 and is needed only there. Nothing in the present section uses it, because the joining curve will be exhibited and the infimum computed by hand, and what makes that possible is that Proposition 3 has already produced the geodesic balls of M. Indeed, that proposition makes $\operatorname{Exp}_{[\boldsymbol{A},s]}$ a diffeomorphism of the ball $\mathcal{E}_{[\boldsymbol{A},s]}$ of radius $\min\left(\sigma_{\min}(\boldsymbol{A}), s\right)$ in $T_{[\boldsymbol{A},s]}M$ onto the open set $U_{[\boldsymbol{A},s]} \subseteq M$, which is precisely what it means for $U_{[\boldsymbol{A},s]}$ to be a *geodesic ball* centred at $[\boldsymbol{A}, s]$ (Lee, 2018, Chapter 6), and (1.15) writes in closed form the radial geodesic reaching any prescribed point of it. Proposition 16 assembles the two ingredients into a single statement: the minimizing geodesic between two points, and the transport along it.

Two preliminaries are needed. The first makes the orthogonal splitting $T_{(\boldsymbol{A},s)}\Theta = V_{(\boldsymbol{A},s)} \oplus H_{(\boldsymbol{A},s)}$ of Proposition 1 explicit, through a Lyapunov equation that will turn out to govern the transport as well; it also shows that the horizontal lift (2.2) depends smoothly on its base point, which is used repeatedly below. The second is the submersion identity, in the form we need.

Lemma 14. Let $\boldsymbol{A} \in \mathbb{R}^{p\times q}$ have rank q and put $\boldsymbol{P}_{\boldsymbol{A}} := \boldsymbol{A}^T\boldsymbol{A}$, a positive definite $q \times q$ matrix.

1. For every $\boldsymbol{K} \in \mathbb{R}^{q\times q}$ the Lyapunov equation

$$\boldsymbol{P}_{\boldsymbol{A}}\boldsymbol{\Omega} + \boldsymbol{\Omega}\boldsymbol{P}_{\boldsymbol{A}} = \boldsymbol{K} \tag{2.7}$$

has a unique solution $\boldsymbol{\Omega} =: \mathcal{L}_{\boldsymbol{A}}(\boldsymbol{K})$. The map $\mathcal{L}_{\boldsymbol{A}}$ is linear, the map $(\boldsymbol{A}, \boldsymbol{K}) \mapsto \mathcal{L}_{\boldsymbol{A}}(\boldsymbol{K})$ is smooth, and $\mathcal{L}_{\boldsymbol{A}}(\boldsymbol{K})$ is skew-symmetric whenever $\boldsymbol{K}$ is.

2. For every $(\boldsymbol{V}, c) \in T_{(\boldsymbol{A},s)}\Theta$ the orthogonal projection $\mathcal{H}_{(\boldsymbol{A},s)}$ onto the horizontal space is

$$\mathcal{H}_{(\boldsymbol{A},s)}\left((\boldsymbol{V}, c)\right) = \left(\boldsymbol{V} - \boldsymbol{A}\, \mathcal{L}_{\boldsymbol{A}}\left(\boldsymbol{A}^T \boldsymbol{V} - \boldsymbol{V}^T \boldsymbol{A}\right),\ c\right), \tag{2.8}$$

the subtracted term being the orthogonal projection of $(\boldsymbol{V}, c)$ onto $V_{(\boldsymbol{A},s)}$.

3. Consequently $\iota_{(\boldsymbol{A},s)}\left(\mathrm{d}\pi_{(\boldsymbol{A},s)}((\boldsymbol{V}, c))\right) = \mathcal{H}_{(\boldsymbol{A},s)}\left((\boldsymbol{V}, c)\right)$, and this expression is smooth in $(\boldsymbol{A}, s, \boldsymbol{V}, c)$.

Proof. **1.** Diagonalize $\boldsymbol{P_A} = \boldsymbol{O}\,\mathrm{diag}(p_1, \dots, p_q)\,\boldsymbol{O}^T$ with $\boldsymbol{O} \in O(q)$ and every $p_i > 0$. Conjugating (2.7) by $\boldsymbol{O}$ turns the linear operator $\boldsymbol{\Omega} \mapsto \boldsymbol{P_A}\boldsymbol{\Omega} + \boldsymbol{\Omega}\boldsymbol{P_A}$ into the entrywise multiplication by $p_i + p_j > 0$, which is invertible; existence, uniqueness and linearity follow. Smoothness holds because the operator depends polynomially on $\boldsymbol{A}$ and the inverse of an invertible linear map depends smoothly on it. If $\boldsymbol{K}^T = -\boldsymbol{K}$, transposing (2.7) gives $\boldsymbol{P_A}\left(-\boldsymbol{\Omega}^T\right) + \left(-\boldsymbol{\Omega}^T\right)\boldsymbol{P_A} = -\boldsymbol{K}^T = \boldsymbol{K}$, so $-\boldsymbol{\Omega}^T$ solves the same equation and uniqueness forces $\boldsymbol{\Omega}^T = -\boldsymbol{\Omega}$.

2. By Proposition 1 the vertical space is $\left\{(\boldsymbol{A}\boldsymbol{\Omega}, 0) : \boldsymbol{\Omega}^T = -\boldsymbol{\Omega}\right\}$, so the orthogonal projection of $(\boldsymbol{V}, c)$ onto it is $(\boldsymbol{A}\boldsymbol{\Omega}_*, 0)$, where $\boldsymbol{\Omega}_*$ minimizes $\|\boldsymbol{V} - \boldsymbol{A}\boldsymbol{\Omega}\|_F^2$ over skew-symmetric $\boldsymbol{\Omega}$; the scalar component contributes the constant c^2 and plays no role. Differentiating this strictly convex quadratic in a skew-symmetric direction $\boldsymbol{K}$ gives the stationarity condition $\mathrm{tr}\left(\boldsymbol{K}^T \boldsymbol{A}^T \left(\boldsymbol{V} - \boldsymbol{A}\boldsymbol{\Omega}_*\right)\right) = 0$ for every skew-symmetric $\boldsymbol{K}$, which says exactly that the skew-symmetric part of $\boldsymbol{A}^T\boldsymbol{V} - \boldsymbol{P_A}\boldsymbol{\Omega}_*$ vanishes:

$$\boldsymbol{A}^T\boldsymbol{V} - \boldsymbol{V}^T\boldsymbol{A} = \boldsymbol{P_A}\boldsymbol{\Omega}_* - \boldsymbol{\Omega}_*^T\boldsymbol{P_A} = \boldsymbol{P_A}\boldsymbol{\Omega}_* + \boldsymbol{\Omega}_*\boldsymbol{P_A}.$$

This is (2.7) with the skew-symmetric right hand side $\boldsymbol{K} = \boldsymbol{A}^T\boldsymbol{V} - \boldsymbol{V}^T\boldsymbol{A}$, and part 1 identifies $\boldsymbol{\Omega}_* = \mathcal{L}_{\boldsymbol{A}}(\boldsymbol{K})$ and confirms that it is skew-symmetric, as it must be; subtracting $(\boldsymbol{A}\boldsymbol{\Omega}_*, 0)$ from $(\boldsymbol{V}, c)$ gives (2.8).

3. The vector $\mathcal{H}_{(\boldsymbol{A},s)}((\boldsymbol{V}, c))$ is horizontal and differs from $(\boldsymbol{V}, c)$ by a vertical vector, so it has the same image under $\mathrm{d}\pi_{(\boldsymbol{A},s)}$, whose kernel is $V_{(\boldsymbol{A},s)}$ by (1.8). Smoothness is that of (2.8). ☐

Lemma 15. Let $c : I \to \Theta$ be a smooth curve with horizontal velocity at every time, let $\gamma := \pi \circ c$, and let $\boldsymbol{x}$ be a smooth vector field along γ, with horizontal lift $\tilde{\boldsymbol{x}}(t) := \iota_{c(t)}\left(\boldsymbol{x}(t)\right)$. Then the horizontal lift of the covariant derivative $D_t\boldsymbol{x}$ of (M, g) is the horizontal part of the ordinary derivative of $\tilde{\boldsymbol{x}}$:

$$\iota_{c(t)}\left(D_t\boldsymbol{x}(t)\right) = \mathcal{H}_{c(t)}\left(\dot{\tilde{\boldsymbol{x}}}(t)\right). \tag{2.9}$$

In particular $\boldsymbol{x}$ is parallel along γ if and only if $\dot{\tilde{\boldsymbol{x}}}(t) \in V_{c(t)}$ for every t.

Proof. Write $A(\boldsymbol{x}) := D_t\boldsymbol{x}$ and $B(\boldsymbol{x}) := \mathrm{d}\pi_{c(t)}\left(\mathcal{H}_{c(t)}\left(\dot{\tilde{\boldsymbol{x}}}\right)\right)$; both are $\mathbb{R}$-linear operators on smooth vector fields along γ. Both satisfy the Leibniz rule $L(f\boldsymbol{x}) = f'\boldsymbol{x} + fL(\boldsymbol{x})$ for $f \in C^\infty(I)$: for A this is part of the definition of the covariant derivative along a curve, and for B it follows from $\iota_c(f\boldsymbol{x}) = f\tilde{\boldsymbol{x}}$, from $\frac{d}{dt}(f\tilde{\boldsymbol{x}}) = f'\tilde{\boldsymbol{x}} + f\dot{\tilde{\boldsymbol{x}}}$ and from $\mathcal{H}_{c(t)}(\tilde{\boldsymbol{x}}) = \tilde{\boldsymbol{x}}$. They agree on extendible fields. Indeed, let Y be a smooth vector field on an open subset of M and $\boldsymbol{x} = Y \circ \gamma$; then $\tilde{\boldsymbol{x}} = \tilde{Y} \circ c$ with $\tilde{Y}$ the basic field lifting Y, so $\dot{\tilde{\boldsymbol{x}}}$ is the Euclidean covariant derivative $\tilde{\nabla}_{\dot{c}}\tilde{Y}$ of Θ, whose horizontal part is the lift of $\nabla_{\mathrm{d}\pi(\dot{c})}Y = \nabla_{\dot{\gamma}}Y = D_t\boldsymbol{x}$ by the fundamental identity for Riemannian submersions (O'Neill, 1967, Section 2; Valencia, 2021, Section 3), which applies because $\dot{c}$ is horizontal. A covariant derivative along a curve is uniquely determined by $\mathbb{R}$-linearity, the Leibniz rule and its values on extendible fields (see Lee, 2018, Chapter 4, the uniqueness clause of the existence theorem for covariant derivatives along curves), so $A = B$, which is (2.9). The last sentence follows because $\iota_{c(t)}$ is injective and $\ker \mathcal{H}_{c(t)} = V_{c(t)}$. ☐

Proposition 16 (The minimizing geodesic and its parallel transport)**.** Let $[\boldsymbol{A}, s] \in M$ and let $[\boldsymbol{B}, t] \in U_{[\boldsymbol{A},s]}$. Put

$$\boldsymbol{Q} := \boldsymbol{Q}(\boldsymbol{A}, \boldsymbol{B}), \qquad \boldsymbol{V} := \boldsymbol{B}\boldsymbol{Q} - \boldsymbol{A}, \qquad c := t - s, \qquad \boldsymbol{A}_\tau := \boldsymbol{A} + \tau\boldsymbol{V}, \tag{2.10}$$

with $\boldsymbol{Q}(\,\cdot\,,\,\cdot\,)$ the orthogonal matrix (1.16), and let

$$\gamma(\tau) := [\boldsymbol{A}_\tau,\ s + \tau c] \in M, \qquad \tilde{\gamma}(\tau) := (\boldsymbol{A}_\tau,\ s + \tau c) \in \Theta, \qquad \tau \in [0,1]. \tag{2.11}$$

1. **The curve.** Neither γ nor its length depends on the representatives chosen for $[\boldsymbol{A}, s]$ and for $[\boldsymbol{B}, t]$. The curve γ is the geodesic of Proposition 3 issued from $\gamma(0) = [\boldsymbol{A}, s]$ with initial velocity

$$\boldsymbol{v} := \mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}\left([\boldsymbol{B}, t]\right) \in \mathcal{E}_{[\boldsymbol{A},s]}, \qquad \iota_{(\boldsymbol{A},s)}(\boldsymbol{v}) = (\boldsymbol{V}, c); \tag{2.12}$$

it has constant speed $|\boldsymbol{v}|$ and ends at $\gamma(1) = [\boldsymbol{B}, t]$, and $\tilde{\gamma}$ is its horizontal lift through $(\boldsymbol{A}, s)$, a straight line of constant horizontal velocity $(\boldsymbol{V}, c)$. Its length realizes the Riemannian distance (2.6):

$$d_M\left([\boldsymbol{A}, s],\ [\boldsymbol{B}, t]\right) = |\boldsymbol{v}| = \sqrt{\|\boldsymbol{B}\boldsymbol{Q}(\boldsymbol{A}, \boldsymbol{B}) - \boldsymbol{A}\|_F^2 + (t-s)^2}, \tag{2.13}$$

and, if $[\boldsymbol{B}, t] \neq [\boldsymbol{A}, s]$, then γ is, up to reparametrization, the *only* curve in M from $[\boldsymbol{A}, s]$ to $[\boldsymbol{B}, t]$ of that length. Finally $\mathrm{rank}(\boldsymbol{A}_\tau) = q$ for every $\tau \in [0,1]$, so Lemma 14 applies to $\boldsymbol{A}_\tau$; we abbreviate $\mathcal{L}_\tau := \mathcal{L}_{\boldsymbol{A}_\tau}$.

2. **The transport equation.** Let $\tau_0 \in [0,1]$ and $(\boldsymbol{X}_0, \beta_0) \in \mathbb{R}^{p\times q} \times \mathbb{R}$. The linear ordinary differential equation

$$\dot{\boldsymbol{X}}(\tau) = \boldsymbol{A}_\tau \boldsymbol{\Omega}(\tau), \qquad \boldsymbol{\Omega}(\tau) := \mathcal{L}_\tau\left(\boldsymbol{X}(\tau)^T \boldsymbol{V} - \boldsymbol{V}^T \boldsymbol{X}(\tau)\right), \qquad \dot{\beta}(\tau) = 0, \tag{2.14}$$

has a unique solution on all of $[0,1]$ subject to $(\boldsymbol{X}(\tau_0), \beta(\tau_0)) = (\boldsymbol{X}_0, \beta_0)$; it is linear in $(\boldsymbol{X}_0, \beta_0)$ and smooth in $(\tau, \tau_0, \boldsymbol{A}, \boldsymbol{V}, \boldsymbol{X}_0, \beta_0)$. If moreover $(\boldsymbol{X}_0, \beta_0) \in H_{\tilde{\gamma}(\tau_0)}$, then the solution stays horizontal, that is $(\boldsymbol{X}(\tau), \beta(\tau)) \in H_{\tilde{\gamma}(\tau)}$ for every $\tau \in [0,1]$, and it preserves the Euclidean norm: $\|\boldsymbol{X}(\tau)\|_F^2 + \beta(\tau)^2$ is constant.

3. **The transport map.** For $\tau_0, \tau_1 \in [0,1]$ write

$$\Pi(\gamma)_{\tau_0}^{\tau_1} : T_{\gamma(\tau_0)}M \longrightarrow T_{\gamma(\tau_1)}M \tag{2.15}$$

for the parallel transport of (M, g) along γ from time τ_0 to time τ_1. It exists, it is a linear isometry, its inverse is $\Pi(\gamma)_{\tau_1}^{\tau_0}$, and, read through the horizontal lifts $\iota_{\tilde{\gamma}(\tau)}$, it is the flow of (2.14):

$$\Pi(\gamma)_{\tau_0}^{\tau_1}\left(\mathrm{d}\pi_{\tilde{\gamma}(\tau_0)}\left((\boldsymbol{X}_0, \beta_0)\right)\right) = \mathrm{d}\pi_{\tilde{\gamma}(\tau_1)}\left((\boldsymbol{X}(\tau_1), \beta(\tau_1))\right), \qquad (\boldsymbol{X}_0, \beta_0) \in H_{\tilde{\gamma}(\tau_0)}. \tag{2.16}$$

By part 1 the curve γ is determined, up to reparametrization, by its two endpoints alone, and parallel transport is unchanged by a reparametrization; the notation

$$\Pi_{[\boldsymbol{B},t]}^{[\boldsymbol{A},s]} := \Pi(\gamma)_1^0 : T_{[\boldsymbol{B},t]}M \longrightarrow T_{[\boldsymbol{A},s]}M \tag{2.17}$$

is therefore unambiguous, and this is the map $\Pi_{x_1}^{x_2}$ of Section II of Sun et al. (2026). Explicitly, let $\boldsymbol{z} \in T_{[\boldsymbol{B},t]}M$, let

$$(\boldsymbol{Y}, \beta) := \iota_{(\boldsymbol{B}\boldsymbol{Q},t)}(\boldsymbol{z}), \qquad (\boldsymbol{B}\boldsymbol{Q},\ t) = (\boldsymbol{A} + \boldsymbol{V},\ s + c) = \tilde{\gamma}(1), \tag{2.18}$$

be the horizontal lift of $\boldsymbol{z}$ at the aligned representative $(\boldsymbol{B}\boldsymbol{Q}, t)$ of $[\boldsymbol{B}, t]$, and let $\boldsymbol{X}$ be the unique solution on $[0,1]$ of the terminal value problem

$$\dot{\boldsymbol{X}}(\tau) = \boldsymbol{A}_\tau\, \mathcal{L}_\tau\left(\boldsymbol{X}(\tau)^T \boldsymbol{V} - \boldsymbol{V}^T \boldsymbol{X}(\tau)\right), \qquad \boldsymbol{X}(1) = \boldsymbol{Y}. \tag{2.19}$$

Then

$$\Pi_{[\boldsymbol{B},t]}^{[\boldsymbol{A},s]}(\boldsymbol{z}) = \mathrm{d}\pi_{(\boldsymbol{A},s)}\left((\boldsymbol{X}(0),\ \beta)\right), \qquad \left|\Pi_{[\boldsymbol{B},t]}^{[\boldsymbol{A},s]}(\boldsymbol{z})\right| = |\boldsymbol{z}|, \tag{2.20}$$

and $(\boldsymbol{A}, \boldsymbol{V}, \boldsymbol{Y}, \beta) \mapsto (\boldsymbol{X}(0), \beta)$ is smooth, and linear in $(\boldsymbol{Y}, \beta)$. In particular $\Pi^{[\boldsymbol{A},s]}_{[\boldsymbol{A},s]} = \mathrm{id}$; and $\boldsymbol{X} \equiv \boldsymbol{V}$ solves (2.19) for $\boldsymbol{Y} = \boldsymbol{V}$, so that $\Pi^{[\boldsymbol{A},s]}_{[\boldsymbol{B},t]}(\dot\gamma(1)) = \dot\gamma(0) = \boldsymbol{v}$, as a geodesic requires.

Proof. **1.** Since $[\boldsymbol{B}, t] \in U_{[\boldsymbol{A},s]}$, part 4 of Proposition 3 gives $\boldsymbol{v} = \mathrm{d}\pi_{(\boldsymbol{A},s)}((\boldsymbol{V}, c))$ with $(\boldsymbol{V}, c)$ as in (2.10), and part 1 of the same proposition, applied to $\boldsymbol{v}$, makes $\boldsymbol{A}^T(\boldsymbol{A}+\boldsymbol{V}) = \boldsymbol{A}^T\boldsymbol{B}\boldsymbol{Q}$ symmetric; hence $\boldsymbol{A}^T\boldsymbol{V}$ is symmetric and $(\boldsymbol{V}, c)$ is horizontal at $(\boldsymbol{A}, s)$ by part 3 of Proposition 1. This is (2.12), and Proposition 3 then identifies $\gamma(\tau) = \mathrm{Exp}_{[\boldsymbol{A},s]}(\tau\boldsymbol{v})$ as the geodesic issued from $[\boldsymbol{A}, s]$ with velocity $\boldsymbol{v}$, with $\gamma(1) = [\boldsymbol{A}+\boldsymbol{V},\ s+c] = [\boldsymbol{B}\boldsymbol{Q}, t] = [\boldsymbol{B}, t]$ because $\boldsymbol{Q} \in O(q)$. Both $\mathrm{Exp}_{[\boldsymbol{A},s]}$ and $\boldsymbol{v}$ are defined without reference to representatives, so γ is too. Since $\boldsymbol{A}_\tau^T\boldsymbol{V} = \boldsymbol{A}^T\boldsymbol{V} + \tau\boldsymbol{V}^T\boldsymbol{V}$ is symmetric for every τ, the constant velocity $(\boldsymbol{V}, c)$ of the straight line $\tilde\gamma$ is horizontal at every time, and $\pi \circ \tilde\gamma = \gamma$; so $\tilde\gamma$ is the horizontal lift of γ through $(\boldsymbol{A}, s)$ and, π being a Riemannian submersion, $|\dot\gamma(\tau)| = |\boldsymbol{v}|$ for every τ and γ has g-length $|\boldsymbol{v}|$.

By part 2 of Proposition 3 the map $\mathrm{Exp}_{[\boldsymbol{A},s]}$ is a diffeomorphism of the ball $\mathcal{E}_{[\boldsymbol{A},s]}$ onto the open set $U_{[\boldsymbol{A},s]}$, so $U_{[\boldsymbol{A},s]}$ is a geodesic ball centred at $[\boldsymbol{A}, s]$ and γ is the radial geodesic of that ball reaching $[\boldsymbol{B}, t]$. Proposition 6.11 of Lee (2018) therefore says that, for $[\boldsymbol{B}, t] \neq [\boldsymbol{A}, s]$, the curve γ is, up to reparametrization, the unique curve of least length in M joining the two points. In particular every competitor in (2.6) is at least as long as γ, so the infimum is attained at γ and equals $|\boldsymbol{v}| = \sqrt{\|\boldsymbol{V}\|_F^2 + c^2}$, the horizontal lift $\iota_{(\boldsymbol{A},s)}$ being a linear isometry for the Euclidean metric (1.4); this is (2.13), and no global property of M was used to obtain it. When $[\boldsymbol{B}, t] = [\boldsymbol{A}, s]$ one has $\boldsymbol{Q} = \boldsymbol{I}_q$, $(\boldsymbol{V}, c) = (\boldsymbol{0}, 0)$ and γ constant, and (2.13) holds trivially.

Finally $\|\boldsymbol{V}\| \leqslant \|\boldsymbol{V}\|_F \leqslant |\boldsymbol{v}| < \min(\sigma_{\min}(\boldsymbol{A}), s)$, so Weyl's perturbation inequality, as quoted in the proof of Proposition 3, gives $\sigma_{\min}(\boldsymbol{A}_\tau) \geqslant \sigma_{\min}(\boldsymbol{A}) - \tau\|\boldsymbol{V}\| > 0$ for $\tau \in [0, 1]$; hence $\boldsymbol{A}_\tau$ has rank q.

2. By part 1 of Lemma 14 the right hand side of (2.14) is linear in $(\boldsymbol{X}, \beta)$, with coefficients that are smooth in τ and in $(\boldsymbol{A}, \boldsymbol{V})$ on the open set where $\boldsymbol{A}_\tau$ has rank q, a set which by part 1 contains $[0, 1]$. Linear systems have solutions that exist and are unique on the whole interval, whatever the time τ_0 at which the data are prescribed, and that depend smoothly on the time and linearly on the data (Lee, 2018, Theorem 4.31); adjoining $(\boldsymbol{A}, \boldsymbol{V})$ to the unknown as extra components with vanishing derivative turns the dependence on those parameters into a dependence on initial conditions, which is smooth as well (Lee, 2013, Theorem D.1 and Theorem D.6). By the same lemma each $\boldsymbol{\Omega}(\tau)$ is skew-symmetric, because $\boldsymbol{X}^T\boldsymbol{V} - \boldsymbol{V}^T\boldsymbol{X}$ is.

For horizontality, put $\boldsymbol{S}(\tau) := \boldsymbol{A}_\tau^T\boldsymbol{X}(\tau) - \boldsymbol{X}(\tau)^T\boldsymbol{A}_\tau$, which by part 3 of Proposition 1 vanishes precisely when $(\boldsymbol{X}(\tau), \beta(\tau))$ is horizontal at $\tilde\gamma(\tau)$. Using $\dot{\boldsymbol{A}}_\tau = \boldsymbol{V}$, $\dot{\boldsymbol{X}} = \boldsymbol{A}_\tau\boldsymbol{\Omega}$ and $\boldsymbol{\Omega}^T = -\boldsymbol{\Omega}$,

$$\dot{\boldsymbol{S}}(\tau) = \boldsymbol{V}^T\boldsymbol{X} + \boldsymbol{A}_\tau^T\boldsymbol{A}_\tau\boldsymbol{\Omega} - \boldsymbol{\Omega}^T\boldsymbol{A}_\tau^T\boldsymbol{A}_\tau - \boldsymbol{X}^T\boldsymbol{V} = \left(\boldsymbol{A}_\tau^T\boldsymbol{A}_\tau\boldsymbol{\Omega} + \boldsymbol{\Omega}\boldsymbol{A}_\tau^T\boldsymbol{A}_\tau\right) - \left(\boldsymbol{X}^T\boldsymbol{V} - \boldsymbol{V}^T\boldsymbol{X}\right) = \boldsymbol{0}, \quad (2.21)$$

the last equality being exactly (2.7) for $\boldsymbol{\Omega} = \boldsymbol{\Omega}(\tau)$. Since $\boldsymbol{S}(\tau_0) = \boldsymbol{0}$ by hypothesis, $\boldsymbol{S} \equiv \boldsymbol{0}$.

For the norm, the horizontality just proved makes $\boldsymbol{A}_\tau^T\boldsymbol{X}(\tau)$ symmetric, so

$$\frac{d}{d\tau}\left[\operatorname{tr}\left(\boldsymbol{X}\boldsymbol{X}^T\right) + \beta^2\right] = 2\operatorname{tr}\left(\boldsymbol{X}^T\dot{\boldsymbol{X}}\right) + 0 = 2\operatorname{tr}\left(\boldsymbol{X}^T\boldsymbol{A}_\tau\boldsymbol{\Omega}\right) = 2\operatorname{tr}\left(\left(\boldsymbol{A}_\tau^T\boldsymbol{X}\right)\boldsymbol{\Omega}\right) = 0,$$

the last equality because the trace of the product of a symmetric and a skew-symmetric matrix vanishes.

3. Let $(\boldsymbol{X}_0, \beta_0) \in H_{\tilde\gamma(\tau_0)}$ and let $(\boldsymbol{X}, \beta)$ be the solution of (2.14) it determines. By part 2 the pair $(\boldsymbol{X}(\tau), \beta(\tau))$ is horizontal at $\tilde\gamma(\tau)$, hence is the horizontal lift of

$$\boldsymbol{x}(\tau) := \mathrm{d}\pi_{\tilde\gamma(\tau)}\left((\boldsymbol{X}(\tau), \beta(\tau))\right),$$

and its derivative $(\boldsymbol{A}_\tau\boldsymbol{\Omega}(\tau),\ 0)$ is vertical by Proposition 1, $\boldsymbol{\Omega}(\tau)$ being skew-symmetric. The curve $\tilde\gamma$ has horizontal velocity at every time by part 1, so Lemma 15 applies and gives $D_\tau\boldsymbol{x} = 0$: the field $\boldsymbol{x}$ is parallel along γ. Existence and uniqueness of the parallel field with a prescribed value at τ_0 (Lee, 2018, Theorem 4.32) identify $\boldsymbol{x}$ with the parallel transport of $\boldsymbol{x}(\tau_0)$, which is (2.16). The map $\Pi(\gamma)^{\tau_1}_{\tau_0}$ is linear because (2.14) is; it is an isometry by the conservation of the norm in part 2 together with the fact that each $\iota_{\tilde\gamma(\tau)}$ is one; and $\Pi(\gamma)^{\tau_0}_{\tau_1}$ inverts it, by uniqueness.

Parallel transport is unchanged if γ is reparametrized. Indeed, let $\phi : [a,b] \to [0,1]$ be a diffeomorphism. The curve $\tilde{\gamma} \circ \phi$ lies over $\gamma \circ \phi$ and has velocity $\dot{\phi}(\varsigma)\,(\boldsymbol{V}, c)$, horizontal at every time; and if $\tilde{\boldsymbol{x}}$ is the horizontal lift of a field $\boldsymbol{x}$ along γ, then $\tilde{\boldsymbol{x}} \circ \phi$ is the horizontal lift of $\boldsymbol{x} \circ \phi$ along $\gamma \circ \phi$, with derivative $\dot{\phi} \cdot \left(\dot{\tilde{\boldsymbol{x}}} \circ \phi\right)$, which is vertical exactly when $\dot{\tilde{\boldsymbol{x}}} \circ \phi$ is. By the criterion closing Lemma 15, the field $\boldsymbol{x} \circ \phi$ is parallel along $\gamma \circ \phi$ if and only if $\boldsymbol{x}$ is parallel along γ. Together with the uniqueness statement of part 1, this makes $\Pi(\gamma)_1^0$ a function of the pair $([\boldsymbol{A}, s],\ [\boldsymbol{B}, t])$ alone, so (2.17) is well posed.

For (2.20), apply the above with $\tau_0 = 1$, $\tau_1 = 0$ and $(\boldsymbol{X}_0, \beta_0) = (\boldsymbol{Y}, \beta)$, which is horizontal at $\tilde{\gamma}(1)$ by (2.18): the second component of (2.14) stays equal to β, the first is (2.19), and part 2 supplies existence, uniqueness, linearity in $(\boldsymbol{Y}, \beta)$, smoothness in $(\boldsymbol{A}, \boldsymbol{V})$ and the preservation of the norm. If $[\boldsymbol{B}, t] = [\boldsymbol{A}, s]$, then $\boldsymbol{V} = \boldsymbol{0}$ and $c = 0$, so (2.19) reads $\dot{\boldsymbol{X}} = \boldsymbol{0}$ and $\boldsymbol{X}(0) = \boldsymbol{Y}$, whence $\Pi_{[\boldsymbol{A},s]}^{[\boldsymbol{A},s]} = \mathrm{d}\pi_{(\boldsymbol{A},s)} \circ \iota_{(\boldsymbol{A},s)} = \mathrm{id}$. Lastly $\boldsymbol{V}^T\boldsymbol{V} - \boldsymbol{V}^T\boldsymbol{V} = \boldsymbol{0}$, so the constant field $\boldsymbol{X} \equiv \boldsymbol{V}$ solves (2.19) with $\boldsymbol{Y} = \boldsymbol{V}$; since $\iota_{\tilde{\gamma}(1)}\left(\dot{\gamma}(1)\right) = (\boldsymbol{V}, c)$, this gives $\Pi_{[\boldsymbol{B},t]}^{[\boldsymbol{A},s]}\left(\dot{\gamma}(1)\right) = \mathrm{d}\pi_{(\boldsymbol{A},s)}\left((\boldsymbol{V}, c)\right) = \boldsymbol{v}$. □

Three features of (2.20) deserve to be isolated. The scalar component is transported trivially, $\dot{\beta} = 0$: the factor of Θ carrying σ^2 is a flat direction on which $O(q)$ acts trivially, and it never mixes with the loading matrix. When $q = 1$ the matrix $\boldsymbol{X}(\tau)^T\boldsymbol{V} - \boldsymbol{V}^T\boldsymbol{X}(\tau)$ is skew-symmetric of size 1×1, hence zero, so (2.19) reads $\dot{\boldsymbol{X}} = \boldsymbol{0}$ and $\Pi_{[\boldsymbol{B},t]}^{[\boldsymbol{A},s]}$ is the identity read in the horizontal lifts; this is as it must be, since for $q = 1$ the group $O(1)$ is discrete, π is a local isometry and M is flat. For $q \geqslant 2$ there is no closed form, and what stands in the way is the curvature of (M, g): writing $\mathcal{K}_\tau$ for the linear operator $\boldsymbol{X} \mapsto \boldsymbol{A}_\tau \mathcal{L}_\tau\left(\boldsymbol{X}^T\boldsymbol{V} - \boldsymbol{V}^T\boldsymbol{X}\right)$ of $\mathbb{R}^{p\times q}$, the terminal value problem (2.19) is equivalent to

$$\boldsymbol{X}(0) = \boldsymbol{Y} - \int_0^1 \mathcal{K}_\sigma\left(\boldsymbol{X}(\sigma)\right)\,\mathrm{d}\sigma, \qquad \|\mathcal{K}_\sigma\| \leqslant \frac{\|\boldsymbol{A}_\sigma\|\,\|\boldsymbol{V}\|_F}{\sigma_{\min}(\boldsymbol{A}_\sigma)^2}, \tag{2.22}$$

the operator norm being the one induced by $\|\cdot\|_F$, and the bound following from $\|\boldsymbol{X}^T\boldsymbol{V} - \boldsymbol{V}^T\boldsymbol{X}\|_F \leqslant 2\|\boldsymbol{X}\|_F\|\boldsymbol{V}\|_F$ together with the observation that, in an orthonormal eigenbasis of $\boldsymbol{A}_\sigma^T\boldsymbol{A}_\sigma$, solving (2.7) divides the (i,j) entry by $p_i + p_j \geqslant 2\sigma_{\min}(\boldsymbol{A}_\sigma)^2$. So $\boldsymbol{X}(0)$ differs from $\boldsymbol{Y}$ by a quantity of the order of $\|\boldsymbol{V}\|_F \leqslant d_M\left([\boldsymbol{A}, s], [\boldsymbol{B}, t]\right)$: parallel transport departs from the naive identification of the two horizontal spaces by a term of the order of the distance travelled. Controlling that term is the entire business of the next subsection.

Definition 17. For $\boldsymbol{u} \in B_\rho$ we write

$$\Gamma_{\boldsymbol{u}} := \Pi_{\mathrm{Exp}_{\theta_0}(\boldsymbol{u})}^{\theta_0} : T_{\mathrm{Exp}_{\theta_0}(\boldsymbol{u})}M \longrightarrow T_{\theta_0}M \tag{2.23}$$

for the map (2.17) of Proposition 16, applied with $[\boldsymbol{A}, s] = \theta_0$ and $[\boldsymbol{B}, t] = \mathrm{Exp}_{\theta_0}(\boldsymbol{u})$. By part 1 of that proposition the curve along which the transport is taken is the radial geodesic $\tau \mapsto \mathrm{Exp}_{\theta_0}(\tau\boldsymbol{u})$ of the normal chart, and it is the unique minimizing one; the subscript $\boldsymbol{u}$ therefore records the second point through that chart rather than directly, which is the form the estimates below need. This is the map written $\Pi_{\psi(\theta_{n,\boldsymbol{h}})}^{\psi(\theta)}$ in Definition III.5 of Sun et al. (2026), in the case $\psi = \mathrm{id}$. Concretely, in the notation (2.3), formula (2.20) reads $\Gamma_{\boldsymbol{u}}(\boldsymbol{z}) = \mathrm{d}\pi_{(\boldsymbol{W}_0, \sigma_0^2)}\left((\boldsymbol{X}(0), \beta)\right)$, where $(\boldsymbol{Y}, \beta) = \iota_{(\boldsymbol{W}_0 + \boldsymbol{U}_{\boldsymbol{u}},\ \sigma_0^2 + b_{\boldsymbol{u}})}(\boldsymbol{z})$ and $\boldsymbol{X}$ solves (2.19) with $\boldsymbol{A}_\tau = \boldsymbol{W}_0 + \tau\boldsymbol{U}_{\boldsymbol{u}}$ and $\boldsymbol{V} = \boldsymbol{U}_{\boldsymbol{u}}$. By Proposition 16, $\Gamma_{\boldsymbol{u}}$ is a linear isometry and $\Gamma_{\boldsymbol{0}} = \mathrm{id}$; and, after both tangent spaces are identified with subspaces of $\mathbb{R}^{p\times q} \times \mathbb{R}$ through the horizontal lifts ι and $\iota_{(\boldsymbol{W}_0 + \boldsymbol{U}_{\boldsymbol{u}},\ \sigma_0^2 + b_{\boldsymbol{u}})}$, the map $(\boldsymbol{u}, \boldsymbol{z}) \mapsto \Gamma_{\boldsymbol{u}}(\boldsymbol{z})$ is smooth on $B_\rho \times (\mathbb{R}^{p\times q} \times \mathbb{R})$, because $\boldsymbol{u} \mapsto (\boldsymbol{U}_{\boldsymbol{u}}, b_{\boldsymbol{u}})$ is linear and $(\boldsymbol{A}, \boldsymbol{V}, \boldsymbol{Y}, \beta) \mapsto (\boldsymbol{X}(0), \beta)$ is smooth by part 3 of that proposition.

The transport comparison

Comparing errors measured at two different base points presupposes that the logarithm may be taken at a *moving* base point, and this is the standard fact we record first: it holds provided both points stay in a small enough neighbourhood of θ_0.

Lemma 18 (A uniformly normal neighbourhood)**.** There exists $r \in \left(0, \frac{\varrho}{4}\right)$ such that, writing

$$\mathcal{V} := \mathrm{Exp}_{\theta_0}(B_r) \subseteq \mathcal{V}^\sharp := \mathrm{Exp}_{\theta_0}(B_{2r}) \subseteq \mathcal{U}, \tag{2.24}$$

the following hold.

1. For all $x, y \in \mathcal{V}^\sharp$ there is exactly one $\boldsymbol{v} \in \mathcal{E}_x$ with $\mathrm{Exp}_x(\boldsymbol{v}) = y$; we denote it by $\mathrm{Exp}_x^{-1}(y)$, and $\mathrm{Exp}_x^{-1}(x) = \mathbf{0}$.
2. The map $(\boldsymbol{u}, \boldsymbol{w}) \mapsto \mathrm{Exp}^{-1}_{\mathrm{Exp}_{\theta_0}(\boldsymbol{u})}\left(\mathrm{Exp}_{\theta_0}(\boldsymbol{w})\right)$ is smooth on $B_{2r} \times B_{2r}$, read through the horizontal lift $\iota_{(\boldsymbol{W}_0+\boldsymbol{U}_{\boldsymbol{u}},\ \sigma_0^2+b_{\boldsymbol{u}})}$ as a map into $\mathbb{R}^{p\times q} \times \mathbb{R}$.

Proof. Let $\mathcal{E} \subseteq TM$ be the set of pairs $(x, \boldsymbol{v})$ with $\boldsymbol{v} \in \mathcal{E}_x$, the ball of Proposition 3. Its radius at $x = [\boldsymbol{A}, s]$ is $\min(\sigma_{\min}(\boldsymbol{A}), s)$, which is a well defined function on M because the singular values of $\boldsymbol{A}$ are unchanged by $\boldsymbol{A} \mapsto \boldsymbol{A}\boldsymbol{Q}$ with $\boldsymbol{Q} \in O(q)$, and is continuous because its composition with the submersion π is. Hence $\mathcal{E}$ is open in TM and contains the zero section, and $\mathrm{Exp} : \mathcal{E} \to M$ is smooth, being affine in the horizontal lift by (1.14) and part 3 of Lemma 14. Consider

$$E : \mathcal{E} \longrightarrow M \times M, \qquad E(x, \boldsymbol{v}) := (x,\ \mathrm{Exp}_x(\boldsymbol{v}))\,.$$

Its differential at $(\theta_0, \mathbf{0})$ is invertible: in the splitting $T_{(\theta_0,\mathbf{0})}TM \cong T_{\theta_0}M \oplus T_{\theta_0}M$ it is $(\boldsymbol{a}, \boldsymbol{b}) \mapsto (\boldsymbol{a}, \boldsymbol{a}+\boldsymbol{b})$, because $\mathrm{d}\left(\mathrm{Exp}_{\theta_0}\right)_{\mathbf{0}} = \mathrm{id}$. The inverse function theorem therefore gives open sets $\mathcal{N} \ni (\theta_0, \mathbf{0})$ in $\mathcal{E}$ and $\mathcal{O} \ni (\theta_0, \theta_0)$ in $M \times M$ such that $E : \mathcal{N} \to \mathcal{O}$ is a diffeomorphism. This is the usual construction of uniformly normal neighbourhoods (Lee, 2018, Theorem 5.14 and its proof); note that completeness of M, which fails here, is nowhere used, only the openness of the domain $\mathcal{E}$ of Exp. Writing $E^{-1}(x, y) = (x, \Xi(x, y))$, the map Ξ is smooth on $\mathcal{O}$ and satisfies $\mathrm{Exp}_x(\Xi(x, y)) = y$ and $\Xi(x, x) = \mathbf{0}$. Since Exp_{θ_0} is continuous and $(\theta_0, \theta_0) \in \mathcal{O}$, we may choose $r \in \left(0, \frac{\varrho}{4}\right)$ so small that $\mathcal{V}^\sharp \times \mathcal{V}^\sharp \subseteq \mathcal{O}$.

For part 1, existence is $\mathrm{Exp}_x(\Xi(x, y)) = y$ together with $\Xi(x, y) \in \mathcal{E}_x$, and uniqueness is the injectivity of Exp_x on $\mathcal{E}_x$, which is part 2 of Proposition 3; so $\mathrm{Exp}_x^{-1} = \Xi(x, \cdot\,)$ unambiguously, and $\mathrm{Exp}_x^{-1}(x) = \mathbf{0}$ because $\Xi(x, x) = \mathbf{0}$. Part 2 follows by composing Ξ with the smooth map $(\boldsymbol{u}, \boldsymbol{w}) \mapsto \left(\mathrm{Exp}_{\theta_0}(\boldsymbol{u}),\ \mathrm{Exp}_{\theta_0}(\boldsymbol{w})\right)$ and with the horizontal lift, which is smooth in its base point by part 3 of Lemma 14. □

We fix such an r once and for all, together with $\mathcal{V}$ and $\mathcal{V}^\sharp$ as in (2.24). On $\mathcal{V}^\sharp$ the map $\mathrm{Exp}_{\theta_0}^{-1}$ in the sense of part 1 of Lemma 18 agrees with the logarithmic map (1.15) of Proposition 3, both being inverses of the injective map Exp_{θ_0} on B_ρ; no ambiguity can therefore arise from the notation. Part 1 of the lemma says exactly that $y \in U_x$ whenever $x, y \in \mathcal{V}^\sharp$, so part 1 of Proposition 16 applies to every such pair: any two points of $\mathcal{V}^\sharp$ are joined by a unique minimizing geodesic, and the transport Π_y^x of (2.17) is defined throughout $\mathcal{V}^\sharp \times \mathcal{V}^\sharp$.

Definition III.5 of Sun et al. (2026) compares $\mathrm{Exp}^{-1}_{\theta_{n,\boldsymbol{h}}}(\hat\theta_n)$, transported back to $T_{\theta_0}M$ by $\Pi^{\theta_0}_{\theta_{n,\boldsymbol{h}}}$, with $\mathrm{Exp}^{-1}_{\theta_0}(\hat\theta_n)$. On a flat space the two would differ by exactly the vector joining the two base points; on M they differ by that vector plus a curvature correction, and the whole content of this subsection is that the correction is quadratically small. Both base points are read through the normal chart at θ_0, which turns the comparison into a function of two chart vectors. Define, using part 1 of Lemma 18 and Definition 17,

$$F : B_{2r} \times B_{2r} \longrightarrow T_{\theta_0}M, \qquad F(\boldsymbol{u}, \boldsymbol{w}) := \Gamma_{\boldsymbol{u}}\left(\mathrm{Exp}^{-1}_{\mathrm{Exp}_{\theta_0}(\boldsymbol{u})}\left(\mathrm{Exp}_{\theta_0}(\boldsymbol{w})\right)\right). \tag{2.25}$$

Lemma 19 (Transport comparison)**.** There is a constant $C < \infty$, depending only on θ_0 and r, such that

$$|F(\boldsymbol{u}, \boldsymbol{w}) - (\boldsymbol{w} - \boldsymbol{u})| \leqslant C\,|\boldsymbol{u}|\,|\boldsymbol{w} - \boldsymbol{u}| \qquad \text{for all } \boldsymbol{u}, \boldsymbol{w} \in B_r. \tag{2.26}$$

Proof. By part 2 of Lemma 18 and Definition 17, F is smooth on $B_{2r} \times B_{2r}$; put

$$R(\boldsymbol{u}, \boldsymbol{w}) := F(\boldsymbol{u}, \boldsymbol{w}) - (\boldsymbol{w} - \boldsymbol{u}),$$

a smooth $T_{\theta_0}M$-valued function on $B_{2r} \times B_{2r}$. It vanishes on two slices:

- $R(\boldsymbol{u},\boldsymbol{u})=\mathbf{0}$ for every $\boldsymbol{u}\in B_{2r}$, because $\operatorname{Exp}_x^{-1}(x)=\mathbf{0}$ and $\Gamma_{\boldsymbol{u}}$ is linear, so $F(\boldsymbol{u},\boldsymbol{u})=\mathbf{0}=\boldsymbol{u}-\boldsymbol{u}$;
- $R(\mathbf{0},\boldsymbol{w})=\mathbf{0}$ for every $\boldsymbol{w}\in B_{2r}$, because $\Gamma_{\mathbf{0}}=\mathrm{id}$ and $\operatorname{Exp}_{\theta_0}^{-1}\left(\operatorname{Exp}_{\theta_0}(\boldsymbol{w})\right)=\boldsymbol{w}$, so $F(\mathbf{0},\boldsymbol{w})=\boldsymbol{w}=\boldsymbol{w}-\mathbf{0}$.

Set

$$C:=\sup_{(\boldsymbol{u},\boldsymbol{w})\in\overline{B_r}\times\overline{B_r}}\|\partial_1\partial_2 R(\boldsymbol{u},\boldsymbol{w})\|<\infty,$$

the norm being the one induced on bilinear maps $T_{\theta_0}M\times T_{\theta_0}M\to T_{\theta_0}M$ by $|\cdot|$; the supremum is finite because $\partial_1\partial_2 R$ is continuous and $\overline{B_r}\times\overline{B_r}$ is a compact subset of the open set $B_{2r}\times B_{2r}$ on which R is smooth.

Let $\boldsymbol{u},\boldsymbol{w}\in B_r$. The ball B_r being convex, the segment from $\boldsymbol{u}$ to $\boldsymbol{w}$ lies in it, and the fundamental theorem of calculus in the second variable, together with $R(\boldsymbol{u},\boldsymbol{u})=\mathbf{0}$, gives

$$R(\boldsymbol{u},\boldsymbol{w})=\int_0^1\partial_2 R\left(\boldsymbol{u},\ \boldsymbol{u}+t(\boldsymbol{w}-\boldsymbol{u})\right)[\boldsymbol{w}-\boldsymbol{u}]\ \mathrm{d}t. \tag{2.27}$$

Fix $\boldsymbol{z}\in B_r$. The segment from $\mathbf{0}$ to $\boldsymbol{u}$ also lies in B_r, and $\partial_2 R(\mathbf{0},\boldsymbol{z})=\mathbf{0}$ because $R(\mathbf{0},\,\cdot\,)$ vanishes identically; the same argument in the first variable therefore gives

$$\partial_2 R(\boldsymbol{u},\boldsymbol{z})=\int_0^1\partial_1\partial_2 R(\varsigma\boldsymbol{u},\boldsymbol{z})[\boldsymbol{u}]\,\mathrm{d}\varsigma,\qquad\text{whence}\qquad\|\partial_2 R(\boldsymbol{u},\boldsymbol{z})\|\leqslant C|\boldsymbol{u}|. \tag{2.28}$$

Substituting (2.28) into (2.27) yields (2.26). □

Remark 20. Lemma 19 is the only place where the curvature of (M,g) enters the asymptotic theory, and (2.26) says exactly how much it costs. In the applications below $\boldsymbol{u}$ is a local perturbation of size $n^{-\frac{1}{2}}$ and $\boldsymbol{w}-\boldsymbol{u}$ is an estimation error of size $O_{\mathbb{P}}(n^{-\frac{1}{2}})$, so the right hand side is $O_{\mathbb{P}}(n^{-1})$; after the multiplication by $\sqrt{n}$ that all these statements carry, it is $O_{\mathbb{P}}(n^{-\frac{1}{2}})$ and disappears. The two vanishing slices used in the proof are the two trivial cases of the comparison, namely no displacement of the base point and no error to compare, and it is their combination, not either alone, that produces the product $|\boldsymbol{u}||\boldsymbol{w}-\boldsymbol{u}|$.

The estimation error in the chart

Proposition 13 speaks of $\operatorname{Exp}_{\theta_0}^{-1}(\hat\theta_n)$, which is defined only when $\hat\theta_n$ lies in the normal neighbourhood. The next proposition is the bookkeeping that makes this a random vector defined everywhere, on an event whose probability tends to one, and identifies its horizontal lift.

Proposition 21 (The error vector)**.** Define

$$\mathcal{A}_n:=\left\{\hat\theta_n\in\mathcal{V}\right\},\qquad\hat{\boldsymbol{u}}_n:=\begin{cases}\operatorname{Exp}_{\theta_0}^{-1}\left(\hat\theta_n\right) & \text{on } \mathcal{A}_n,\\ \mathbf{0} & \text{otherwise.}\end{cases} \tag{2.29}$$

Then, almost surely, $\mathcal{A}_n$ holds for all n large enough; in particular $\mathbb{P}^n_{\theta_0}(\mathcal{A}_n)\to 1$, and the same holds with $\mathcal{V}$ replaced by $\operatorname{Exp}_{\theta_0}(B_\epsilon)$ for any $\epsilon\in(0,r]$. On $\mathcal{A}_n$ the horizontal lift of $\hat{\boldsymbol{u}}_n$ is the *aligned error*

$$\hat{\boldsymbol{e}}_n:=\iota\left(\hat{\boldsymbol{u}}_n\right)=\left(\hat{\boldsymbol{W}}_n\boldsymbol{Q}_n-\boldsymbol{W}_0,\ \hat\sigma_n^2-\sigma_0^2\right),\qquad\boldsymbol{Q}_n:=\boldsymbol{Q}\left(\boldsymbol{W}_0,\hat{\boldsymbol{W}}_n\right)\in O(q), \tag{2.30}$$

with $\boldsymbol{Q}(\,\cdot\,,\,\cdot\,)$ the orthogonal matrix (1.16). Consequently Proposition 13 may be read off the representatives: with $\boldsymbol{Z}\sim N\left(\mathbf{0},G^{-1}\right)$ on $T_{\theta_0}M$,

$$\sqrt{n}\left(\hat{\boldsymbol{W}}_n\boldsymbol{Q}_n-\boldsymbol{W}_0,\ \hat\sigma_n^2-\sigma_0^2\right)\rightsquigarrow\iota(\boldsymbol{Z})\qquad\text{in } H_{(\boldsymbol{W}_0,\sigma_0^2)}, \tag{2.31}$$

the left hand side being $\sqrt{n}\,\hat{\boldsymbol{e}}_n$ on $\mathcal{A}_n$.

Proof. $\mathrm{Exp}_{\theta_0}(B_\epsilon)$ is an open neighbourhood of θ_0 in M for every $\epsilon \in (0,\rho]$, by part 2 of Proposition 3; since $\hat{\theta}_n \to \theta_0$ almost surely by Theorem 9, almost every realization eventually enters it. The identity (2.30) is (1.15) applied with $[\boldsymbol{A}, s] = \theta_0$ and $[\boldsymbol{B}, t] = \hat{\theta}_n$, together with the fact that $\mathrm{d}\pi_{(\boldsymbol{W}_0,\sigma_0^2)}$ and ι are mutually inverse on horizontal vectors. Finally, Proposition 13 gives $\sqrt{n}\,\mathrm{Exp}_{\theta_0}^{-1}(\hat{\theta}_n) \rightsquigarrow \boldsymbol{Z}$, the convention on $\mathcal{A}_n^c$ being harmless because two random vectors that agree on events of probability tending to one have the same weak limits; applying the continuous mapping theorem to the linear isometry ι and using (2.30) gives (2.31). □

Remark 22. Three comments on Proposition 13 orient everything that follows.

1. It is an intrinsic statement. The quantity that converges is the point $\hat{\theta}_n$ of M, read through the logarithm at θ_0. The orthogonal alignment $\boldsymbol{Q}_n$ appearing in (2.31) is not an ad hoc Procrustes device introduced to make a limit exist: by Proposition 21 it *is* the logarithmic map of the first chapter, written in a representative.

2. It concerns one law. The expansion (1.42) and the ensuing weak convergence hold under the single law $\mathbb{P}_{\theta_0}^n$, and Remark 27 will show that the local regularity proved below does not follow from them, not even if they are assumed at every other point of M.

3. No assumption is made on the multiplicities of the $\sigma_j(\boldsymbol{W}_0)$. The result is valid when some, or all, of $\sigma_1(\boldsymbol{W}_0), \dots, \sigma_q(\boldsymbol{W}_0)$ coincide, a case in which the individual eigenvectors of $\boldsymbol{S}_n$ are *not* asymptotically normal. There is no contradiction: an individual eigenvector is not a function of the point $\hat{\theta}_n \in M$, whereas everything in Proposition 13 is. Section 2.3 makes this quantitative by exhibiting coordinates in which the limiting covariance is computed without ever dividing by a difference of singular values.

Coordinates, and the local experiments

We keep the g_{θ_0}-orthonormal basis $\boldsymbol{e}_1, \dots, \boldsymbol{e}_d$ of $T_{\theta_0}M$, the identification $T_{\theta_0}M \cong \mathbb{R}^d$ it provides, the matrix $\boldsymbol{G}$ with entries $\boldsymbol{G}_{jk} = G(\boldsymbol{e}_j, \boldsymbol{e}_k)$ and the coordinate vector $\nabla l(\boldsymbol{x}; \theta_0) \in \mathbb{R}^d$ of the score, with entries $s(\boldsymbol{x}; \theta_0)(\boldsymbol{e}_j)$, all fixed in Proposition 13 and its proof. Under this identification the Riemannian norm $|\cdot|$ becomes the Euclidean norm, and bilinearity gives $G(\boldsymbol{h}_1, \boldsymbol{h}_2) = \boldsymbol{h}_1^T \boldsymbol{G} \boldsymbol{h}_2$ and $s(\boldsymbol{x}; \theta_0)(\boldsymbol{h}) = \nabla l(\boldsymbol{x}; \theta_0)^T \boldsymbol{h}$; the matrix $\boldsymbol{G}$ is positive definite by Theorem 7. We abbreviate the normalized score

$$\boldsymbol{\Delta}_n := \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \nabla l(\boldsymbol{X}_i; \theta_0) \in \mathbb{R}^d, \tag{2.32}$$

so that (1.43) reads

$$\sqrt{n}\,\hat{\boldsymbol{u}}_n = \boldsymbol{G}^{-1} \boldsymbol{\Delta}_n + o_{\mathbb{P}}(1) \qquad \text{under } \mathbb{P}_{\theta_0}^n. \tag{2.33}$$

By Theorem 7 the vectors $\nabla l(\boldsymbol{X}_1; \theta_0), \nabla l(\boldsymbol{X}_2; \theta_0), \dots$ are independent, identically distributed, centred and square integrable with covariance matrix $\boldsymbol{G}$, so the multivariate central limit theorem gives

$$\boldsymbol{\Delta}_n \rightsquigarrow N(\boldsymbol{0}, \boldsymbol{G}) \qquad \text{under } \mathbb{P}_{\theta_0}^n. \tag{2.34}$$

Finally, for $\boldsymbol{h} \in T_{\theta_0}M$ and n large enough that $\frac{\boldsymbol{h}}{\sqrt{n}} \in B_r$ we write

$$\theta_{n,\boldsymbol{h}} := \mathrm{Exp}_{\theta_0}\left(\frac{\boldsymbol{h}}{\sqrt{n}}\right) \in \mathcal{V}, \qquad \theta_{n,\boldsymbol{0}} = \theta_0, \tag{2.35}$$

for the *geodesic local alternative* in the direction $\boldsymbol{h}$. In this notation the local asymptotic normality expansion (1.33) of Proposition 8 reads

$$\Lambda_n(\boldsymbol{h}) := \sum_{i=1}^{n} \log \frac{p(\boldsymbol{X}_i;\ \theta_{n,\boldsymbol{h}})}{p(\boldsymbol{X}_i; \theta_0)} = \boldsymbol{\Delta}_n^T \boldsymbol{h} - \frac{1}{2} \boldsymbol{h}^T \boldsymbol{G} \boldsymbol{h} + o_{\mathbb{P}}(1), \tag{2.36}$$

because the score is a linear functional, so that $\sum_{i=1}^{n} s(\boldsymbol{X}_i; \theta_0)\left(\frac{\boldsymbol{h}}{\sqrt{n}}\right) = \boldsymbol{\Delta}_n^T \boldsymbol{h}$.

Local regularity

We can now state the definition. Given an M-valued estimator sequence T_n we put, with $\theta_{n,\boldsymbol{h}}$ the geodesic local alternative (2.35),

$$\hat{\boldsymbol{u}}_n[T_n] := \begin{cases} \operatorname{Exp}_{\theta_0}^{-1}(T_n) & \text{on } \{T_n \in \mathcal{V}\}, \\ \mathbf{0} & \text{otherwise,} \end{cases} \qquad \hat{\boldsymbol{u}}_n^{\boldsymbol{h}}[T_n] := \begin{cases} \Gamma_{\frac{\boldsymbol{h}}{\sqrt{n}}}\left(\operatorname{Exp}_{\theta_{n,\boldsymbol{h}}}^{-1}(T_n)\right) & \text{on } \{T_n \in \mathcal{V}\}, \\ \mathbf{0} & \text{otherwise,} \end{cases} \tag{2.37}$$

both taking values in the single vector space $T_{\theta_0}M$; the second is well defined as soon as $\frac{\boldsymbol{h}}{\sqrt{n}} \in B_r$, by part 1 of Lemma 18. For the maximum likelihood estimator the first of these is exactly the vector $\hat{\boldsymbol{u}}_n$ of (2.29), and we keep that notation for it. Following Sun et al. (2026) we call an estimator sequence T_n *localizable at* θ_0 when $\mathbb{P}^n_{\theta_0}(T_n \in \mathcal{V}) \to 1$, which by Proposition 21 the maximum likelihood estimator is; only for such sequences do the quantities (2.37) carry any information in the limit.

Definition 23. A localizable estimator sequence T_n is *locally regular at* θ_0 *with limit law* L, where L is a Borel probability measure on $T_{\theta_0}M$, if for every $\boldsymbol{h} \in T_{\theta_0}M$

$$\sqrt{n}\, \hat{\boldsymbol{u}}_n^{\boldsymbol{h}}[T_n] \rightsquigarrow L \qquad \text{under } \mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}, \tag{2.38}$$

the limit L being the same for every $\boldsymbol{h}$. This is Definition III.5 of Sun et al. (2026) with $\psi = \mathrm{id}$ and $\Psi = M$: by (2.23) the map $\Gamma_{\boldsymbol{h}/\sqrt{n}}$ of (2.37) is their $\Pi^{\theta_0}_{\theta_{n,\boldsymbol{h}}}$, so (2.38) is their display (16) verbatim.

Lemma 24 (Contiguity)**.** For every $\boldsymbol{h} \in T_{\theta_0}M$ the sequences $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$ and $\mathbb{P}^n_{\theta_0}$ are mutually contiguous. Consequently, if a sequence of random vectors is $o_{\mathbb{P}}(1)$, respectively $O_{\mathbb{P}}(1)$, under $\mathbb{P}^n_{\theta_0}$, then it is so under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$ as well.

Proof. By (1.34) of Proposition 8 the log-likelihood ratio $\Lambda_n(\boldsymbol{h})$ of (2.36) converges in distribution under $\mathbb{P}^n_{\theta_0}$ to $Z \sim N\left(-\frac{1}{2}G(\boldsymbol{h},\boldsymbol{h}),\ G(\boldsymbol{h},\boldsymbol{h})\right)$, and $\mathbb{E}\left[e^Z\right] = e^{-\frac{1}{2}G(\boldsymbol{h},\boldsymbol{h})+\frac{1}{2}G(\boldsymbol{h},\boldsymbol{h})} = 1$, so Le Cam's first lemma in the form of Example 6.5 of van der Vaart (1998) gives mutual contiguity. The transfer of $o_{\mathbb{P}}(1)$ is immediate from the definition of contiguity. For $O_{\mathbb{P}}(1)$, suppose $\boldsymbol{Z}_n$ were not bounded in probability under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$: there would be $\varepsilon > 0$, a subsequence n_k and constants $M_k \to \infty$ with $\mathbb{P}^{n_k}_{\theta_{n_k,\boldsymbol{h}}}(|\boldsymbol{Z}_{n_k}| > M_k) > \varepsilon$ for all k, whereas $\mathbb{P}^{n_k}_{\theta_0}(|\boldsymbol{Z}_{n_k}| > M_k) \to 0$ because $\boldsymbol{Z}_n = O_{\mathbb{P}}(1)$ under $\mathbb{P}^n_{\theta_0}$ and $M_k \to \infty$; contiguity makes these two incompatible. $\square$

The next lemma is the bridge: on this manifold, regularity in the intrinsic sense of Definition 23 is the same thing as regularity of the ordinary Euclidean kind, read in the normal chart at θ_0. It is what allows the classical Euclidean theory to be applied verbatim here.

Lemma 25 (Chart form of regularity)**.** Let T_n be a localizable estimator sequence with $\sqrt{n}\, \hat{\boldsymbol{u}}_n[T_n] = O_{\mathbb{P}}(1)$ under $\mathbb{P}^n_{\theta_0}$. Then, for every $\boldsymbol{h} \in T_{\theta_0}M$,

$$\sqrt{n}\, \hat{\boldsymbol{u}}_n^{\boldsymbol{h}}[T_n] = \sqrt{n}\, \hat{\boldsymbol{u}}_n[T_n] - \boldsymbol{h} + o_{\mathbb{P}}(1) \qquad \text{under } \mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}. \tag{2.39}$$

Consequently T_n is locally regular at θ_0 with limit law L if and only if

$$\sqrt{n}\, \hat{\boldsymbol{u}}_n[T_n] - \boldsymbol{h} \rightsquigarrow L \qquad \text{under } \mathbb{P}^n_{\theta_{n,\boldsymbol{h}}} \text{ for every } \boldsymbol{h} \in T_{\theta_0}M. \tag{2.40}$$

Moreover the boundedness hypothesis is automatic: every locally regular sequence satisfies it.

Proof. Fix $\boldsymbol{h}$ and abbreviate $\hat{\boldsymbol{u}}_n := \hat{\boldsymbol{u}}_n[T_n]$, $\boldsymbol{u}_n := \frac{\boldsymbol{h}}{\sqrt{n}}$ and $\mathcal{B}_n := \{T_n \in \mathcal{V}\}$. By hypothesis $\mathbb{P}^n_{\theta_0}(\mathcal{B}_n) \to 1$, hence $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}(\mathcal{B}_n) \to 1$ by Lemma 24, and by the same lemma $\sqrt{n}\hat{\boldsymbol{u}}_n = O_{\mathbb{P}}(1)$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$ as well. On $\mathcal{B}_n$, and for n large enough that $\boldsymbol{u}_n \in B_r$, we have $T_n = \operatorname{Exp}_{\theta_0}(\hat{\boldsymbol{u}}_n)$ with $\hat{\boldsymbol{u}}_n \in B_r$, so (2.25) and (2.37) give

$$\hat{\boldsymbol{u}}_n^{\boldsymbol{h}}[T_n] = \Gamma_{\boldsymbol{u}_n}\left(\mathrm{Exp}^{-1}_{\mathrm{Exp}_{\theta_0}(\boldsymbol{u}_n)}\left(\mathrm{Exp}_{\theta_0}(\hat{\boldsymbol{u}}_n)\right)\right) = F(\boldsymbol{u}_n, \hat{\boldsymbol{u}}_n).$$

Lemma 19 therefore gives, on $\mathcal{B}_n$,

$$\left|\sqrt{n}\,\hat{\boldsymbol{u}}_n^{\boldsymbol{h}}[T_n] - \left(\sqrt{n}\hat{\boldsymbol{u}}_n - \boldsymbol{h}\right)\right| = \sqrt{n}\left|F(\boldsymbol{u}_n, \hat{\boldsymbol{u}}_n) - (\hat{\boldsymbol{u}}_n - \boldsymbol{u}_n)\right| \leqslant \sqrt{n}\,C\,\frac{|\boldsymbol{h}|}{\sqrt{n}}\left|\hat{\boldsymbol{u}}_n - \frac{\boldsymbol{h}}{\sqrt{n}}\right| = \frac{C|\boldsymbol{h}|}{\sqrt{n}}\left|\sqrt{n}\hat{\boldsymbol{u}}_n - \boldsymbol{h}\right|. \tag{2.41}$$

The right hand side is $O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right) = o_{\mathbb{P}}(1)$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$, and $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}(\mathcal{B}_n^c) \to 0$, which proves (2.39). The equivalence of (2.38) and (2.40) is then Slutsky's lemma.

For the last assertion, let T_n be locally regular with limit law L and take $\boldsymbol{h} = \boldsymbol{0}$ in (2.38): since $\theta_{n,\boldsymbol{0}} = \theta_0$ and $\Gamma_{\boldsymbol{0}} = \mathrm{id}$, this reads $\sqrt{n}\hat{\boldsymbol{u}}_n[T_n] \rightsquigarrow L$ under $\mathbb{P}^n_{\theta_0}$, and a weakly convergent sequence is bounded in probability by Prohorov's theorem. □

Theorem 26 (Local regularity of the maximum likelihood estimator)**.** *Let $1 \leqslant q < p$ and let $\theta_0 = [\boldsymbol{W}_0, \sigma_0^2] \in M$. Then the maximum likelihood estimator $\hat{\theta}_n$ of Section 1.4 is locally regular at θ_0 in the sense of Definition 23, with limit law*

$$L = N\left(\boldsymbol{0},\ G^{-1}\right) \qquad \text{on } T_{\theta_0}M, \tag{2.42}$$

which does not depend on $\boldsymbol{h}$. Explicitly, for every $\boldsymbol{h} \in T_{\theta_0}M$,

$$\sqrt{n}\,\Gamma_{\frac{\boldsymbol{h}}{\sqrt{n}}}\left(Exp^{-1}_{\theta_{n,\boldsymbol{h}}}\left(\hat{\theta}_n\right)\right) \rightsquigarrow N\left(\boldsymbol{0},\ G^{-1}\right) \qquad \text{under } \mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}. \tag{2.43}$$

Proof. We work in the coordinates (2.32) and fix $\boldsymbol{h} \in T_{\theta_0}M$.

Step 1: a single linear statistic drives everything. The two expansions (2.33) and (2.36), both valid under $\mathbb{P}^n_{\theta_0}$, are linear in the *same* vector $\boldsymbol{\Delta}_n$ of (2.32), so no separate joint central limit theorem is needed: by (2.34), $\boldsymbol{\Delta}_n \rightsquigarrow N(\boldsymbol{0}, \boldsymbol{G})$, and the continuous mapping theorem applied to the linear map $\boldsymbol{\delta} \mapsto \left(\boldsymbol{G}^{-1}\boldsymbol{\delta},\ \boldsymbol{\delta}^T\boldsymbol{h} - \frac{1}{2}\boldsymbol{h}^T\boldsymbol{G}\boldsymbol{h}\right)$, together with Slutsky's lemma, yields

$$\left(\sqrt{n}\hat{\boldsymbol{u}}_n,\ \Lambda_n(\boldsymbol{h})\right) \rightsquigarrow N\left(\begin{pmatrix}\boldsymbol{0}\\ -\frac{1}{2}\boldsymbol{h}^T\boldsymbol{G}\boldsymbol{h}\end{pmatrix}, \begin{pmatrix}\boldsymbol{G}^{-1} & \boldsymbol{h}\\ \boldsymbol{h}^T & \boldsymbol{h}^T\boldsymbol{G}\boldsymbol{h}\end{pmatrix}\right) \qquad \text{under } \mathbb{P}^n_{\theta_0}, \tag{2.44}$$

the off-diagonal block being $\mathrm{Cov}\left(\boldsymbol{G}^{-1}\boldsymbol{\Delta},\ \boldsymbol{\Delta}^T\boldsymbol{h}\right) = \boldsymbol{G}^{-1}\boldsymbol{G}\boldsymbol{h} = \boldsymbol{h}$ for $\boldsymbol{\Delta} \sim N(\boldsymbol{0}, \boldsymbol{G})$.

Step 2: Le Cam's third lemma. The limit in (2.44) has exactly the form required by Example 6.7 of van der Vaart (1998), the Gaussian case of Le Cam's third lemma (Theorem 6.6 there): the last coordinate is the log-likelihood ratio and is $N\left(-\frac{1}{2}\boldsymbol{h}^T\boldsymbol{G}\boldsymbol{h},\ \boldsymbol{h}^T\boldsymbol{G}\boldsymbol{h}\right)$, with $\boldsymbol{h}^T\boldsymbol{G}\boldsymbol{h} = G(\boldsymbol{h}, \boldsymbol{h})$. The cited result therefore gives

$$\sqrt{n}\,\hat{\boldsymbol{u}}_n \rightsquigarrow N\left(\boldsymbol{h}, \boldsymbol{G}^{-1}\right) \qquad \text{under } \mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}: \tag{2.45}$$

the limit under the contiguous alternative is the limit under the null shifted by the covariance $\boldsymbol{h}$ between the statistic and the log-likelihood ratio. Equivalently $\sqrt{n}\hat{\boldsymbol{u}}_n - \boldsymbol{h} \rightsquigarrow N(\boldsymbol{0}, \boldsymbol{G}^{-1})$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$, a law free of $\boldsymbol{h}$.

Step 3: from the chart to the transported error. The hypotheses of Lemma 25 hold for $T_n = \hat{\theta}_n$: it is localizable by Proposition 21, and $\sqrt{n}\hat{\boldsymbol{u}}_n = O_{\mathbb{P}}(1)$ under $\mathbb{P}^n_{\theta_0}$ by Proposition 13. The equivalence (2.40) of that lemma therefore turns (2.45) into (2.43), with limit law $N(\boldsymbol{0}, \boldsymbol{G}^{-1})$, which in the g_{θ_0}-orthonormal basis $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_d$ represents $N\left(\boldsymbol{0}, G^{-1}\right)$. □

Remark 27. It is worth being explicit about what (2.43) excludes, since Proposition 13 alone excludes nothing of the kind. Fix a point $\theta_* \in M$, known to the statistician, and let $\hat{\theta}_n^{\mathrm{H}}$ be the Hodges-type modification of $\hat{\theta}_n$ that replaces it by θ_* whenever $\hat{\theta}_n \in \mathrm{Exp}_{\theta_*}(B)$, with B the ball of radius $n^{-\frac{1}{4}}$ in $T_{\theta_*}M$, and leaves it unchanged otherwise.

For any fixed true parameter $\theta_0 \neq \theta_*$ the modification occurs on an event of probability tending to zero, by Theorem 9, so $\hat{\theta}_n^{\mathrm{H}}$ satisfies the exact analogue of Proposition 13 there. For $\theta_0 = \theta_*$ it does better: since $|\hat{\boldsymbol{u}}_n| = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right) \ll n^{-\frac{1}{4}}$ by Proposition 13, we have $\hat{\theta}_n^{\mathrm{H}} = \theta_*$ with probability tending to one, so that $\sqrt{n}\hat{\boldsymbol{u}}_n\left[\hat{\theta}_n^{\mathrm{H}}\right] \rightsquigarrow \delta_{\mathbf{0}}$, a limit strictly more concentrated than $N\left(\mathbf{0}, G^{-1}\right)$. This is the superefficiency phenomenon, and it shows that no bound whatsoever can be extracted from pointwise statements of the type of Proposition 13 alone.

And yet, at $\theta_0 = \theta_*$ and $\boldsymbol{h} \neq \mathbf{0}$, the same computation gives $\sqrt{n}\hat{\boldsymbol{u}}_n\left[\hat{\theta}_n^{\mathrm{H}}\right] \rightsquigarrow \delta_{\mathbf{0}}$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$ as well — the estimator is pinned to θ_* on an event of probability tending to one, since $|\hat{\boldsymbol{u}}_n| = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right) \ll n^{-\frac{1}{4}}$ under the alternative too, by Lemma 24 — so by (2.39) its transported error converges to $\delta_{-\boldsymbol{h}}$, which *does* depend on $\boldsymbol{h}$. Hence $\hat{\theta}_n^{\mathrm{H}}$ is superefficient at θ_* and is not locally regular there. Local regularity is precisely the statement that the asymptotics of Proposition 13 do not degrade as the truth is moved at the scale $n^{-\frac{1}{2}}$, and it is a strictly stronger property.

2.2 Consequences

What local regularity adds

Proposition 13 and Theorem 26 end in the same limit law $N\left(\mathbf{0}, G^{-1}\right)$, and the second was deduced from the first in three short steps; it is therefore worth saying plainly, before anything is derived from it, what the second asserts that the first does not. The difference is not in the limit. It is in the *set of laws under which the limit is asserted*: Proposition 13 is a statement about the single sequence $\mathbb{P}^n_{\theta_0}$, whereas Theorem 26 is a statement about the whole shrinking family $\left\{\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}} : \boldsymbol{h} \in T_{\theta_0}M\right\}$ of laws that no experiment of size n can tell apart from it. Everything in this section is a consequence of that one difference.

Why the difference is not a technicality. Lemma 24 says that $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$ and $\mathbb{P}^n_{\theta_0}$ are mutually contiguous: for any fixed $\boldsymbol{h}$, no sequence of tests whose level under $\mathbb{P}^n_{\theta_0}$ tends to zero can have power against $\theta_{n,\boldsymbol{h}}$ tending to one, and symmetrically with the two parameters exchanged. The qualification about the level is not decorative — the test that always rejects has power 1 — but it is exactly the qualification under which a testing problem is nontrivial. The statistician who has observed $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$ is therefore never in a position to assert that the truth is θ_0 rather than some $\theta_{n,\boldsymbol{h}}$; the best that can ever be known is that the truth lies within $O\left(n^{-\frac{1}{2}}\right)$ of the estimate, which is exactly the resolution at which the family $\{\theta_{n,\boldsymbol{h}}\}$ lives. An asymptotic property that holds at θ_0 but fails at $\theta_{n,\boldsymbol{h}}$ is thus *empirically unverifiable*: it is attached to a point that the data cannot single out. Proposition 13 is such a statement as it stands, and Remark 27 shows that it is not vacuously stable — the Hodges-type estimator constructed there satisfies the exact analogue of Proposition 13 at every point of M *other than* θ_*, does strictly better than it at θ_* itself, and is nevertheless worthless near θ_*, because the point at which its behaviour is anomalous moves with the sample size in precisely the way the statistician cannot track. Local regularity is the repair: it upgrades a claim indexed by a point to a claim indexed by a neighbourhood of the resolution scale, and it is exactly the strength needed, no more. We now list what this buys.

1. There is a lower bound at all. This is the largest single gain, and it is a gain of *kind*, not of degree. From Proposition 13 alone *no* optimality statement about $\hat{\theta}_n$ can be extracted, at θ_0 or anywhere else: Remark 27 exhibits a competitor whose limit law at θ_* is $\delta_{\mathbf{0}}$, infinitely more concentrated than $N\left(\mathbf{0}, G^{-1}\right)$ for every reasonable measure of concentration, and the same device can be run at any finite set of points at once. The class of all estimator sequences carries no nontrivial asymptotic lower bound whatsoever. Once attention is restricted to the locally regular ones, the Hájek–Le Cam convolution theorem in the manifold form of Theorem III.1 of Sun et al. (2026) applies to the present model — its hypotheses are Theorem 7 and Definition 23, both verified here — and yields, for every locally regular T_n with limit law L,

$$L = N\left(\mathbf{0}, G^{-1}\right) * \mu \qquad \text{for some Borel probability measure } \mu \text{ on } T_{\theta_0}M. \tag{2.46}$$

Nothing below depends on the manifold form of the theorem being available. Lemma 25 turns Definition 23 into ordinary Euclidean regularity read in the normal chart, where (2.36) exhibits a locally asymptotically normal sequence of experiments with nonsingular information matrix $\boldsymbol{G}$; the classical convolution theorem, Chapter 8 of van der Vaart (1998), applies to that sequence and returns (2.46) directly. Either way, every locally regular limit law is the law of $\boldsymbol{Z}+\boldsymbol{N}$ with $\boldsymbol{Z} \sim N\left(\mathbf{0}, G^{-1}\right)$ independent of $\boldsymbol{N} \sim \mu$: relative to the limit of Theorem 26, an independent noise term has been *added*, never subtracted. Two consequences of (2.46) deserve to be named separately, because they are the statements a practitioner actually uses; a third item records, for contrast and for orientation only, the intrinsic Cramér–Rao inequality that the convolution theorem replaces here.

- *Efficiency.* Theorem 26 exhibits $\hat{\theta}_n$ as the case $\mu=\delta_{\mathbf{0}}$ of (2.46), so the maximum likelihood estimator is asymptotically efficient in the class of locally regular estimators of $\theta_0 \in M$. This is Theorem V.1(i) of Sun et al. (2026), whose proof consists of exactly the verification carried out in Theorem 26.

- *A risk bound for every bowl-shaped loss.* Call $\ell: T_{\theta_0} M \rightarrow[0, \infty)$ bowl-shaped if its sublevel sets $\{\ell \leqslant c\}$ are closed, convex and symmetric about $\mathbf{0}$; equivalently, if ℓ is nonnegative, lower semicontinuous and quasiconvex with $\ell(-\boldsymbol{v})=\ell(\boldsymbol{v})$. Anderson's lemma applied to (2.46) gives $\mathbb{E}[\ell(\boldsymbol{Z}+\boldsymbol{N})] \geqslant \mathbb{E}[\ell(\boldsymbol{Z})]$, and the portmanteau theorem for nonnegative lower semicontinuous functions — this is where the sublevel sets are required to be *closed* — turns the convergence $\sqrt{n} \hat{\boldsymbol{u}}_n\left[T_n\right] \rightsquigarrow L$ into

$$\liminf_n \mathbb{E}_{\mathbb{P}_{\theta_0}^n}\left[\ell\left(\sqrt{n}\, \hat{\boldsymbol{u}}_n\left[T_n\right]\right)\right] \geqslant \mathbb{E}[\ell(\boldsymbol{Z})] \qquad \text{for every locally regular } T_n, \tag{2.47}$$

  and equally with $\mathbb{P}_{\theta_0}^n$ replaced by any $\mathbb{P}_{\theta_{n, \boldsymbol{h}}}^n$, the limit law being the same under all of them. See Chapter 8 of van der Vaart (1998) for the Euclidean argument, which Lemma 25 imports verbatim. Squared error is the case $\ell=|\cdot|^2$, but so is $\ell=\mathbf{1}_{A^c}$ for a fixed closed convex symmetric A, that is the probability of missing a fixed region, which is what a confidence statement is about.

- *The asymptotic Cramér–Rao bound.* The intrinsic Cramér–Rao inequality on a parameter manifold, Lemma III.2 and Corollary III.1 of Sun et al. (2026), bounds below the covariance of the residual of an *unbiased* estimator in the sense of their Definition III.3, and carries at finite n two extra terms built from the curvature tensor of (M, g) which have no Euclidean counterpart; by the linearity of the curvature operator recorded there, those terms are $O\left(n^{-1}\right)$ while the main term G^{-1} is $O(1)$, so that the bound reduces asymptotically to the curvature-free G^{-1}. We use it nowhere. Converting it into a statement about $\hat{\theta}_n$ would require an unbiasedness hypothesis, which nothing here gives any reason to expect of the maximum likelihood estimator at finite n, and the convergence $\operatorname{Cov}\left(\sqrt{n} \hat{\boldsymbol{u}}_n\right) \rightarrow \boldsymbol{G}^{-1}$, which does *not* follow from Proposition 13: weak convergence to $N\left(\mathbf{0}, \boldsymbol{G}^{-1}\right)$ controls second moments only through the $\liminf$ inequality (2.47), and upgrading that to convergence needs the uniform integrability of $\left|\sqrt{n} \hat{\boldsymbol{u}}_n\right|^2$, which is nowhere established here and is not innocuous, $\hat{\boldsymbol{u}}_n$ being defined by truncation to $\mathbf{0}$ off $\mathcal{A}_n$ in (2.29). The comparison we do make is the one the convolution theorem supplies, namely (2.46) and (2.47), which are statements about laws and need no moment hypothesis at all.

None of this is available to someone holding Proposition 13 alone, for the simple reason that (2.46) is false without the word *regular*: the Hodges sequence has $L=\delta_{\mathbf{0}}$ at θ_*, and $\delta_{\mathbf{0}}$ is not of the form $N\left(\mathbf{0}, G^{-1}\right) * \mu$, such a convolution having a nonsingular Gaussian factor and hence a density.

2. The influence operator is pinned down by a pointwise identity. Asymptotic linearity says that the estimator is, to first order, an average of a fixed function of the data; it places no constraint at all on *which* function. Local regularity places exactly one constraint, and it is a constraint that can be checked at a single point of the model, with no asymptotics at all. Proposition 32 below states it: an asymptotically linear sequence is locally regular *if and only if* its influence operator IF′ satisfies

$$\mathbb{E}\left[s\left(\boldsymbol{X} ; \theta_0\right)(\boldsymbol{z}) \operatorname{IF}'(\boldsymbol{X})\right]=\boldsymbol{z} \qquad \text{for every } \boldsymbol{z} \in T_{\theta_0} M, \tag{2.48}$$

that is, if and only if the cross-covariance between the influence operator and the score is the identity operator of $T_{\theta_0}M$. This is the specialization to $\psi = \mathrm{id}$ of the calculus of influence functions of Lemma III.3 of Sun et al. (2026); the proposition below adds the converse, which is what makes (2.48) usable as a criterion. The gain is concrete. A statistician who has computed an influence operator for a competing estimator, by whatever means, may decide whether that estimator is locally regular by evaluating one expectation under the true law, and may then read off its asymptotic covariance and compare it with G^{-1} — and, by part 2 of that proposition, the comparison always goes the same way. Without local regularity the influence operator is merely *some* centred square integrable function subject to no normalization, and no comparison between two estimators is possible at all.

3. The estimator becomes equivariant in the limit experiment. The expansion (2.36) of Proposition 8 says that the local experiments $\left(\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}} : \boldsymbol{h} \in T_{\theta_0}M\right)$ converge, in Le Cam's sense, to the Gaussian shift experiment $\left(N\left(\boldsymbol{h}, \boldsymbol{G}^{-1}\right) : \boldsymbol{h} \in \mathbb{R}^d\right)$, in which one observes a single Gaussian vector whose mean is the unknown local parameter and whose covariance is known (van der Vaart, 1998, Chapter 9). Proposition 13 identifies the behaviour of $\sqrt{n}\hat{\boldsymbol{u}}_n$ at the *single* point $\boldsymbol{h} = \boldsymbol{0}$ of that limit experiment, which is no more informative than knowing the value of a function at one point of its domain. Equation (2.45) identifies it at every $\boldsymbol{h}$ at once, and the answer, $N\left(\boldsymbol{h}, \boldsymbol{G}^{-1}\right)$, is exactly the law of the identity estimator $\boldsymbol{y} \mapsto \boldsymbol{y}$ in the limit experiment: in the limit, $\sqrt{n}\hat{\boldsymbol{u}}_n$ *is* the natural shift-equivariant estimator of the local parameter. Local regularity is the assertion that the map sending $\boldsymbol{h}$ to the law of $\sqrt{n}\hat{\boldsymbol{u}}_n$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$ converges to an equivariant family, and equivariance is the property that transfers the optimality theory of the limit experiment — where it is a finite dimensional Gaussian computation — back to the sequence. This is the structural reason behind item 1: (2.46) is the image, under the convergence of experiments, of the elementary fact that an equivariant estimator of a Gaussian mean cannot improve on the observation itself.

4. The property survives smooth transformation. Let φ be a smooth map from a neighbourhood of θ_0 in M into a fixed finite dimensional vector space; shrinking the radius r of (2.24) if necessary, we may assume that φ is defined on all of $\mathcal{V}$. As in (2.37), the random vector $\varphi(T_n)$ is to be read with the convention that it is set to a fixed value, say $\varphi(\theta_0)$, off the event $\{T_n \in \mathcal{V}\}$; the convention is harmless because that event has probability tending to one under every $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$, by localizability and Lemma 24, and two random vectors agreeing on events of probability tending to one have the same weak limits. With that convention, if T_n is locally regular at θ_0 with limit law L, then $\varphi(T_n)$ satisfies

$$\sqrt{n}\left(\varphi(T_n) - \varphi\left(\theta_{n,\boldsymbol{h}}\right)\right) \rightsquigarrow \left(\mathrm{d}\varphi_{\theta_0}\right)_{\#} L \qquad \text{under } \mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}, \tag{2.49}$$

with a limit free of $\boldsymbol{h}$. This is the delta method applied in the normal chart to (2.40). Indeed, put $\tilde{\varphi} := \varphi \circ \mathrm{Exp}_{\theta_0}$, smooth on B_r and with differential $\mathrm{d}\varphi_{\theta_0}$ at $\boldsymbol{0}$ because $\mathrm{d}\left(\mathrm{Exp}_{\theta_0}\right)_{\boldsymbol{0}} = \mathrm{id}$; Taylor's theorem, with the second derivative of $\tilde{\varphi}$ bounded on the compact set $\overline{B_{r/2}}$, supplies a constant $c < \infty$ with

$$|\tilde{\varphi}(\boldsymbol{a}) - \tilde{\varphi}(\boldsymbol{b}) - \mathrm{d}\varphi_{\theta_0}(\boldsymbol{a} - \boldsymbol{b})| \leqslant c\left(|\boldsymbol{a}|^2 + |\boldsymbol{b}|^2\right) \qquad \text{for } \boldsymbol{a}, \boldsymbol{b} \in \overline{B_{r/2}}.$$

Apply this with $\boldsymbol{a} = \hat{\boldsymbol{u}}_n[T_n]$ and $\boldsymbol{b} = \frac{\boldsymbol{h}}{\sqrt{n}}$, both in $\overline{B_{r/2}}$ with probability tending to one, and multiply by $\sqrt{n}$: the left hand side of (2.49) equals $\mathrm{d}\varphi_{\theta_0}\left(\sqrt{n}\hat{\boldsymbol{u}}_n[T_n] - \boldsymbol{h}\right)$ up to a remainder bounded by $c\,n^{-\frac{1}{2}}\left(\left|\sqrt{n}\hat{\boldsymbol{u}}_n[T_n]\right|^2 + |\boldsymbol{h}|^2\right)$, which is $O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right)$ because $\sqrt{n}\hat{\boldsymbol{u}}_n[T_n] = O_{\mathbb{P}}(1)$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$ by Lemmas 24 and 25. Part 3 of Corollary 42 below is the instance $\varphi = \boldsymbol{\Sigma}$, carried out in full and with an exact remainder, and Remark 43 records that part 3 is the only part of that corollary requiring anything from Section 2.1. The point is that (2.49) has no analogue starting from Proposition 13: the delta method applied to a pointwise central limit theorem returns a pointwise central limit theorem, and a family of pointwise statements indexed by θ carries no information about how they fit together, as Remark 27 demonstrates. Local regularity is stable under the operations one performs on estimators in practice; pointwise asymptotic normality is not.

5. Confidence sets that cover what they claim to cover. This is the operational payoff. Corollary 40 constructs the Wald region $\mathcal{C}_{n,1-\alpha} \subseteq M$ and shows that its coverage tends to $1 - \alpha$ along every sequence of local alternatives, not merely at θ_0. Remark 41 shows what the failure looks like when regularity is absent: the Hodges region covers $\theta_{n,\boldsymbol{h}}$ with limiting probability 1 for small $\boldsymbol{h}$ and 0 for large $\boldsymbol{h}$, and $1-\alpha$ for no $\boldsymbol{h}$ whatsoever, $\boldsymbol{h}=\mathbf{0}$ included. Since the statistician does not know $\boldsymbol{h}$ and, by contiguity, cannot come to know it, a region whose coverage depends on $\boldsymbol{h}$ reports a confidence level that is not the level it achieves.

The rest of the section carries this out. The next subsection collects the three elementary facts that the arguments need. The two subsections after it prove the characterization announced in item 2 and answer the question raised by item 3, namely in what sense $\hat{\theta}_n$ is asymptotically normal *on the manifold*. The last two contain the consequences themselves: the confidence sets of item 5, and the instance of item 4 at the level of the fitted covariance matrix.

Three preliminaries

The first preliminary is that the covariance matrix is a *quadratic* function of the local parameter, exactly and with no remainder, which is what makes every delta method argument below an identity rather than an approximation. Throughout the section we write $\boldsymbol{\Sigma}_{\boldsymbol{u}} := \boldsymbol{\Sigma}\left(\operatorname{Exp}_{\theta_0}(\boldsymbol{u})\right)$ for the covariance matrix at the point of M with normal coordinate $\boldsymbol{u}\in B_\rho$.

Lemma 28. Let $\mathcal{D}_0 : T_{\theta_0}M \to \operatorname{Sym}(p)$ be the linear map

$$\mathcal{D}_0(\boldsymbol{h}) := D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\theta_0) = \boldsymbol{U}_{\boldsymbol{h}}\boldsymbol{W}_0^T + \boldsymbol{W}_0\boldsymbol{U}_{\boldsymbol{h}}^T + b_{\boldsymbol{h}}\boldsymbol{I}_p \tag{2.50}$$

of (1.23). Then, for every $\boldsymbol{u}\in B_\rho$, one has the *exact* identity

$$\boldsymbol{\Sigma}_{\boldsymbol{u}} = \boldsymbol{\Sigma}_0 + \mathcal{D}_0(\boldsymbol{u}) + \boldsymbol{U}_{\boldsymbol{u}}\boldsymbol{U}_{\boldsymbol{u}}^T, \tag{2.51}$$

with no remainder. Moreover, with r the radius fixed in (2.24), there are constants $0<c_1\leqslant c_2<\infty$, depending only on θ_0 and r, such that

$$c_1\boldsymbol{I}_p \preceq \boldsymbol{\Sigma}_{\boldsymbol{u}} \preceq c_2\boldsymbol{I}_p \qquad \text{for every } \boldsymbol{u}\in\overline{B_r}. \tag{2.52}$$

In particular $\boldsymbol{u}\mapsto\boldsymbol{\Sigma}_{\boldsymbol{u}}^{-1}$ is smooth and bounded on $\overline{B_r}$.

Proof. By (1.14) and (1.6),

$$\boldsymbol{\Sigma}_{\boldsymbol{u}} = (\boldsymbol{W}_0+\boldsymbol{U}_{\boldsymbol{u}})(\boldsymbol{W}_0+\boldsymbol{U}_{\boldsymbol{u}})^T + \left(\sigma_0^2+b_{\boldsymbol{u}}\right)\boldsymbol{I}_p = \boldsymbol{\Sigma}_0 + \boldsymbol{U}_{\boldsymbol{u}}\boldsymbol{W}_0^T + \boldsymbol{W}_0\boldsymbol{U}_{\boldsymbol{u}}^T + b_{\boldsymbol{u}}\boldsymbol{I}_p + \boldsymbol{U}_{\boldsymbol{u}}\boldsymbol{U}_{\boldsymbol{u}}^T,$$

which is (2.51). For the bounds, the first summand of the middle expression is positive semidefinite, and $|b_{\boldsymbol{u}}|\leqslant|\boldsymbol{u}|\leqslant r<\rho\leqslant\sigma_0^2$ by (2.4) and (2.5); hence $\boldsymbol{\Sigma}_{\boldsymbol{u}}\succeq\left(\sigma_0^2-r\right)\boldsymbol{I}_p$ and we may take $c_1:=\sigma_0^2-r>0$. Similarly $\|\boldsymbol{U}_{\boldsymbol{u}}\|\leqslant\|\boldsymbol{U}_{\boldsymbol{u}}\|_F\leqslant r$, so that $\|\boldsymbol{\Sigma}_{\boldsymbol{u}}\|\leqslant(\|\boldsymbol{W}_0\|+r)^2+\sigma_0^2+r=:c_2$. $\square$

The second preliminary is the general notion of asymptotic linearity, of which (1.42) is the instance proved in Section 1.4. Naming it lets us record that the influence operator of a sequence is determined by the sequence, so that (1.41) is *the* influence operator of $\hat{\theta}_n$ and not merely one of them.

Definition 29 (Asymptotic linearity)**.** A localizable estimator sequence T_n is *asymptotically linear at θ_0 with influence operator* $\mathrm{IF}' : \mathbb{R}^p \to T_{\theta_0}M$ if IF' is Borel measurable, $\mathrm{IF}'(\boldsymbol{X}_1)$ is centred and square integrable, and

$$\sqrt{n}\,\hat{\boldsymbol{u}}_n[T_n] = \frac{1}{\sqrt{n}}\sum_{i=1}^n \mathrm{IF}'(\boldsymbol{X}_i) + o_{\mathbb{P}}(1) \qquad \text{under } \mathbb{P}_{\theta_0}^n, \tag{2.53}$$

with $\hat{\boldsymbol{u}}_n[T_n]$ the chart error of (2.37). Proposition 13 says that $\hat{\theta}_n$ is asymptotically linear in this sense, with influence operator IF of (1.41).

Proposition 30 (Uniqueness of the influence operator)**.** If IF' and IF'' both satisfy (2.53) for the same estimator sequence T_n, then $\mathrm{IF}'(\boldsymbol{X}_1)=\mathrm{IF}''(\boldsymbol{X}_1)$ almost surely.

Proof. Subtracting the two expansions (2.53) gives $n^{-\frac{1}{2}}\sum_{i=1}^{n}\boldsymbol{D}(\boldsymbol{X}_i) = o_{\mathbb{P}}(1)$ with $\boldsymbol{D} := \mathrm{IF}' - \mathrm{IF}''$, which is centred and square integrable by Definition 29. The multivariate central limit theorem gives $n^{-\frac{1}{2}}\sum_{i=1}^{n}\boldsymbol{D}(\boldsymbol{X}_i) \rightsquigarrow N\left(\mathbf{0}, \mathrm{Cov}\left(\boldsymbol{D}(\boldsymbol{X}_1)\right)\right)$, and a sequence converging in distribution to a limit and also to $\mathbf{0}$ in probability forces that limit to be $\delta_{\mathbf{0}}$; hence $\mathrm{Cov}\left(\boldsymbol{D}(\boldsymbol{X}_1)\right) = \mathbf{0}$ and, $\boldsymbol{D}(\boldsymbol{X}_1)$ being centred, $\boldsymbol{D}(\boldsymbol{X}_1) = \mathbf{0}$ almost surely. □

The third preliminary is that the Fisher information operator may be evaluated at a nearby point, and in particular at the estimator, without losing smoothness or positive definiteness. To compare the operators at two points we read them in one and the same frame, transported along the radial geodesic.

Lemma 31. For $\boldsymbol{u} \in B_r$ let $\boldsymbol{G}(\boldsymbol{u})$ be the $d \times d$ matrix

$$\boldsymbol{G}(\boldsymbol{u})_{jk} := G_{\mathrm{Exp}_{\theta_0}(\boldsymbol{u})}\left(\Gamma_{\boldsymbol{u}}^{-1}(\boldsymbol{e}_j),\ \Gamma_{\boldsymbol{u}}^{-1}(\boldsymbol{e}_k)\right), \qquad j,k = 1,\ldots,d, \tag{2.54}$$

the matrix of the Fisher information operator at $\mathrm{Exp}_{\theta_0}(\boldsymbol{u})$ in the frame obtained by transporting $\boldsymbol{e}_1,\ldots,\boldsymbol{e}_d$ along the radial geodesic. Then $\boldsymbol{u} \mapsto \boldsymbol{G}(\boldsymbol{u})$ is smooth on B_r, positive definite everywhere, and $\boldsymbol{G}(\mathbf{0}) = \boldsymbol{G}$.

Proof. Write $(\boldsymbol{W}_{\boldsymbol{u}}, \sigma_{\boldsymbol{u}}^2) := (\boldsymbol{W}_0 + \boldsymbol{U}_{\boldsymbol{u}},\ \sigma_0^2 + b_{\boldsymbol{u}})$ for the representative of $\mathrm{Exp}_{\theta_0}(\boldsymbol{u})$ used in (1.14), which is affine in $\boldsymbol{u}$, and let $\left(\tilde{\boldsymbol{U}}_j(\boldsymbol{u}),\ \tilde{b}_j(\boldsymbol{u})\right) := \iota_{(\boldsymbol{W}_{\boldsymbol{u}},\sigma_{\boldsymbol{u}}^2)}\left(\Gamma_{\boldsymbol{u}}^{-1}(\boldsymbol{e}_j)\right)$ be the horizontal lift there. By (1.24) and (1.23) applied at the point $\mathrm{Exp}_{\theta_0}(\boldsymbol{u})$,

$$\boldsymbol{G}(\boldsymbol{u})_{jk} = \frac{1}{2}\,\mathrm{tr}\left\{\boldsymbol{\Sigma}_{\boldsymbol{u}}^{-1}\boldsymbol{N}_j(\boldsymbol{u})\boldsymbol{\Sigma}_{\boldsymbol{u}}^{-1}\boldsymbol{N}_k(\boldsymbol{u})\right\}, \qquad \boldsymbol{N}_j(\boldsymbol{u}) := \tilde{\boldsymbol{U}}_j(\boldsymbol{u})\boldsymbol{W}_{\boldsymbol{u}}^T + \boldsymbol{W}_{\boldsymbol{u}}\tilde{\boldsymbol{U}}_j(\boldsymbol{u})^T + \tilde{b}_j(\boldsymbol{u})\boldsymbol{I}_p.$$

Each ingredient is smooth in $\boldsymbol{u}$: the matrix $\boldsymbol{\Sigma}_{\boldsymbol{u}}$ is polynomial in $\boldsymbol{u}$ by (2.51) and is bounded below by $c_1\boldsymbol{I}_p$ by (2.52), so $\boldsymbol{\Sigma}_{\boldsymbol{u}}^{-1}$ is smooth; $\Gamma_{\boldsymbol{u}}^{-1}$ is smooth by Proposition 16 and Definition 17; and $\iota_{(\boldsymbol{W}_{\boldsymbol{u}},\sigma_{\boldsymbol{u}}^2)}$ is smooth in $\boldsymbol{u}$ by part 3 of Lemma 14. That $\boldsymbol{G}(\mathbf{0}) = \boldsymbol{G}$ is $\Gamma_{\mathbf{0}} = \mathrm{id}$, and positive definiteness is Theorem 7 at the point $\mathrm{Exp}_{\theta_0}(\boldsymbol{u})$, the change of frame being invertible. □

The influence operator of a locally regular sequence

Proposition 32 (Regularity as a normalization of the influence operator)**.** Let T_n be a localizable estimator sequence which is asymptotically linear at θ_0 in the sense of Definition 29, with influence operator $\mathrm{IF}' : \mathbb{R}^p \to T_{\theta_0}M$. In the coordinates fixed in (2.32), let $\boldsymbol{V} := \mathrm{Cov}\left(\mathrm{IF}'(\boldsymbol{X}_1)\right)$ and $\boldsymbol{K} := \mathbb{E}\left[\mathrm{IF}'(\boldsymbol{X}_1)\,\nabla l(\boldsymbol{X}_1;\theta_0)^T\right]$ be its covariance matrix and its cross-covariance matrix with the score.

1. For every $\boldsymbol{h} \in T_{\theta_0}M$,
$$\sqrt{n}\,\hat{\boldsymbol{u}}_n[T_n] \rightsquigarrow N\left(\boldsymbol{K}\boldsymbol{h},\ \boldsymbol{V}\right) \qquad \text{under } \mathbb{P}_{\theta_{n,\boldsymbol{h}}}^n. \tag{2.55}$$
Consequently T_n is locally regular at θ_0 if and only if $\boldsymbol{K} = \boldsymbol{I}_d$, that is, if and only if (2.48) holds; and in that case the limit law is $L = N\left(\mathbf{0}, \boldsymbol{V}\right)$.

2. If T_n is locally regular then $\boldsymbol{V} \succeq \boldsymbol{G}^{-1}$, with equality if and only if $\mathrm{IF}'(\boldsymbol{X}_1) = \mathrm{IF}(\boldsymbol{X}_1)$ almost surely, IF being the influence function (1.41) of $\hat{\theta}_n$.

Proof. **1.** Write $\boldsymbol{\psi}_i := \mathrm{IF}'(\boldsymbol{X}_i) \in \mathbb{R}^d$ and $\boldsymbol{\Psi}_n := n^{-\frac{1}{2}}\sum_{i=1}^{n}\boldsymbol{\psi}_i$, so that Definition 29 reads $\sqrt{n}\hat{\boldsymbol{u}}_n[T_n] = \boldsymbol{\Psi}_n + o_{\mathbb{P}}(1)$ under $\mathbb{P}_{\theta_0}^n$. The vectors $(\boldsymbol{\psi}_i,\ \nabla l(\boldsymbol{X}_i;\theta_0)) \in \mathbb{R}^{2d}$ are independent, identically distributed, centred and square integrable, the second block by Theorem 7 and the first by Definition 29, so the multivariate central limit theorem gives

$$(\boldsymbol{\Psi}_n,\ \boldsymbol{\Delta}_n) \rightsquigarrow N\left(\mathbf{0},\ \begin{pmatrix} \boldsymbol{V} & \boldsymbol{K} \\ \boldsymbol{K}^T & \boldsymbol{G} \end{pmatrix}\right) \qquad \text{under } \mathbb{P}_{\theta_0}^n,$$

with $\boldsymbol{\Delta}_n$ the normalized score (2.32). Applying the continuous mapping theorem to the linear map $(\boldsymbol{a}, \boldsymbol{\delta}) \mapsto \left(\boldsymbol{a},\ \boldsymbol{\delta}^T\boldsymbol{h} - \frac{1}{2}\boldsymbol{h}^T\boldsymbol{G}\boldsymbol{h}\right)$, and then Slutsky's lemma together with the expansion (2.36) of $\Lambda_n(\boldsymbol{h})$,

$$\left(\boldsymbol{\Psi}_n,\ \Lambda_n(\boldsymbol{h})\right) \rightsquigarrow N\left(\begin{pmatrix}\mathbf{0}\\ -\frac{1}{2}\boldsymbol{h}^T\boldsymbol{G}\boldsymbol{h}\end{pmatrix},\ \begin{pmatrix}\boldsymbol{V} & \boldsymbol{K}\boldsymbol{h}\\ \boldsymbol{h}^T\boldsymbol{K}^T & \boldsymbol{h}^T\boldsymbol{G}\boldsymbol{h}\end{pmatrix}\right) \qquad \text{under } \mathbb{P}^n_{\theta_0},$$

the off-diagonal block being Cov $\left(\boldsymbol{\psi}_1,\ \nabla l(\boldsymbol{X}_1;\theta_0)^T\boldsymbol{h}\right) = \boldsymbol{K}\boldsymbol{h}$. This is again the situation of Example 6.7 of van der Vaart (1998), so Le Cam's third lemma gives $\boldsymbol{\Psi}_n \rightsquigarrow N\left(\boldsymbol{K}\boldsymbol{h}, \boldsymbol{V}\right)$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$; and $\sqrt{n}\hat{\boldsymbol{u}}_n[T_n] - \boldsymbol{\Psi}_n = o_{\mathbb{P}}(1)$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$ as well, by Lemma 24. This proves (2.55).

In particular $\sqrt{n}\hat{\boldsymbol{u}}_n[T_n] = O_{\mathbb{P}}(1)$ under $\mathbb{P}^n_{\theta_0}$, so Lemma 25 applies and makes local regularity equivalent to the requirement that $\sqrt{n}\hat{\boldsymbol{u}}_n[T_n] - \boldsymbol{h}$ have a limit law free of $\boldsymbol{h}$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$. By (2.55) that limit is $N\left(\left(\boldsymbol{K} - \boldsymbol{I}_d\right)\boldsymbol{h},\ \boldsymbol{V}\right)$, whose mean must therefore not depend on $\boldsymbol{h}$; taking $\boldsymbol{h} = \mathbf{0}$ shows that the constant value is $\mathbf{0}$, so $\left(\boldsymbol{K} - \boldsymbol{I}_d\right)\boldsymbol{h} = \mathbf{0}$ for every $\boldsymbol{h}$ and $\boldsymbol{K} = \boldsymbol{I}_d$. Conversely $\boldsymbol{K} = \boldsymbol{I}_d$ makes that limit $N(\mathbf{0}, \boldsymbol{V})$ for every $\boldsymbol{h}$. Finally, for $\boldsymbol{z} \in T_{\theta_0}M$ the j-th coordinate of $\mathbb{E}\left[s(\boldsymbol{X};\theta_0)(\boldsymbol{z})\,\mathrm{IF}'(\boldsymbol{X})\right]$ is $\mathbb{E}\left[\left(\nabla l(\boldsymbol{X};\theta_0)^T\boldsymbol{z}\right)\boldsymbol{\psi}_{1j}\right] = \left(\boldsymbol{K}\boldsymbol{z}\right)_j$, so $\boldsymbol{K} = \boldsymbol{I}_d$ and (2.48) are the same statement.

2. Assume $\boldsymbol{K} = \boldsymbol{I}_d$ and put $\boldsymbol{D} := \boldsymbol{\psi}_1 - \boldsymbol{G}^{-1}\nabla l(\boldsymbol{X}_1;\theta_0)$, which is centred and square integrable. Then $\mathbb{E}\left[\boldsymbol{D}\,\nabla l(\boldsymbol{X}_1;\theta_0)^T\right] = \boldsymbol{K} - \boldsymbol{G}^{-1}\boldsymbol{G} = \mathbf{0}$, so $\boldsymbol{D}$ is uncorrelated with $\boldsymbol{G}^{-1}\nabla l(\boldsymbol{X}_1;\theta_0)$ and

$$\boldsymbol{V} = \mathrm{Cov}\left(\boldsymbol{G}^{-1}\nabla l(\boldsymbol{X}_1;\theta_0)\right) + \mathrm{Cov}(\boldsymbol{D}) = \boldsymbol{G}^{-1} + \mathrm{Cov}(\boldsymbol{D}) \succeq \boldsymbol{G}^{-1},$$

using $\mathrm{Cov}\left(\boldsymbol{G}^{-1}\nabla l(\boldsymbol{X}_1;\theta_0)\right) = \boldsymbol{G}^{-1}\boldsymbol{G}\boldsymbol{G}^{-1} = \boldsymbol{G}^{-1}$. Equality forces $\mathrm{Cov}(\boldsymbol{D}) = \mathbf{0}$ and hence, $\boldsymbol{D}$ being centred, $\boldsymbol{D} = \mathbf{0}$ almost surely, which by (1.41) and the identification made at the end of the proof of Proposition 13 says $\mathrm{IF}'(\boldsymbol{X}_1) = \mathrm{IF}(\boldsymbol{X}_1)$ almost surely. □

Remark 33. Part 2 is the convolution theorem (2.46) restricted to the asymptotically linear class, proved from scratch in five lines rather than cited: the decomposition $\boldsymbol{\psi}_1 = \boldsymbol{G}^{-1}\nabla l(\boldsymbol{X}_1;\theta_0) + \boldsymbol{D}$ with $\boldsymbol{D}$ uncorrelated with the score is exactly the decomposition $L = N\left(\mathbf{0}, G^{-1}\right) * \mu$, read at the level of influence operators rather than of limit laws. Part 1 should be read alongside Proposition 30: taken together they say that an asymptotically linear sequence has exactly one influence operator, that local regularity is the single linear condition (2.48) on it, and that among the influence operators satisfying that condition the score-proportional one (1.41) is the unique minimizer of the asymptotic covariance.

Note also where local regularity is *not* needed. Part 1 shows that (2.55) holds for *any* asymptotically linear sequence, so the behaviour under local alternatives is always computable, from $\boldsymbol{K}$ alone; what regularity adds is not the computability but the answer $\boldsymbol{K} = \boldsymbol{I}_d$. The Hodges sequence of Remark 27 is the natural illustration. At $\theta_0 = \theta_*$ it is localizable, being equal to $\theta_* \in \mathcal{V}$ with probability tending to one, and on that same event $\hat{\boldsymbol{u}}_n\left[\hat{\theta}^{\mathrm{H}}_n\right] = \mathrm{Exp}^{-1}_{\theta_*}(\theta_*) = \mathbf{0}$, so that $\sqrt{n}\,\hat{\boldsymbol{u}}_n\left[\hat{\theta}^{\mathrm{H}}_n\right] = o_{\mathbb{P}}(1)$ under $\mathbb{P}^n_{\theta_*}$. It is therefore asymptotically linear at θ_* with influence operator $\mathrm{IF}' \equiv \mathbf{0}$, which is centred and square integrable; hence $\boldsymbol{K} = \mathbf{0}$ and $\boldsymbol{V} = \mathbf{0}$, and part 1 predicts $\sqrt{n}\hat{\boldsymbol{u}}_n\left[\hat{\theta}^{\mathrm{H}}_n\right] \rightsquigarrow N\left(\boldsymbol{K}\boldsymbol{h}, \boldsymbol{V}\right) = N(\mathbf{0}, \mathbf{0}) = \delta_{\mathbf{0}}$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$ for every $\boldsymbol{h}$, which is exactly what Remark 27 computes by hand. It violates (2.48) as flagrantly as possible, $\boldsymbol{K} = \mathbf{0}$ being as far from $\boldsymbol{I}_d$ as a matrix can be. So the Hodges sequence is no counterexample to (2.48), which it simply violates while failing to be locally regular; it is a counterexample to dispensing with the hypothesis of local regularity in (2.46). Part 2 is untouched by it, since part 2 assumes $\boldsymbol{K} = \boldsymbol{I}_d$ and it is precisely that assumption which fails here; no contradiction arises from $\boldsymbol{V} = \mathbf{0} \prec \boldsymbol{G}^{-1}$.

Asymptotic normality as a random point of the manifold

Proposition 13 and Theorem 26 are statements about a random *vector*, $\sqrt{n}\,\mathrm{Exp}^{-1}_{\theta_0}\left(\hat{\theta}_n\right)$, living in the fixed vector space $T_{\theta_0}M$. It is natural to ask what they say about the random *point* $\hat{\theta}_n \in M$ itself, in the spirit of Example III.1 of Sun et al. (2026), where a Gaussian law on a manifold is a law whose density is proportional to $\exp\left(-\frac{d(x,\mu)^2}{2\tau^2}\right)$ and is centred at a point μ of the manifold. The obstruction to asking the question naively is that M carries no scalar multiplication, so the rescaling by $\sqrt{n}$ that all these statements involve cannot be performed on M;

and the dilation $\theta \mapsto \mathrm{Exp}_{\theta_0}\left(\sqrt{n}\,\mathrm{Exp}_{\theta_0}^{-1}(\theta)\right)$, which one might substitute for it, does not map M into itself, the domain B_ρ of Proposition 3 being a bounded ball. The rescaling must therefore be moved onto the *metric*: one asks not that the law of $\hat{\theta}_n$ converge — it converges to δ_{θ_0}, which says only that $\hat{\theta}_n$ is consistent — but that it be indistinguishable, at the resolution $n^{-\frac{1}{2}}$, from an explicit Gaussian-shaped law on M centred at the true parameter. That is the formulation below, and it turns out to be exactly local regularity.

The formulation uses the Riemannian distance of (M, g), so we first record that this distance is finite — a fact needed nowhere earlier and therefore not yet stated.

Lemma 34 (M is connected)**.** The manifold M is connected; consequently its Riemannian distance d_M is finite on $M \times M$.

Proof. Write $\mathrm{St} := \left\{\boldsymbol{U} \in \mathbb{R}^{p\times q} : \boldsymbol{U}^T\boldsymbol{U} = \boldsymbol{I}_q\right\}$ and let $\mathrm{Sym}^{++}(q)$ be the set of positive definite $q \times q$ symmetric matrices, which is convex and hence connected. The continuous map $(\boldsymbol{U}, \boldsymbol{P}) \mapsto \boldsymbol{U}\boldsymbol{P}$ from $\mathrm{St} \times \mathrm{Sym}^{++}(q)$ into $\mathbb{R}^{p\times q}$ has as image exactly the set of rank q matrices: any such $\boldsymbol{A}$ is $\boldsymbol{U}\boldsymbol{P}$ with $\boldsymbol{P} := \left(\boldsymbol{A}^T\boldsymbol{A}\right)^{\frac{1}{2}}$, which is positive definite, and $\boldsymbol{U} := \boldsymbol{A}\boldsymbol{P}^{-1}$, which satisfies $\boldsymbol{U}^T\boldsymbol{U} = \boldsymbol{P}^{-1}\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{P}^{-1} = \boldsymbol{I}_q$. The set St is itself connected, being the image of the connected group $SO(p)$ under $\boldsymbol{R} \mapsto \boldsymbol{R}\boldsymbol{E}$, where $\boldsymbol{E} \in \mathbb{R}^{p\times q}$ consists of the first q columns of $\boldsymbol{I}_p$: given $\boldsymbol{U} \in \mathrm{St}$, complete it to $\left[\boldsymbol{U}\ \ \boldsymbol{U}^\perp\right] \in O(p)$ and, if the determinant is -1, change the sign of one column of $\boldsymbol{U}^\perp$, which is possible precisely because $q < p$. So the set of rank q matrices is connected, hence so is Θ, the product of that set with $(0, \infty)$, and hence so is its continuous image $M = \pi(\Theta)$. A connected Riemannian manifold is path connected, and the infimum of the lengths of the joining curves is finite, so $d_M < \infty$ throughout. $\square$

Definition 35 (Wrapped normal laws and the rescaled metric)**.** Let d_M denote the Riemannian distance of the connected Riemannian manifold (M, g), and write $\mathrm{law}_{\mathbb{P}}(Y)$ for the law of a random element Y under $\mathbb{P}$, the subscript being omitted when the law is clear.

1. For $\theta \in M$, a centred Gaussian law ν on $T_\theta M$ and $n \geqslant 1$, let $\nu^{(n)}$ be the law of $n^{-\frac{1}{2}}\boldsymbol{Z}$ with $\boldsymbol{Z} \sim \nu$, conditioned on the event $\left\{n^{-\frac{1}{2}}\boldsymbol{Z} \in \mathcal{E}_\theta\right\}$, which has positive probability because $\mathcal{E}_\theta$ is an open ball about the origin; and put
$$\mathcal{N}_n(\theta, \nu) := \left(\mathrm{Exp}_\theta\right)_{\#}\, \nu^{(n)}, \tag{2.56}$$
the push-forward of $\nu^{(n)}$ by the exponential map at θ: the *wrapped normal law* on M with centre θ, dispersion n^{-1} and shape ν. We abbreviate $\mathcal{N}_n(\theta) := \mathcal{N}_n\left(\theta,\ N\left(\mathbf{0}, G_\theta^{-1}\right)\right)$, the shape being given by the inverse Fisher information operator at θ itself.

2. For Borel probability measures P, Q on M put
$$\mathrm{BL}_n(P, Q) := \sup\left\{\left|\int_M f\,\mathrm{d}P - \int_M f\,\mathrm{d}Q\right| \ :\ \|f\|_\infty \leqslant 1,\ |f(x) - f(y)| \leqslant \sqrt{n}\,d_M(x, y)\ \ \forall\, x, y \in M\right\}, \tag{2.57}$$
the bounded Lipschitz distance for the rescaled metric $\sqrt{n}\,d_M$, a genuine metric by Lemma 34; and write BL for the ordinary bounded Lipschitz distance on $T_{\theta_0}M \cong \mathbb{R}^d$, which metrizes weak convergence there.

Two comments on the definition. First, the conditioning in part 1 is a formality for every θ in the compact set $\mathcal{K} := \mathrm{Exp}_{\theta_0}\left(\overline{B_{r/2}}\right) \subseteq \mathrm{Exp}_{\theta_0}(B_r)$, and this is all we shall need, since the centres occurring below are θ_0 and $\theta_{n,\boldsymbol{h}}$, for which $|\boldsymbol{h}|/\sqrt{n} \leqslant \frac{r}{2}$ once n is large. For $\theta = [\boldsymbol{A}, s]$ ranging in $\mathcal{K}$ the radius $\min\left(\sigma_{\min}(\boldsymbol{A}), s\right)$ of $\mathcal{E}_\theta$ is a positive continuous function of θ, as observed in the proof of Lemma 18, hence is bounded below by some $\rho' > 0$; and the operator norm of G_θ^{-1} equals that of $\boldsymbol{G}\left(\mathrm{Exp}_{\theta_0}^{-1}(\theta)\right)^{-1}$, the frame of (2.54) being orthonormal, which is bounded above on $\overline{B_{r/2}}$ by Lemma 31. The Gaussian tail bound then gives $\mathbb{P}\left(n^{-\frac{1}{2}}\boldsymbol{Z} \notin \mathcal{E}_\theta\right) \leqslant \mathbb{P}\left(|\boldsymbol{Z}| > \sqrt{n}\rho'\right) = O\left(e^{-cn}\right)$

for some $c > 0$, uniformly over $\theta \in \mathcal{K}$, so the conditioning perturbs the law by $O\left(e^{-cn}\right)$ in total variation and is invisible in everything that follows. The same bound holds, a fortiori, whenever the shape ν has a covariance whose operator norm is bounded along the sequence, which is the only property of it used below. Second, BL_n is nondecreasing in n, the class of competing f growing with n, and $\mathrm{BL}_n(P, Q) \to 0$ is a strictly stronger requirement than weak convergence: it says that P and Q cannot be told apart by any bounded test function varying on the scale $n^{-\frac{1}{2}}$, which is the scale on which $\hat{\theta}_n$ and θ_0 differ. Two laws both collapsing to δ_{θ_0} are automatically close in the weak topology and the closeness says nothing; BL_n is what removes that triviality.

Lemma 36 (The normal chart is bi-Lipschitz)**.** There exist $\varepsilon \in (0, r]$ and $L \geqslant 1$, depending only on θ_0, such that

$$L^{-1}|\boldsymbol{u}-\boldsymbol{w}| \;\leqslant\; d_M\left(\mathrm{Exp}_{\theta_0}(\boldsymbol{u}),\ \mathrm{Exp}_{\theta_0}(\boldsymbol{w})\right) \;\leqslant\; L\,|\boldsymbol{u}-\boldsymbol{w}| \qquad \text{for all } \boldsymbol{u}, \boldsymbol{w} \in B_\varepsilon. \tag{2.58}$$

Moreover $d_M\left(\theta_0,\ \mathrm{Exp}_{\theta_0}(\boldsymbol{u})\right) = |\boldsymbol{u}|$ for every $\boldsymbol{u} \in B_\rho$.

Proof. Fix $\varepsilon_0 \in (0, r]$. By part 2 of Proposition 3 the map Exp_{θ_0} is a diffeomorphism of B_ρ onto the open set $\mathcal{U}$, so $\mathrm{d}\left(\mathrm{Exp}_{\theta_0}\right)_{\boldsymbol{u}}$ is a linear isomorphism for every $\boldsymbol{u} \in B_\rho$; by continuity and compactness of $\overline{B_{\varepsilon_0}} \subseteq B_\rho$ there is $L_0 \geqslant 1$ with

$$L_0^{-1}|\boldsymbol{\xi}| \leqslant \left|\mathrm{d}\left(\mathrm{Exp}_{\theta_0}\right)_{\boldsymbol{u}}(\boldsymbol{\xi})\right|_g \leqslant L_0|\boldsymbol{\xi}| \qquad \text{for all } \boldsymbol{u} \in \overline{B_{\varepsilon_0}} \text{ and } \boldsymbol{\xi} \in T_{\theta_0}M, \tag{2.59}$$

where $|\cdot|_g$ is the norm of g at the relevant point.

The upper bound in (2.58) follows with $L = L_0$ for all $\boldsymbol{u}, \boldsymbol{w} \in B_{\varepsilon_0}$: the segment $t \mapsto \boldsymbol{u} + t(\boldsymbol{w}-\boldsymbol{u})$ stays in the convex set B_{ε_0}, its image under Exp_{θ_0} joins the two points and has, by (2.59), g-length at most $L_0|\boldsymbol{u}-\boldsymbol{w}|$, and d_M is the infimum of the lengths of such curves.

For the lower bound put $\varepsilon := \frac{\varepsilon_0}{4}$ and take $\boldsymbol{u}, \boldsymbol{w} \in B_\varepsilon$, so that $|\boldsymbol{u}-\boldsymbol{w}| < \frac{\varepsilon_0}{2}$. Let c be any piecewise smooth curve in M from $\mathrm{Exp}_{\theta_0}(\boldsymbol{u})$ to $\mathrm{Exp}_{\theta_0}(\boldsymbol{w})$. If c stays inside $\mathrm{Exp}_{\theta_0}\left(B_{\frac{\varepsilon_0}{2}}\right)$, then $\mathrm{Exp}_{\theta_0}^{-1} \circ c$ is a curve in $B_{\frac{\varepsilon_0}{2}}$ from $\boldsymbol{u}$ to $\boldsymbol{w}$, of Euclidean length at least $|\boldsymbol{u}-\boldsymbol{w}|$, so (2.59) gives $\mathrm{length}(c) \geqslant L_0^{-1}|\boldsymbol{u}-\boldsymbol{w}|$. Otherwise, let t_* be the first time at which c leaves $\mathrm{Exp}_{\theta_0}\left(B_{\frac{\varepsilon_0}{2}}\right)$; the restriction of $\mathrm{Exp}_{\theta_0}^{-1} \circ c$ to $[0, t_*]$ joins $\boldsymbol{u}$, of norm less than $\frac{\varepsilon_0}{4}$, to a point of norm $\frac{\varepsilon_0}{2}$, hence has Euclidean length at least $\frac{\varepsilon_0}{4} > \frac{1}{2}|\boldsymbol{u}-\boldsymbol{w}|$, and $\mathrm{length}(c) \geqslant \frac{1}{2}L_0^{-1}|\boldsymbol{u}-\boldsymbol{w}|$. Taking the infimum over c and setting $L := 2L_0$ gives (2.58).

The last assertion is the Gauss lemma. Since Exp_{θ_0} is a diffeomorphism of the ball $B_\rho \subseteq T_{\theta_0}M$ onto $\mathcal{U}$, the set $\mathcal{U}$ is a geodesic ball centred at θ_0, and inside a geodesic ball the radial geodesic $t \mapsto \mathrm{Exp}_{\theta_0}(t\boldsymbol{u})$, $t \in [0,1]$, is the unique minimizing curve in M from θ_0 to $\mathrm{Exp}_{\theta_0}(\boldsymbol{u})$ (Lee, 2018, Chapter 6). Its length is $|\boldsymbol{u}|$, geodesics having constant speed and the initial speed being $|\boldsymbol{u}|$. □

We fix ε and L as in Lemma 36 for the rest of this subsection and write $\mathcal{V}_\varepsilon := \mathrm{Exp}_{\theta_0}(B_\varepsilon) \subseteq \mathcal{V}$. The next lemma is the translation device: it says that comparing two M-valued sequences in BL_n is the same as comparing their rescaled chart images in the ordinary bounded Lipschitz distance, so that every statement about random points of M made below reduces to a statement about random vectors of $T_{\theta_0}M$, where the preceding results live.

Lemma 37 (Rescaled comparison)**.** For each n let Y_n and Y_n' be M-valued random elements, possibly defined under different laws, with $\mathbb{P}\left(Y_n \in \mathcal{V}_\varepsilon\right) \to 1$ and $\mathbb{P}\left(Y_n' \in \mathcal{V}_\varepsilon\right) \to 1$, and put

$$\boldsymbol{y}_n := \begin{cases} \sqrt{n}\,\mathrm{Exp}_{\theta_0}^{-1}(Y_n) & \text{on } \{Y_n \in \mathcal{V}_\varepsilon\}, \\ \mathbf{0} & \text{otherwise,} \end{cases}$$

and likewise $\boldsymbol{y}_n'$ from Y_n'. Then, with $P_n := \mathrm{law}(Y_n)$ and $P_n' := \mathrm{law}(Y_n')$ on M,

$$\mathrm{BL}_n\left(P_n, P_n'\right) \longrightarrow 0 \qquad \Longleftrightarrow \qquad \mathrm{BL}\left(\mathrm{law}(\boldsymbol{y}_n),\ \mathrm{law}(\boldsymbol{y}_n')\right) \longrightarrow 0. \tag{2.60}$$

In particular, if $\boldsymbol{y}_n$ and $\boldsymbol{y}_n'$ converge in distribution to the same limit, then $\mathrm{BL}_n\left(P_n, P_n'\right) \to 0$.

Proof. Abbreviate $\eta_n := \mathbb{P}(Y_n \notin \mathcal{V}_\varepsilon) + \mathbb{P}(Y_n' \notin \mathcal{V}_\varepsilon) \to 0$.

($\Leftarrow$) Let $f : M \to \mathbb{R}$ satisfy $\|f\|_\infty \leqslant 1$ and $|f(x) - f(y)| \leqslant \sqrt{n}\, d_M(x,y)$, and define g on $\sqrt{n}B_\varepsilon$ by $g(\boldsymbol{z}) := f\left(\mathrm{Exp}_{\theta_0}\left(n^{-\frac{1}{2}}\boldsymbol{z}\right)\right)$. By the upper bound in (2.58), $|g(\boldsymbol{z}) - g(\boldsymbol{z}')| \leqslant \sqrt{n}\, L\, n^{-\frac{1}{2}}|\boldsymbol{z} - \boldsymbol{z}'| = L|\boldsymbol{z} - \boldsymbol{z}'|$, and $\|g\|_\infty \leqslant 1$. Extend g to all of $T_{\theta_0}M$ by the McShane formula $\tilde{g}(\boldsymbol{z}) := \inf_{\boldsymbol{z}' \in \sqrt{n}B_\varepsilon}(g(\boldsymbol{z}') + L|\boldsymbol{z} - \boldsymbol{z}'|)$, truncated to $[-1, 1]$; both bounds are preserved and $\tilde{g} = g$ on $\sqrt{n}B_\varepsilon$. On $\{Y_n \in \mathcal{V}_\varepsilon\}$ we have $f(Y_n) = \tilde{g}(\boldsymbol{y}_n)$, and both are bounded by 1 everywhere, so

$$\left|\int_M f\,\mathrm{d}P_n - \int_M f\,\mathrm{d}P_n'\right| \leqslant |\mathbb{E}[\tilde{g}(\boldsymbol{y}_n)] - \mathbb{E}[\tilde{g}(\boldsymbol{y}_n')]| + 2\eta_n \leqslant L\,\mathrm{BL}(\mathrm{law}(\boldsymbol{y}_n),\ \mathrm{law}(\boldsymbol{y}_n')) + 2\eta_n,$$

the last step because $L^{-1}\tilde{g}$ is bounded by 1 and 1-Lipschitz, L being at least 1. Taking the supremum over f gives $\mathrm{BL}_n(P_n, P_n') \leqslant L\,\mathrm{BL} + 2\eta_n$.

($\Rightarrow$) Symmetrically, let $g : T_{\theta_0}M \to \mathbb{R}$ satisfy $\|g\|_\infty \leqslant 1$ and $\mathrm{Lip}(g) \leqslant 1$, and define f on $\mathcal{V}_\varepsilon$ by $f(x) := g\left(\sqrt{n}\,\mathrm{Exp}_{\theta_0}^{-1}(x)\right)$. For $x, y \in \mathcal{V}_\varepsilon$ the lower bound in (2.58) gives $|f(x) - f(y)| \leqslant \sqrt{n}\left|\mathrm{Exp}_{\theta_0}^{-1}(x) - \mathrm{Exp}_{\theta_0}^{-1}(y)\right| \leqslant L\sqrt{n}\, d_M(x,y)$, so f is L-Lipschitz for the metric $\sqrt{n}d_M$ on $\mathcal{V}_\varepsilon$; extend it to M by the McShane formula for that metric — finite by Lemma 34, which is what makes the formula meaningful — again truncated to $[-1, 1]$. The same two displays, read in the other direction, give $\mathrm{BL}(\mathrm{law}(\boldsymbol{y}_n), \mathrm{law}(\boldsymbol{y}_n')) \leqslant L\,\mathrm{BL}_n(P_n, P_n') + 2\eta_n$.

The final assertion holds because BL metrizes weak convergence on $T_{\theta_0}M \cong \mathbb{R}^d$. ∎

Theorem 38 (The maximum likelihood estimator as an asymptotically wrapped normal point of M)**.** *Let $1 \leqslant q < p$ and $\theta_0 = [\boldsymbol{W}_0, \sigma_0^2] \in M$, and let $\mathcal{N}_n(\,\cdot\,)$ and BL_n be as in Definition 35.*

1. *(Fixed centre.)*

$$BL_n\left(law_{\mathbb{P}^n_{\theta_0}}\left(\hat{\theta}_n\right),\ \mathcal{N}_n(\theta_0)\right) \longrightarrow 0. \tag{2.61}$$

That is, at the resolution $n^{-\frac{1}{2}}$ the maximum likelihood estimator is a wrapped normal random point of M, centred at $[\boldsymbol{W}_0, \sigma_0^2]$, with dispersion n^{-1} and with shape the inverse Fisher information operator.

2. *(Moving centre.) For every $\boldsymbol{h} \in T_{\theta_0}M$,*

$$BL_n\left(law_{\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}}\left(\hat{\theta}_n\right),\ \mathcal{N}_n\left(\theta_{n,\boldsymbol{h}}\right)\right) \longrightarrow 0, \tag{2.62}$$

the approximating law being the wrapped normal centred at the true parameter $\theta_{n,\boldsymbol{h}}$, with shape the inverse Fisher information operator at that same point.

3. *(The moving centre statement is local regularity.) Let T_n be a localizable estimator sequence with $\sqrt{n}\hat{\boldsymbol{u}}_n[T_n] = O_{\mathbb{P}}(1)$ under $\mathbb{P}^n_{\theta_0}$, let ν be a centred Gaussian law on $T_{\theta_0}M$, and for $\boldsymbol{u} \in B_\rho$ let $\nu_{\boldsymbol{u}}$ denote its transport to $T_{\mathrm{Exp}_{\theta_0}(\boldsymbol{u})}M$ by the linear isometry $\Gamma_{\boldsymbol{u}}^{-1}$ of Definition 17. Then T_n is locally regular at θ_0 with limit law ν, in the sense of Definition 23, if and only if*

$$BL_n\left(law_{\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}}(T_n),\ \mathcal{N}_n\left(\theta_{n,\boldsymbol{h}},\ \nu_{\boldsymbol{h}/\sqrt{n}}\right)\right) \longrightarrow 0 \qquad \text{for every } \boldsymbol{h} \in T_{\theta_0}M. \tag{2.63}$$

Proof. Fix $\boldsymbol{h} \in T_{\theta_0}M$ and abbreviate $\boldsymbol{u}_n := \frac{\boldsymbol{h}}{\sqrt{n}}$ and $\theta_n := \theta_{n,\boldsymbol{h}} = \mathrm{Exp}_{\theta_0}(\boldsymbol{u}_n)$. Parts 1 and 2 are proved together, part 1 being the case $\boldsymbol{h} = \boldsymbol{0}$, in which $\boldsymbol{u}_n = \boldsymbol{0}$, $\theta_n = \theta_0$ and $\Gamma_{\boldsymbol{0}} = \mathrm{id}$; part 3 is then read off the same computation. Throughout, $\boldsymbol{G}(\,\cdot\,)$ is the transported information matrix (2.54) of Lemma 31.

Step 1: the estimator side. By Proposition 21 and Lemma 24, $\mathbb{P}^n_{\theta_n}\left(\hat{\theta}_n \in \mathcal{V}_\varepsilon\right) \to 1$, and (2.45) gives

$$\sqrt{n}\,\hat{\boldsymbol{u}}_n \rightsquigarrow N\left(\boldsymbol{h}, \boldsymbol{G}^{-1}\right) \qquad \text{under } \mathbb{P}^n_{\theta_n}. \tag{2.64}$$

For $\boldsymbol{h} = \boldsymbol{0}$ this is Proposition 13 itself, and neither Lemma 24 nor Theorem 26 is used.

Step 2: the wrapped normal side. Let $\Theta_n \sim \mathcal{N}_n(\theta_n, \lambda_n)$, where $\lambda_n := N\left(\boldsymbol{0}, G_{\theta_n}^{-1}\right)$ for parts 1 and 2 and $\lambda_n := \nu_{\boldsymbol{u}_n}$ for part 3. In both cases λ_n is a centred Gaussian law on $T_{\theta_n}M$, and we compute its covariance matrix in the frame $\Gamma_{\boldsymbol{u}_n}^{-1}(\boldsymbol{e}_1), \ldots, \Gamma_{\boldsymbol{u}_n}^{-1}(\boldsymbol{e}_d)$ of $T_{\theta_n}M$, which is g_{θ_n}-orthonormal because $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_d$ is g_{θ_0}-orthonormal and $\Gamma_{\boldsymbol{u}_n}$ is a linear isometry. The coordinates of a vector $\boldsymbol{v} \in T_{\theta_n}M$ in that frame are by construction the $\boldsymbol{e}$-coordinates of $\Gamma_{\boldsymbol{u}_n}(\boldsymbol{v}) \in T_{\theta_0}M$. For parts 1 and 2 the matrix of the bilinear form G_{θ_n} in that frame is $\boldsymbol{G}(\boldsymbol{u}_n)$, by the very definition (2.54), so the covariance matrix of λ_n is $\boldsymbol{G}(\boldsymbol{u}_n)^{-1}$; for part 3 it is the fixed covariance matrix $\boldsymbol{C}_\nu$ of ν in the basis $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_d$, transport being an isometry. Either way the covariance stays bounded, so the conditioning built into $\lambda_n^{(n)}$ costs $O\left(e^{-cn}\right)$ in total variation, as recorded after Definition 35, and may be discarded. Writing $\Theta_n = \operatorname{Exp}_{\theta_n}(\boldsymbol{v}_n)$ with $\boldsymbol{v}_n \sim \lambda_n^{(n)}$, the vector $\sqrt{n}\,\Gamma_{\boldsymbol{u}_n}(\boldsymbol{v}_n)$ therefore has, in the basis $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_d$, the law $N\left(\mathbf{0}, \boldsymbol{G}(\boldsymbol{u}_n)^{-1}\right)$, respectively $N\left(\mathbf{0}, \boldsymbol{C}_\nu\right)$; in particular $\sqrt{n}|\boldsymbol{v}_n| = \sqrt{n}|\Gamma_{\boldsymbol{u}_n}(\boldsymbol{v}_n)| = O_{\mathbb{P}}(1)$.

We check next that $\Theta_n \in \mathcal{V}_\varepsilon$ with probability tending to one. The set of pairs $(x, \boldsymbol{v})$ in TM with $\boldsymbol{v} \in \mathcal{E}_x$ and $\operatorname{Exp}_x(\boldsymbol{v}) \in \mathcal{V}_\varepsilon$ is open, the domain of Exp being open in TM and Exp continuous on it as established in the proof of Lemma 18, and it contains $(\theta_0, \mathbf{0})$; hence there is $\delta > 0$ such that $\operatorname{Exp}_x(\boldsymbol{v}) \in \mathcal{V}_\varepsilon$ whenever $x \in \operatorname{Exp}_{\theta_0}(B_\delta)$ and $|\boldsymbol{v}| < \delta$. Since $|\boldsymbol{u}_n| \to 0$ and $\mathbb{P}\left(|\boldsymbol{v}_n| < \delta\right) \to 1$, the claim follows. Put $\boldsymbol{w}_n := \operatorname{Exp}_{\theta_0}^{-1}(\Theta_n)$ on that event and $\boldsymbol{w}_n := \mathbf{0}$ off it.

On that event $\boldsymbol{u}_n$ and $\boldsymbol{w}_n$ lie in B_r, so θ_n and Θ_n lie in $\mathcal{V}^\sharp$; since moreover $\boldsymbol{v}_n \in \mathcal{E}_{\theta_n}$ and $\operatorname{Exp}_{\theta_n}(\boldsymbol{v}_n) = \Theta_n$, the uniqueness clause of part 1 of Lemma 18 identifies $\boldsymbol{v}_n$ with $\operatorname{Exp}_{\theta_n}^{-1}\left(\operatorname{Exp}_{\theta_0}(\boldsymbol{w}_n)\right)$, and the definition (2.25) of F then reads $\Gamma_{\boldsymbol{u}_n}(\boldsymbol{v}_n) = F\left(\boldsymbol{u}_n, \boldsymbol{w}_n\right)$. Lemma 19 gives

$$\left|F\left(\boldsymbol{u}_n, \boldsymbol{w}_n\right) - \left(\boldsymbol{w}_n - \boldsymbol{u}_n\right)\right| \leqslant C|\boldsymbol{u}_n||\boldsymbol{w}_n - \boldsymbol{u}_n|, \tag{2.65}$$

whence first, by the triangle inequality and because $\Gamma_{\boldsymbol{u}_n}$ is an isometry, $|\boldsymbol{w}_n - \boldsymbol{u}_n| \leqslant \frac{|\boldsymbol{v}_n|}{1 - C|\boldsymbol{u}_n|}$ for n large enough that $C|\boldsymbol{u}_n| < 1$, so that $\sqrt{n}|\boldsymbol{w}_n - \boldsymbol{u}_n| = O_{\mathbb{P}}(1)$; and then, feeding this back into (2.65) multiplied by $\sqrt{n}$,

$$\sqrt{n}\,\boldsymbol{w}_n = \boldsymbol{h} + \sqrt{n}\,\Gamma_{\boldsymbol{u}_n}(\boldsymbol{v}_n) + O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right).$$

By Lemma 31, $\boldsymbol{G}(\boldsymbol{u}_n) \to \boldsymbol{G}(\mathbf{0}) = \boldsymbol{G}$, so $N\left(\mathbf{0}, \boldsymbol{G}(\boldsymbol{u}_n)^{-1}\right) \rightsquigarrow N\left(\mathbf{0}, \boldsymbol{G}^{-1}\right)$; Slutsky's lemma therefore gives

$$\sqrt{n}\,\boldsymbol{w}_n \rightsquigarrow N\left(\boldsymbol{h}, \boldsymbol{G}^{-1}\right) \quad \text{in parts 1 and 2,} \qquad \sqrt{n}\,\boldsymbol{w}_n \rightsquigarrow N\left(\boldsymbol{h}, \boldsymbol{C}_\nu\right) \quad \text{in part 3.} \tag{2.66}$$

Step 3: conclusion. Parts 1 and 2 follow from Lemma 37 applied with $Y_n := \hat{\theta}_n$ under $\mathbb{P}_{\theta_n}^n$ and $Y_n' := \Theta_n$: the localization hypotheses were verified in Steps 1 and 2, the chart error of Lemma 37 agrees with $\sqrt{n}\hat{\boldsymbol{u}}_n$ on $\left\{\hat{\theta}_n \in \mathcal{V}_\varepsilon\right\}$, an event of probability tending to one, and the two rescaled chart images have the same weak limit $N\left(\boldsymbol{h}, \boldsymbol{G}^{-1}\right)$ by (2.64) and (2.66).

For part 3, apply the equivalence (2.60) of the same lemma with $Y_n := T_n$ under $\mathbb{P}_{\theta_n}^n$ and $Y_n' := \Theta_n$; the hypothesis $\sqrt{n}\hat{\boldsymbol{u}}_n[T_n] = O_{\mathbb{P}}(1)$ together with localizability and Lemma 24 gives $\mathbb{P}_{\theta_n}^n\left(T_n \in \mathcal{V}_\varepsilon\right) \to 1$ and makes the two versions of the chart error agree there. The left hand side of (2.60) is then (2.63) for the given $\boldsymbol{h}$, and by (2.66) and the fact that BL metrizes weak convergence its right hand side holds if and only if $\sqrt{n}\hat{\boldsymbol{u}}_n[T_n] \rightsquigarrow N\left(\boldsymbol{h}, \boldsymbol{C}_\nu\right)$ under $\mathbb{P}_{\theta_n}^n$, that is, if and only if $\sqrt{n}\hat{\boldsymbol{u}}_n[T_n] - \boldsymbol{h} \rightsquigarrow \nu$ there. Requiring this for every $\boldsymbol{h}$ is exactly condition (2.40) of Lemma 25, which that lemma shows to be equivalent to local regularity with limit law ν; and its last assertion makes the boundedness hypothesis automatic on the regularity side. □

Remark 39 (Reading the theorem)**. 1. What each part costs.** Part 1 uses Proposition 13 and the chart estimate Lemma 36, and nothing else; in particular it does *not* use local regularity. It does not really use the transport comparison Lemma 19 either: at $\boldsymbol{h} = \mathbf{0}$ one has $\boldsymbol{u}_n = \mathbf{0}$, $\Gamma_{\mathbf{0}} = \mathrm{id}$ and therefore $\boldsymbol{w}_n = \boldsymbol{v}_n$ exactly, so that (2.65) is invoked only in the trivial slice $R(\mathbf{0}, \cdot\,) = \mathbf{0}$ already recorded in the proof of that lemma. Corollary 40 below is different in this respect: it needs the transport comparison even at $\boldsymbol{h} = \mathbf{0}$, because there the moving base point is $\hat{\theta}_n$ and not $\theta_{n,\boldsymbol{h}}$. Part 2 is part 1 with the centre moved, and the only ingredient that changes is (2.64), which for $\boldsymbol{h} \neq \mathbf{0}$ is precisely (2.45), the substance of Theorem 26. Part 3 says that this is not an artefact of the proof: for a localizable sequence with the boundedness property, the moving centre statement and local regularity are the *same* statement. So the answer to the question whether local regularity is needed for asymptotic normality at the level of the manifold is: not for the version centred at a fixed, known θ_0, which is a restatement of Proposition 13; and for the version centred at the true parameter, whatever it is, local regularity is not merely sufficient but necessary.

2. The shape may be perturbed. Step 2 uses about the shape only that its covariance, read in the transported frame, is bounded and converges to $\boldsymbol{G}^{-1}$. Part 2 therefore also holds with $\mathcal{N}_n\left(\theta_{n,\boldsymbol{h}}\right)$ replaced by $\mathcal{N}_n\left(\theta_{n,\boldsymbol{h}},\ \left(N\left(\mathbf{0}, G_{\theta_0}^{-1}\right)\right)_{\boldsymbol{h}/\sqrt{n}}\right)$, the transport of the information at θ_0 itself: at the resolution $n^{-\frac{1}{2}}$ the shape need not be re-evaluated at the moving centre. What may *not* be moved is the centre, which is the content of the next comment.

3. Why the moving centre version is the one that means something. The centre of the approximating law in (2.61) is a point the statistician must be told. The centre in (2.62) is the true parameter, whichever member of the contiguous family it is. By Lemma 24 no procedure can distinguish these members, so only the second version supports the sentence one actually wants to say, namely that $\hat{\theta}_n$ is distributed around the truth like a wrapped normal law with dispersion n^{-1} and shape G^{-1}; and it is that sentence which Corollary 40 below converts into a confidence region. The Hodges-type sequence of Remark 27 satisfies the analogue of part 1 at every $\theta_0 \neq \theta_*$, and at θ_* satisfies a degenerate analogue with $\mathcal{N}_n$ replaced by δ_{θ_*}; it satisfies no version of part 2 at θ_*, since by (2.39) its transported error converges to $\delta_{-\boldsymbol{h}}$, which depends on $\boldsymbol{h}$. Part 3 is the formal statement that this is the only way things can go wrong.

4. The link with Example III.1 of Sun et al. (2026). On $\operatorname{Exp}_\theta\left(\mathcal{E}_\theta\right)$ the wrapped law $\mathcal{N}_n(\theta)$ has a density with respect to the Riemannian volume measure of (M, g) proportional to

$$\exp\left(-\frac{n}{2}\, G_\theta\left(\operatorname{Exp}_\theta^{-1}(x),\ \operatorname{Exp}_\theta^{-1}(x)\right)\right)\cdot J_\theta(x)^{-1}, \qquad J_\theta(x) := \left|\det \mathrm{d}\left(\operatorname{Exp}_\theta\right)_{\operatorname{Exp}_\theta^{-1}(x)}\right|, \tag{2.67}$$

the Jacobian factor converting the Lebesgue measure of the tangent space into the volume measure. Two observations. First, the exponent is the Riemannian Gaussian exponent of Example III.1 of Sun et al. (2026), with the quadratic form G_θ in place of $\tau^{-2} g_\theta$: if the information operator happened to be isotropic, $G_\theta = \tau^{-2} g_\theta$, then by the last assertion of Lemma 36 the exponent would be $-\frac{n}{2\tau^2}\, d_M(\theta, x)^2$, and (2.67) would be, up to J_θ, the density displayed in that example with variance parameter $\frac{\tau^2}{n}$. The Fisher information of the probabilistic principal component analysis model is not isotropic — Section 2.3 computes it, and (2.74) already shows that different directions of $T_{\theta_0} M$ are treated very differently — so the honest limiting shape is the anisotropic wrapped law rather than the isotropic Riemannian Gaussian. Second, the Jacobian is invisible at the relevant scale: on the support of $\mathcal{N}_n(\theta)$ one has $\left|\operatorname{Exp}_\theta^{-1}(x)\right| = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right)$, and the standard expansion of the volume density in normal coordinates gives $J_\theta(x) = 1 + O\left(\left|\operatorname{Exp}_\theta^{-1}(x)\right|^2\right) = 1 + O_{\mathbb{P}}\left(n^{-1}\right)$. At the resolution $n^{-\frac{1}{2}}$ at which BL_n measures, the distinction between the wrapped normal law, the intrinsic Riemannian Gaussian density and the Gaussian read in the normal chart is therefore immaterial, and the theorem is insensitive to which of the three is chosen; we have chosen the wrapped law because it requires no curvature computation. Nothing in this observation is used anywhere below; it is recorded only to say what (2.62) amounts to.

5. What is not claimed. The metric BL_n is a weak metric, and nothing above asserts convergence in total variation, nor any control of densities; that would be a genuinely stronger statement, and none of the consequences drawn below needs it. Nor is any uniformity in $\boldsymbol{h}$ asserted: the statements hold along each fixed sequence of local alternatives, which is the exact strength of Definition 23, and the genuinely uniform version would require the local asymptotic normality expansion along arbitrary bounded sequences $\boldsymbol{h}_n$, which Proposition 8 as stated does not provide. Remark 41 says the same thing where it matters.

Locally uniform confidence sets

The first consequence proper is the inferential payoff, and it is the one that distinguishes local regularity from Proposition 13 in practice.

Corollary 40 (Locally uniform Wald confidence sets)**.** Let $\alpha \in (0,1)$, let $\chi^2_{d,1-\alpha}$ be the $(1-\alpha)$ quantile of the chi-squared law with d degrees of freedom, and define the random subset of M

$$\mathcal{C}_{n,1-\alpha} := \operatorname{Exp}_{\hat{\theta}_n}\left(\left\{\boldsymbol{y} \in T_{\hat{\theta}_n} M \ :\ n\, G_{\hat{\theta}_n}(\boldsymbol{y}, \boldsymbol{y}) \leqslant \chi^2_{d,1-\alpha}\right\}\right), \tag{2.68}$$

which is well defined on an event of probability tending to one. Then, for every $\boldsymbol{h} \in T_{\theta_0}M$,

$$\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}\left(\theta_{n,\boldsymbol{h}} \in \mathcal{C}_{n,1-\alpha}\right) \longrightarrow 1-\alpha. \tag{2.69}$$

In particular the coverage is asymptotically correct not only at θ_0, which is the case $\boldsymbol{h} = \boldsymbol{0}$ and needs no more probabilistic input than Proposition 13, but along every sequence of local alternatives at the estimation scale $n^{-\frac{1}{2}}$.

Proof. Fix $\boldsymbol{h}$ and work under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$, which by Lemma 24 is contiguous to $\mathbb{P}^n_{\theta_0}$; all the $o_{\mathbb{P}}(1)$ and $O_{\mathbb{P}}(1)$ statements of Proposition 13 and Proposition 21 therefore remain valid. In particular $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}(\mathcal{A}_n) \to 1$, $\hat{\boldsymbol{u}}_n \to \boldsymbol{0}$ in probability and $\sqrt{n}\hat{\boldsymbol{u}}_n = O_{\mathbb{P}}(1)$.

Well definedness. By Lemma 31 and the continuous mapping theorem, $\boldsymbol{G}(\hat{\boldsymbol{u}}_n) \to \boldsymbol{G}$ in probability, so that $\lambda_{\min}\left(\boldsymbol{G}(\hat{\boldsymbol{u}}_n)\right)$ is bounded away from zero in probability. For any $\boldsymbol{y} \in T_{\hat{\theta}_n}M$, the vector $\boldsymbol{z} := \Gamma_{\hat{\boldsymbol{u}}_n}(\boldsymbol{y})$ has the same norm as $\boldsymbol{y}$, $\Gamma_{\hat{\boldsymbol{u}}_n}$ being a linear isometry, and its coordinates in $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_d$ are the coordinates of $\boldsymbol{y}$ in the frame transported along the radial geodesic; hence, by (2.54),

$$G_{\hat{\theta}_n}(\boldsymbol{y}, \boldsymbol{y}) = \boldsymbol{z}^T \boldsymbol{G}(\hat{\boldsymbol{u}}_n)\boldsymbol{z} \geqslant \lambda_{\min}\left(\boldsymbol{G}(\hat{\boldsymbol{u}}_n)\right)|\boldsymbol{y}|^2. \tag{2.70}$$

The set inside (2.68) is therefore contained in a ball of radius $O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right)$ about $\boldsymbol{0}$. Meanwhile the radius of the domain $\mathcal{E}_{\hat{\theta}_n}$ of $\operatorname{Exp}_{\hat{\theta}_n}$, namely $\min\left(\sigma_{\min}(\hat{\boldsymbol{W}}_n), \hat{\sigma}_n^2\right)$, is a continuous function of the point $\hat{\theta}_n \in M$, as observed in the proof of Lemma 18, and therefore converges in probability to $\rho > 0$ by Theorem 9. Hence, with probability tending to one, that set is a subset of $\mathcal{E}_{\hat{\theta}_n}$ and $\mathcal{C}_{n,1-\alpha}$ is defined.

Reduction to a quadratic form. Both $\hat{\theta}_n$ and $\theta_{n,\boldsymbol{h}}$ lie in $\mathcal{V}$ with probability tending to one, so part 1 of Lemma 18 makes $\boldsymbol{y}_n := \operatorname{Exp}^{-1}_{\hat{\theta}_n}(\theta_{n,\boldsymbol{h}})$ well defined there, and the injectivity of $\operatorname{Exp}_{\hat{\theta}_n}$ on $\mathcal{E}_{\hat{\theta}_n}$ shows that, on an event of probability tending to one, membership $\theta_{n,\boldsymbol{h}} \in \mathcal{C}_{n,1-\alpha}$ is equivalent to

$$n\, G_{\hat{\theta}_n}(\boldsymbol{y}_n, \boldsymbol{y}_n) \leqslant \chi^2_{d,1-\alpha}. \tag{2.71}$$

Put $\boldsymbol{z}_n := \Gamma_{\hat{\boldsymbol{u}}_n}(\boldsymbol{y}_n) \in T_{\theta_0}M$, so that the first equality of (2.70), applied with $\boldsymbol{y} = \boldsymbol{y}_n$, reads $G_{\hat{\theta}_n}(\boldsymbol{y}_n, \boldsymbol{y}_n) = \boldsymbol{z}_n^T\boldsymbol{G}(\hat{\boldsymbol{u}}_n)\boldsymbol{z}_n$.

The limit. Apply Lemma 19 with $\boldsymbol{u} = \hat{\boldsymbol{u}}_n$ and $\boldsymbol{w} = \frac{\boldsymbol{h}}{\sqrt{n}}$, both in B_r with probability tending to one, noting that $F\left(\hat{\boldsymbol{u}}_n, \frac{\boldsymbol{h}}{\sqrt{n}}\right) = \boldsymbol{z}_n$ by (2.25):

$$\left|\sqrt{n}\boldsymbol{z}_n - \left(\boldsymbol{h} - \sqrt{n}\hat{\boldsymbol{u}}_n\right)\right| \leqslant \sqrt{n}\, C\, |\hat{\boldsymbol{u}}_n| \left|\frac{\boldsymbol{h}}{\sqrt{n}} - \hat{\boldsymbol{u}}_n\right| = \frac{C}{\sqrt{n}}\left|\sqrt{n}\hat{\boldsymbol{u}}_n\right|\left|\boldsymbol{h} - \sqrt{n}\hat{\boldsymbol{u}}_n\right| = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right),$$

so that $\sqrt{n}\boldsymbol{z}_n = -(\sqrt{n}\hat{\boldsymbol{u}}_n - \boldsymbol{h}) + o_{\mathbb{P}}(1)$. By (2.45), $\sqrt{n}\hat{\boldsymbol{u}}_n - \boldsymbol{h} \rightsquigarrow N(\boldsymbol{0}, \boldsymbol{G}^{-1})$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$, and this law is symmetric, so $\sqrt{n}\boldsymbol{z}_n \rightsquigarrow N(\boldsymbol{0}, \boldsymbol{G}^{-1})$. Combining with $\boldsymbol{G}(\hat{\boldsymbol{u}}_n) \to \boldsymbol{G}$ in probability, Slutsky's lemma and the continuous mapping theorem applied to $(\boldsymbol{z}, \boldsymbol{A}) \mapsto \boldsymbol{z}^T\boldsymbol{A}\boldsymbol{z}$ give, through the first equality of (2.70),

$$n\, G_{\hat{\theta}_n}(\boldsymbol{y}_n, \boldsymbol{y}_n) = \left(\sqrt{n}\boldsymbol{z}_n\right)^T \boldsymbol{G}(\hat{\boldsymbol{u}}_n)\left(\sqrt{n}\boldsymbol{z}_n\right) \rightsquigarrow \boldsymbol{Z}^T\boldsymbol{G}\boldsymbol{Z}, \qquad \boldsymbol{Z} \sim N\left(\boldsymbol{0}, \boldsymbol{G}^{-1}\right),$$

and $\boldsymbol{Z}^T\boldsymbol{G}\boldsymbol{Z} = \|\boldsymbol{G}^{\frac{1}{2}}\boldsymbol{Z}\|^2 \sim \chi^2_d$ because $\boldsymbol{G}^{\frac{1}{2}}\boldsymbol{Z} \sim N(\boldsymbol{0}, \boldsymbol{I}_d)$. The distribution function of χ^2_d being continuous, the probability of (2.71) converges to $\mathbb{P}\left(\chi^2_d \leqslant \chi^2_{d,1-\alpha}\right) = 1-\alpha$, which is (2.69). $\square$

Remark 41. Three comments.

1. The region is a statistic. Nothing in (2.68) refers to θ_0: the exponential map and the information operator are evaluated at $\hat{\theta}_n$, and Section 2.3 gives both in closed form, so $\mathcal{C}_{n,1-\alpha}$ can be computed from the data alone.

2. Only local regularity is used. The proof uses about $\hat{\theta}_n$ only that it is locally regular at θ_0 with limit law $N\left(\boldsymbol{0}, G^{-1}\right)$. Localizability is part of Definition 23, and consistency in probability follows, since local regularity forces $\sqrt{n}\hat{\boldsymbol{u}}_n = O_{\mathbb{P}}(1)$ and hence $\hat{\boldsymbol{u}}_n \to \boldsymbol{0}$ in probability. Corollary 40 therefore holds verbatim for any locally regular sequence with that limit law, $\hat{\theta}_n$ being replaced by it in (2.68) and $\mathcal{A}_n$ by the corresponding localization event; by (2.46) these are exactly the locally regular sequences that attain the convolution bound.

3. What "locally uniform" does and does not mean. Statement (2.69) is made along each fixed sequence of local alternatives, which is the exact strength of Definition 23 and is already enough to rule out the failure exhibited in Remark 27. Indeed, at $\theta_0 = \theta_*$ the Hodges-type estimator is pinned to θ_* with $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$-probability tending to one, so that on that event the analogous region is the *deterministic* set $\mathrm{Exp}_{\theta_*}\left(\left\{\boldsymbol{v} : n\, G_{\theta_*}(\boldsymbol{v}, \boldsymbol{v}) \leqslant \chi^2_{d,1-\alpha}\right\}\right)$, which by the injectivity of Exp_{θ_*} contains $\theta_{n,\boldsymbol{h}} = \mathrm{Exp}_{\theta_*}\left(\frac{\boldsymbol{h}}{\sqrt{n}}\right)$ precisely when $G(\boldsymbol{h}, \boldsymbol{h}) \leqslant \chi^2_{d,1-\alpha}$, a condition free of n. Its coverage of $\theta_{n,\boldsymbol{h}}$ therefore tends to 1 when $G(\boldsymbol{h}, \boldsymbol{h}) \leqslant \chi^2_{d,1-\alpha}$ and to 0 otherwise, and equals $1-\alpha$ in the limit for no $\boldsymbol{h}$ at all, $\boldsymbol{h} = \boldsymbol{0}$ included. The genuinely uniform statement, $\sup_{|\boldsymbol{h}| \leqslant c} \left|\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}\left(\theta_{n,\boldsymbol{h}} \in \mathcal{C}_{n,1-\alpha}\right) - (1-\alpha)\right| \to 0$, would require the expansion (2.36) along arbitrary bounded sequences $\boldsymbol{h}_n$ rather than along fixed $\boldsymbol{h}$, which Proposition 8 as stated does not provide; we do not need it anywhere.

The fitted covariance matrix

The second consequence records what all of this says about the object the model is really about, namely the fitted covariance matrix $\hat{\boldsymbol{\Sigma}}_n$ of (1.40), which, unlike $\hat{\theta}_n$ itself, is an honest element of the fixed vector space $\mathrm{Sym}(p)$. It is the instance $\varphi = \boldsymbol{\Sigma}$ of item 4 above, and the linear map $\mathcal{D}_0$ of (2.50) plays the role of $\mathrm{d}\varphi_{\theta_0}$: it is the differential $\mathrm{d}\boldsymbol{\Sigma}_{\theta_0}$ of the map $\boldsymbol{\Sigma}$ of (1.6), read through the exponential chart.

Corollary 42 (Covariance level central limit theorem)**.** Let $1 \leqslant q < p$ and $\theta_0 \in M$.

1. Under $\mathbb{P}^n_{\theta_0}$ the fitted covariance matrix is asymptotically linear in $\mathrm{Sym}(p)$ with influence function $\mathcal{D}_0 \circ \mathrm{IF}$,

$$\sqrt{n}\left(\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{\Sigma}_0\right) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \mathcal{D}_0\left(\mathrm{IF}(\boldsymbol{X}_i)\right) + o_{\mathbb{P}}(1), \tag{2.72}$$

and consequently

$$\sqrt{n}\left(\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{\Sigma}_0\right) \rightsquigarrow \mathcal{D}_0(\boldsymbol{Z}), \qquad \boldsymbol{Z} \sim N\left(\boldsymbol{0},\ G^{-1}\right) \text{ on } T_{\theta_0}M. \tag{2.73}$$

2. The map $\mathcal{D}_0$ is injective, and, in the notation of Lemma 44, its image is

$$\mathcal{D}_0\left(T_{\theta_0}M\right) = \left\{\boldsymbol{J}_0 \begin{pmatrix} \boldsymbol{M}_{11} & \boldsymbol{M}_{21}^T \\ \boldsymbol{M}_{21} & \beta \boldsymbol{I}_{p-q} \end{pmatrix} \boldsymbol{J}_0^T \ :\ \boldsymbol{M}_{11} \in \mathrm{Sym}(q),\ \boldsymbol{M}_{21} \in \mathbb{R}^{(p-q)\times q},\ \beta \in \mathbb{R}\right\}, \tag{2.74}$$

where $\boldsymbol{J}_0 := \left[\boldsymbol{U}_0\ \ \boldsymbol{U}_0^{\perp}\right] \in O(p)$ as in (2.81) below. Hence the limit in (2.73) is a centred Gaussian law carried by a d-dimensional subspace of $\mathrm{Sym}(p)$, namely the tangent space at $\boldsymbol{\Sigma}_0$ to the immersed submanifold $\boldsymbol{\Sigma}(M)$ of Proposition 1; it is degenerate in $\mathrm{Sym}(p)$ exactly when $p - q \geqslant 2$, the codimension being

$$\frac{p(p+1)}{2} - d = \frac{(p-q)(p-q+1)}{2} - 1 = \frac{(p-q+2)(p-q-1)}{2}. \tag{2.75}$$

3. The convergence (2.73) is locally regular: for every $\boldsymbol{h} \in T_{\theta_0}M$,

$$\sqrt{n}\left(\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{\Sigma}\left(\theta_{n,\boldsymbol{h}}\right)\right) \rightsquigarrow \mathcal{D}_0(\boldsymbol{Z}) \qquad \text{under } \mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}, \tag{2.76}$$

with the same limit law, free of $\boldsymbol{h}$.

Proof. **1.** On $\mathcal{A}_n$ we have $\hat{\theta}_n = \mathrm{Exp}_{\theta_0}(\hat{\boldsymbol{u}}_n)$, so (2.51) at $\boldsymbol{u} = \hat{\boldsymbol{u}}_n$ is the exact identity

$$\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{\Sigma}_0 = \mathcal{D}_0(\hat{\boldsymbol{u}}_n) + \boldsymbol{U}_{\hat{\boldsymbol{u}}_n} \boldsymbol{U}_{\hat{\boldsymbol{u}}_n}^T \qquad \text{on } \mathcal{A}_n, \tag{2.77}$$

with no remainder, because $\operatorname{Exp}_{\theta_0}$ is affine in the horizontal lift. Since ι is a linear isometry, $\|\boldsymbol{U}_{\hat{\boldsymbol{u}}_n}\boldsymbol{U}_{\hat{\boldsymbol{u}}_n}^T\|_F \leqslant \|\boldsymbol{U}_{\hat{\boldsymbol{u}}_n}\|\,\|\boldsymbol{U}_{\hat{\boldsymbol{u}}_n}\|_F \leqslant \|\boldsymbol{U}_{\hat{\boldsymbol{u}}_n}\|_F^2 \leqslant |\hat{\boldsymbol{u}}_n|^2$, and $|\hat{\boldsymbol{u}}_n| = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right)$ by Proposition 13; hence $\sqrt{n}\,\|\boldsymbol{U}_{\hat{\boldsymbol{u}}_n}\boldsymbol{U}_{\hat{\boldsymbol{u}}_n}^T\|_F = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right) = o_{\mathbb{P}}(1)$. Multiplying (2.77) by $\sqrt{n}$ therefore gives

$$\sqrt{n}\left(\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{\Sigma}_0\right) = \mathcal{D}_0\left(\sqrt{n}\,\hat{\boldsymbol{u}}_n\right) + o_{\mathbb{P}}(1) \qquad \text{on } \mathcal{A}_n, \tag{2.78}$$

and the restriction to $\mathcal{A}_n$ is harmless: the two sides of (2.78) differ only on $\mathcal{A}_n^c$, whose probability tends to zero by Proposition 21, so their difference is $o_{\mathbb{P}}(1)$ and they have the same weak limits. Now (2.72) follows from (1.42) and the linearity of $\mathcal{D}_0$, and (2.73) from Proposition 13, Slutsky's lemma and the continuous mapping theorem applied to the linear, hence continuous, map $\mathcal{D}_0$.

2. If $\mathcal{D}_0(\boldsymbol{h}) = \mathbf{0}$ then $G(\boldsymbol{h},\boldsymbol{h}) = 0$ by (1.24), whence $\boldsymbol{h} = \mathbf{0}$ by Theorem 7; so $\mathcal{D}_0$ is injective and its image has dimension d. That image is the one described in (2.74) by (2.84) of Lemma 45 below, together with part 3 of that lemma and Lemma 46: as $\boldsymbol{h}$ ranges over $T_{\theta_0}M$, the triple $(2\boldsymbol{\Psi} + b_{\boldsymbol{h}}\boldsymbol{I}_q,\ \boldsymbol{C}\boldsymbol{D}_0,\ b_{\boldsymbol{h}})$ ranges over all of $\operatorname{Sym}(q) \times \mathbb{R}^{(p-q)\times q} \times \mathbb{R}$, the matrix $\boldsymbol{D}_0$ being invertible. The image is that of $\mathrm{d}\boldsymbol{\Sigma}_{\theta_0}$ because $\mathcal{D}_0(\boldsymbol{h}) = \frac{d}{dt}\Big|_{t=0}\boldsymbol{\Sigma}\left(\operatorname{Exp}_{\theta_0}(t\boldsymbol{h})\right)$ by (1.23) and $t \mapsto \operatorname{Exp}_{\theta_0}(t\boldsymbol{h})$ has velocity $\boldsymbol{h}$ at $t = 0$. Reading (2.74) off, the only constraint that $\mathcal{D}_0(T_{\theta_0}M)$ imposes on a symmetric matrix is that its trailing $(p-q)\times(p-q)$ block, in the basis $\boldsymbol{J}_0$, be a multiple of the identity; the codimension is therefore $\dim \operatorname{Sym}(p-q) - 1 = \frac{(p-q)(p-q+1)}{2} - 1$. Writing $m := p-q$, this is $\frac{m^2+m-2}{2} = \frac{(m+2)(m-1)}{2}$, and it vanishes if and only if $m = 1$, since $m + 2 \geqslant 3$. The same number is $\frac{p(p+1)}{2} - d$: by (1.2),

$$\frac{p(p+1)}{2} - d = \frac{p(p+1)}{2} - pq - 1 + \frac{q(q-1)}{2} = \frac{p^2 - 2pq + q^2 + p - q - 2}{2} = \frac{m^2 + m - 2}{2}.$$

3. Fix $\boldsymbol{h}$ and work under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$, which is contiguous to $\mathbb{P}^n_{\theta_0}$ by Lemma 24; in particular $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}(\mathcal{A}_n) \to 1$ and the $O_{\mathbb{P}}$ statements above persist, as in the proof of Corollary 40. Applying (2.51) at $\boldsymbol{u} = \frac{\boldsymbol{h}}{\sqrt{n}}$ gives the second exact identity

$$\boldsymbol{\Sigma}\left(\theta_{n,\boldsymbol{h}}\right) - \boldsymbol{\Sigma}_0 = \frac{1}{\sqrt{n}}\mathcal{D}_0(\boldsymbol{h}) + \frac{1}{n}\boldsymbol{U}_{\boldsymbol{h}}\boldsymbol{U}_{\boldsymbol{h}}^T,$$

so subtracting it from (2.77) and multiplying by $\sqrt{n}$,

$$\sqrt{n}\left(\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{\Sigma}\left(\theta_{n,\boldsymbol{h}}\right)\right) = \mathcal{D}_0\left(\sqrt{n}\hat{\boldsymbol{u}}_n - \boldsymbol{h}\right) + \sqrt{n}\,\boldsymbol{U}_{\hat{\boldsymbol{u}}_n}\boldsymbol{U}_{\hat{\boldsymbol{u}}_n}^T - \frac{1}{\sqrt{n}}\boldsymbol{U}_{\boldsymbol{h}}\boldsymbol{U}_{\boldsymbol{h}}^T \qquad \text{on } \mathcal{A}_n.$$

The last term is deterministic and $O\left(n^{-\frac{1}{2}}\right)$. By (2.45), $\sqrt{n}\hat{\boldsymbol{u}}_n - \boldsymbol{h} \rightsquigarrow N(\mathbf{0}, \boldsymbol{G}^{-1})$ under $\mathbb{P}^n_{\theta_{n,\boldsymbol{h}}}$; in particular $\sqrt{n}\hat{\boldsymbol{u}}_n = O_{\mathbb{P}}(1)$ there, so the middle term is again $O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right)$ by the bound of part 1. The continuous mapping theorem applied to $\mathcal{D}_0$ now gives (2.76), the limit being the image of $N(\mathbf{0}, \boldsymbol{G}^{-1})$, that is $\mathcal{D}_0(\boldsymbol{Z})$, and free of $\boldsymbol{h}$. □

Remark 43 (What is used where)**.** Part 1 of Corollary 42 uses nothing from Section 2.1 beyond the localization of Proposition 21, and even there any normal neighbourhood of θ_0 would serve in place of $\mathcal{V}$. Its proof invokes Proposition 13, the exact expansion (2.51) and the continuous mapping theorem, and no more; neither the parallel transport of Proposition 16, nor the transport comparison of Lemma 19, nor the contiguity of Lemma 24, nor the local regularity of Theorem 26 appears anywhere in it. It could have been stated at the end of Section 1.4; it is placed here only because part 3 belongs here, and because it is the covariance level reading of the whole chapter.

The reason no transport is needed is the converse of the phenomenon that made Section 2.1 necessary in the first place. The intrinsic error $\operatorname{Exp}^{-1}_{\theta_{n,\boldsymbol{h}}}(\hat{\theta}_n)$ of Definition 23 lives in $T_{\theta_{n,\boldsymbol{h}}}M$, a vector space that *moves with* n, which is exactly why $\Gamma_{\boldsymbol{h}/\sqrt{n}}$ has to be inserted before any limit can be taken. The covariance level error $\hat{\boldsymbol{\Sigma}}_n - \boldsymbol{\Sigma}(\theta)$, by contrast, is a difference of two elements of the single fixed vector space $\operatorname{Sym}(p)$, for every θ at once; there is nothing to transport, the delta method applies verbatim, and curvature is invisible because $\boldsymbol{\Sigma}(M) \subseteq \operatorname{Sym}(p)$ is being viewed extrinsically. That the delta method is here an identity rather than an approximation is (2.51): the quadratic remainder is written out and cancelled, so (2.76) is a sharper form of the general (2.49).

What part 3 costs is correspondingly small: given (2.45), which is the substance of Theorem 26 and does use Lemma 24 and Le Cam's third lemma, the passage to (2.76) is three lines of exact algebra. One should resist reading more into part 3 than it says: it asserts that the *maximum likelihood* fitted covariance is locally regular as an estimator of $\mathbf{\Sigma}(\theta)$, and nothing about other estimators.

This is the place to close the account opened at the start of the section. Theorem 26 is a statement about *other parameters* than θ_0, and none of the geometry it rests on — Lemma 14, Proposition 16 and Lemmas 18 and 19, which are results about (M, g) in their own right, independent of any statistics — was needed by Proposition 13, which lives entirely inside the single vector space $T_{\theta_0}M$. They are what the two-base-point comparison costs, and Lemma 24 is the one probabilistic ingredient added alongside them. In the terminology of Sun et al. (2026), Theorem 26 verifies their Definition III.5 for the probabilistic principal component analysis model. We have proved it through the chart reduction of Lemma 25 and the classical Euclidean results of van der Vaart (1998), rather than by citing the manifold version, because Lemma 19 makes that reduction exact, and because of the localization of the curvature recorded in Remark 20.

2.3 The Fisher Information Operator in Closed Form

Proposition 13 identifies the limiting covariance of the maximum likelihood estimator as $G_{\theta_0}^{-1}$, and Theorem 26 shows that this limit is not an accident of the point θ_0; but nothing so far says what $G_{\theta_0}^{-1}$ *is*. This section computes it. The only inputs are (1.23), (1.22) and (1.24) of Section 1.3, the spectral decomposition (2.79) of Lemma 44 and the horizontal lift (2.2); no asymptotics are used until Theorem 49, which merely reads Proposition 13 in the coordinates constructed here.

Two features of the answer deserve to be announced. First, the whole calculation is a 2×2 block computation in the singular basis of $\boldsymbol{W}_0$, and it produces a set of coordinates in which G_{θ_0} is block diagonal with blocks of size 1, 1 and $q+1$; the only genuinely coupled block is the one that mixes the q leading directions with the noise variance, and it is inverted by a one-line Schur complement. Second, no assumption whatsoever is made on the multiplicities of the singular values of $\boldsymbol{W}_0$: the formulas below are valid when $\sigma_1(\boldsymbol{W}_0) = \cdots = \sigma_q(\boldsymbol{W}_0)$, a case in which individual eigenvectors of $\boldsymbol{S}_n$ are not asymptotically normal at all. This is the quantitative form of the third comment of Remark 22.

Block coordinates at the true parameter

Everything rests on the spectral decomposition of $\mathbf{\Sigma}_0$, which the first chapter computes eigenvalue by eigenvalue in (1.36) but does not assemble into a basis. We assemble it now.

Lemma 44 (Spectral decomposition of the true covariance)**.** Fix a singular value decomposition $\boldsymbol{W}_0 = \boldsymbol{U}_0\boldsymbol{D}_0\boldsymbol{V}_0^T$, in which $\boldsymbol{U}_0 \in \mathbb{R}^{p\times q}$ has orthonormal columns, $\boldsymbol{V}_0 \in O(q)$, and $\boldsymbol{D}_0 = \operatorname{diag}(\sigma_1(\boldsymbol{W}_0), \dots, \sigma_q(\boldsymbol{W}_0))$ has strictly positive diagonal because $\operatorname{rank}(\boldsymbol{W}_0) = q$. Put $\boldsymbol{P}_0 := \boldsymbol{U}_0\boldsymbol{U}_0^T$, the orthogonal projection onto the column space of $\boldsymbol{W}_0$. Then

$$\mathbf{\Sigma}_0 = \boldsymbol{U}_0\left(\boldsymbol{D}_0^2 + \sigma_0^2\boldsymbol{I}_q\right)\boldsymbol{U}_0^T + \sigma_0^2\left(\boldsymbol{I}_p - \boldsymbol{P}_0\right), \tag{2.79}$$

the two summands acting on orthogonal complementary subspaces; this is the spectral decomposition realizing the eigenvalues (1.36), and the gap separating the signal eigenvalues from the noise eigenvalue is

$$\lambda_q(\mathbf{\Sigma}_0) - \lambda_{q+1}(\mathbf{\Sigma}_0) = \sigma_q(\boldsymbol{W}_0)^2 > 0. \tag{2.80}$$

Passing to another representative $(\boldsymbol{W}_0\boldsymbol{Q}, \sigma_0^2)$ of θ_0, with $\boldsymbol{Q} \in O(q)$, leaves $\boldsymbol{U}_0$, $\boldsymbol{D}_0$ and $\boldsymbol{P}_0$ unchanged and replaces $\boldsymbol{V}_0$ by $\boldsymbol{Q}^T\boldsymbol{V}_0$.

Proof. We have $\boldsymbol{W}_0\boldsymbol{W}_0^T = \boldsymbol{U}_0\boldsymbol{D}_0^2\boldsymbol{U}_0^T$ and $\boldsymbol{I}_p = \boldsymbol{P}_0 + (\boldsymbol{I}_p - \boldsymbol{P}_0)$, whence (2.79); the ranges of $\boldsymbol{P}_0$ and $\boldsymbol{I}_p - \boldsymbol{P}_0$ are orthogonal complements. Equation (2.80) follows from (1.36) because $\sigma_q(\boldsymbol{W}_0) = \sigma_{\min}(\boldsymbol{W}_0) > 0$. The last sentence is immediate from $\boldsymbol{W}_0\boldsymbol{Q} = \boldsymbol{U}_0\boldsymbol{D}_0\left(\boldsymbol{Q}^T\boldsymbol{V}_0\right)^T$. □

We fix the singular value decomposition $\boldsymbol{W}_0 = \boldsymbol{U}_0\boldsymbol{D}_0\boldsymbol{V}_0^T$ of Lemma 44 once and for all, and we complete $\boldsymbol{U}_0$ to an orthogonal matrix: we let $\boldsymbol{U}_0^\perp \in \mathbb{R}^{p\times(p-q)}$ have orthonormal columns spanning the orthogonal complement of the column space of $\boldsymbol{W}_0$, so that

$$\boldsymbol{J}_0 := \left[\boldsymbol{U}_0 \ \ \boldsymbol{U}_0^\perp\right] \in O(p), \qquad \boldsymbol{I}_p - \boldsymbol{P}_0 = \boldsymbol{U}_0^\perp\left(\boldsymbol{U}_0^\perp\right)^T. \tag{2.81}$$

We abbreviate the signal eigenvalues of $\boldsymbol{\Sigma}_0$ supplied by Lemma 44 as

$$\tau_j := \lambda_j(\boldsymbol{\Sigma}_0) = \sigma_j(\boldsymbol{W}_0)^2 + \sigma_0^2 > \sigma_0^2 \qquad (j = 1,\dots,q), \qquad \boldsymbol{T}_0 := \boldsymbol{D}_0^2 + \sigma_0^2\boldsymbol{I}_q = \operatorname{diag}(\tau_1,\dots,\tau_q). \tag{2.82}$$

The letters $\boldsymbol{B}$, $\boldsymbol{C}$, $\boldsymbol{\Psi}$, $\boldsymbol{y}$, $\boldsymbol{z}$, $\boldsymbol{J}_0$, $\boldsymbol{T}_0$, τ_j, κ, $\boldsymbol{K}$ and $\mathcal{G}$ are local to this section.

Lemma 45 (Block form of the differential)**.** Let $\boldsymbol{h} \in T_{\theta_0}M$ have horizontal lift $\iota(\boldsymbol{h}) = (\boldsymbol{U}_{\boldsymbol{h}}, b_{\boldsymbol{h}})$ as in (2.3), and set

$$\boldsymbol{B} := \boldsymbol{U}_0^T\boldsymbol{U}_{\boldsymbol{h}}\boldsymbol{V}_0 \in \mathbb{R}^{q\times q}, \qquad \boldsymbol{C} := \left(\boldsymbol{U}_0^\perp\right)^T\boldsymbol{U}_{\boldsymbol{h}}\boldsymbol{V}_0 \in \mathbb{R}^{(p-q)\times q}, \qquad \boldsymbol{\Psi} := \frac{1}{2}\left(\boldsymbol{B}\boldsymbol{D}_0 + \boldsymbol{D}_0\boldsymbol{B}^T\right) \in \mathrm{Sym}(q). \tag{2.83}$$

Then:

1. $\boldsymbol{J}_0^T\boldsymbol{\Sigma}_0\boldsymbol{J}_0 = \begin{pmatrix}\boldsymbol{T}_0 & \boldsymbol{0}\\ \boldsymbol{0} & \sigma_0^2\boldsymbol{I}_{p-q}\end{pmatrix}$ and $\boldsymbol{J}_0^T\boldsymbol{\Sigma}_0^{-1}\boldsymbol{J}_0 = \begin{pmatrix}\boldsymbol{T}_0^{-1} & \boldsymbol{0}\\ \boldsymbol{0} & \sigma_0^{-2}\boldsymbol{I}_{p-q}\end{pmatrix}$.
2. $\boldsymbol{U}_{\boldsymbol{h}} = \boldsymbol{U}_0\boldsymbol{B}\boldsymbol{V}_0^T + \boldsymbol{U}_0^\perp\boldsymbol{C}\boldsymbol{V}_0^T$ and $|\boldsymbol{h}|^2 = \|\boldsymbol{B}\|_F^2 + \|\boldsymbol{C}\|_F^2 + b_{\boldsymbol{h}}^2$.
3. $\boldsymbol{D}_0\boldsymbol{B} \in \mathrm{Sym}(q)$; conversely every triple $(\boldsymbol{B}, \boldsymbol{C}, b)$ with $\boldsymbol{D}_0\boldsymbol{B}$ symmetric arises from exactly one $\boldsymbol{h} \in T_{\theta_0}M$.
4. In the basis (2.81),
$$\boldsymbol{J}_0^T\,\mathcal{D}_0(\boldsymbol{h})\,\boldsymbol{J}_0 = \begin{pmatrix}2\boldsymbol{\Psi} + b_{\boldsymbol{h}}\boldsymbol{I}_q & \boldsymbol{D}_0\boldsymbol{C}^T\\ \boldsymbol{C}\boldsymbol{D}_0 & b_{\boldsymbol{h}}\boldsymbol{I}_{p-q}\end{pmatrix}, \tag{2.84}$$
with $\mathcal{D}_0(\boldsymbol{h}) = D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\theta_0)$ as in (2.50).

Proof. **1.** Substitute $\boldsymbol{P}_0 = \boldsymbol{U}_0\boldsymbol{U}_0^T$ and $\boldsymbol{I}_p - \boldsymbol{P}_0 = \boldsymbol{U}_0^\perp(\boldsymbol{U}_0^\perp)^T$ into (2.79) and use $\boldsymbol{U}_0^T\boldsymbol{U}_0 = \boldsymbol{I}_q$, $(\boldsymbol{U}_0^\perp)^T\boldsymbol{U}_0^\perp = \boldsymbol{I}_{p-q}$ and $\boldsymbol{U}_0^T\boldsymbol{U}_0^\perp = \boldsymbol{0}$.

2. Since $\boldsymbol{J}_0 \in O(p)$ and $\boldsymbol{V}_0 \in O(q)$, the matrix $\boldsymbol{J}_0^T\boldsymbol{U}_{\boldsymbol{h}}\boldsymbol{V}_0$ has $\boldsymbol{B}$ and $\boldsymbol{C}$ stacked as its blocks, which is the first identity; the Frobenius norm is invariant under multiplication by orthogonal matrices on either side, so $\|\boldsymbol{U}_{\boldsymbol{h}}\|_F^2 = \|\boldsymbol{B}\|_F^2 + \|\boldsymbol{C}\|_F^2$, and the second identity is (2.4).

3. By part 3 of Proposition 1 the pair $(\boldsymbol{U}_{\boldsymbol{h}}, b_{\boldsymbol{h}})$ is horizontal at $(\boldsymbol{W}_0, \sigma_0^2)$ if and only if $\boldsymbol{U}_{\boldsymbol{h}}^T\boldsymbol{W}_0 = \boldsymbol{W}_0^T\boldsymbol{U}_{\boldsymbol{h}}$. Using part 2 and $\boldsymbol{W}_0 = \boldsymbol{U}_0\boldsymbol{D}_0\boldsymbol{V}_0^T$,

$$\boldsymbol{W}_0^T\boldsymbol{U}_{\boldsymbol{h}} = \boldsymbol{V}_0\boldsymbol{D}_0\boldsymbol{U}_0^T\left(\boldsymbol{U}_0\boldsymbol{B}\boldsymbol{V}_0^T + \boldsymbol{U}_0^\perp\boldsymbol{C}\boldsymbol{V}_0^T\right) = \boldsymbol{V}_0\left(\boldsymbol{D}_0\boldsymbol{B}\right)\boldsymbol{V}_0^T,$$

which is symmetric if and only if $\boldsymbol{D}_0\boldsymbol{B}$ is. Conversely, given such a triple, part 2 defines $\boldsymbol{U}_{\boldsymbol{h}}$, the displayed identity makes $(\boldsymbol{U}_{\boldsymbol{h}}, b)$ horizontal, and $\boldsymbol{h} := \mathrm{d}\pi_{(\boldsymbol{W}_0,\sigma_0^2)}\left((\boldsymbol{U}_{\boldsymbol{h}}, b)\right)$ is the unique preimage because ι is a bijection onto $H_{(\boldsymbol{W}_0,\sigma_0^2)}$.

4. By (2.50), $\mathcal{D}_0(\boldsymbol{h}) = \boldsymbol{U}_{\boldsymbol{h}}\boldsymbol{W}_0^T + \boldsymbol{W}_0\boldsymbol{U}_{\boldsymbol{h}}^T + b_{\boldsymbol{h}}\boldsymbol{I}_p$, and by part 2,

$$\boldsymbol{U}_{\boldsymbol{h}}\boldsymbol{W}_0^T = \left(\boldsymbol{U}_0\boldsymbol{B}\boldsymbol{V}_0^T + \boldsymbol{U}_0^{\perp}\boldsymbol{C}\boldsymbol{V}_0^T\right)\boldsymbol{V}_0\boldsymbol{D}_0\boldsymbol{U}_0^T = \boldsymbol{U}_0\boldsymbol{B}\boldsymbol{D}_0\boldsymbol{U}_0^T + \boldsymbol{U}_0^{\perp}\boldsymbol{C}\boldsymbol{D}_0\boldsymbol{U}_0^T,$$

so that $\boldsymbol{J}_0^T\boldsymbol{U}_{\boldsymbol{h}}\boldsymbol{W}_0^T\boldsymbol{J}_0 = \begin{pmatrix}\boldsymbol{B}\boldsymbol{D}_0 & \mathbf{0}\\ \boldsymbol{C}\boldsymbol{D}_0 & \mathbf{0}\end{pmatrix}$. Adding the transpose of this matrix and $b_{\boldsymbol{h}}\boldsymbol{I}_p$, and using $\boldsymbol{B}\boldsymbol{D}_0+(\boldsymbol{B}\boldsymbol{D}_0)^T = 2\boldsymbol{\Psi}$, gives (2.84). ☐

The three blocks have direct meanings, and it is worth recording them before any computation. By (2.84) the matrix $2\boldsymbol{\Psi} + b_{\boldsymbol{h}}\boldsymbol{I}_q$ is the first order change of $\boldsymbol{\Sigma}$ inside the signal block, the matrix $\boldsymbol{C}\boldsymbol{D}_0$ is its change in the signal–noise block, and $b_{\boldsymbol{h}}\boldsymbol{I}_{p-q}$ is its change in the noise block — where the model forces the change to be a *multiple of the identity*, which is the whole source of the efficiency gain computed below, and which is the only constraint, as (2.74) recorded. Note also that $\boldsymbol{\Psi}$, not $\boldsymbol{B}$, is the natural unknown: the map $\boldsymbol{B} \mapsto \boldsymbol{\Psi}$ is exactly the operation that quotients out the $\frac{q(q-1)}{2}$ directions along which $\boldsymbol{\Sigma}$ does not move.

Lemma 46 (Linear coordinates on $T_{\theta_0}M$)**.** The map

$$\kappa : T_{\theta_0}M \longrightarrow \mathrm{Sym}(q) \times \mathbb{R}^{(p-q)\times q} \times \mathbb{R}, \qquad \kappa(\boldsymbol{h}) := (\boldsymbol{\Psi},\ \boldsymbol{C},\ b_{\boldsymbol{h}}) \tag{2.85}$$

of (2.83) is a linear isomorphism.

Proof. Linearity is clear: ι is linear by (2.2), and $\boldsymbol{U}_{\boldsymbol{h}} \mapsto (\boldsymbol{B}, \boldsymbol{C}) \mapsto \boldsymbol{\Psi}$ is a composition of linear maps.

For injectivity, suppose $\kappa(\boldsymbol{h}) = \mathbf{0}$. Then $b_{\boldsymbol{h}} = 0$ and $\boldsymbol{C} = \mathbf{0}$, while $\boldsymbol{\Psi} = \mathbf{0}$ reads $\boldsymbol{B}\boldsymbol{D}_0 = -\boldsymbol{D}_0\boldsymbol{B}^T$. Since $\mathrm{rank}(\boldsymbol{W}_0) = q$, all $\sigma_j(\boldsymbol{W}_0)$ are strictly positive and $\boldsymbol{D}_0$ is invertible, so part 3 of Lemma 45, that is $\boldsymbol{D}_0\boldsymbol{B} = \boldsymbol{B}^T\boldsymbol{D}_0$, gives $\boldsymbol{B}^T = \boldsymbol{D}_0\boldsymbol{B}\boldsymbol{D}_0^{-1}$. Substituting this into $\boldsymbol{B}\boldsymbol{D}_0 = -\boldsymbol{D}_0\boldsymbol{B}^T$ yields $\boldsymbol{B}\boldsymbol{D}_0 = -\boldsymbol{D}_0^2\boldsymbol{B}\boldsymbol{D}_0^{-1}$, and multiplying on the right by $\boldsymbol{D}_0$,

$$\boldsymbol{B}\boldsymbol{D}_0^2 + \boldsymbol{D}_0^2\boldsymbol{B} = \mathbf{0}.$$

Both $\boldsymbol{D}_0^2$ factors are diagonal, so the (i,j) entry of the left hand side is $(\sigma_i(\boldsymbol{W}_0)^2 + \sigma_j(\boldsymbol{W}_0)^2)\,\boldsymbol{B}_{ij}$, and the bracket is strictly positive; hence $\boldsymbol{B} = \mathbf{0}$. By part 2 of Lemma 45 this forces $\boldsymbol{U}_{\boldsymbol{h}} = \mathbf{0}$, so $\iota(\boldsymbol{h}) = (\mathbf{0}, 0)$ and $\boldsymbol{h} = \mathbf{0}$.

For surjectivity it suffices to compare dimensions, κ being injective and linear:

$$\dim\left(\mathrm{Sym}(q) \times \mathbb{R}^{(p-q)\times q} \times \mathbb{R}\right) = \frac{q(q+1)}{2} + (p-q)q + 1 = pq + 1 - \frac{q(q-1)}{2} = d$$

by (1.2). ☐

Lemma 46 lets us treat $(\boldsymbol{\Psi}, \boldsymbol{C}, b)$ as coordinates on $T_{\theta_0}M$. We use throughout the *scalar* coordinates

$$(\boldsymbol{\Psi}_{ij})_{1\leqslant i\leqslant j\leqslant q}, \qquad (\boldsymbol{C}_{ki})_{1\leqslant k\leqslant p-q,\ 1\leqslant i\leqslant q}, \qquad b, \tag{2.86}$$

which identify the target of κ with $\mathbb{R}^d$, and we let $\mathcal{G}$ denote the $d \times d$ symmetric matrix representing the Fisher form in them, that is

$$G_{\theta_0}(\boldsymbol{h}_1, \boldsymbol{h}_2) = \kappa(\boldsymbol{h}_1)^T\,\mathcal{G}\,\kappa(\boldsymbol{h}_2) \qquad \text{for all } \boldsymbol{h}_1, \boldsymbol{h}_2 \in T_{\theta_0}M. \tag{2.87}$$

Note that κ is *not* a g_{θ_0}-isometry, so $\mathcal{G}$ is not the matrix $\boldsymbol{G}$ of (2.32); the relation between the two is recorded in the proof of Theorem 49 and is the only place where it is needed.

The Fisher form and its inverse

Proposition 47 (Explicit Fisher form)**.** For every $\boldsymbol{h} \in T_{\theta_0} M$, in the notation (2.83),

$$G_{\theta_0}(\boldsymbol{h},\boldsymbol{h}) = \frac{1}{2}\,\mathrm{tr}\left[\left(\boldsymbol{T}_0^{-1}\left(2\boldsymbol{\Psi} + b_{\boldsymbol{h}}\boldsymbol{I}_q\right)\right)^2\right] + \frac{1}{\sigma_0^2}\,\mathrm{tr}\left(\boldsymbol{T}_0^{-1}\boldsymbol{D}_0^2\boldsymbol{C}^T\boldsymbol{C}\right) + \frac{(p-q)\,b_{\boldsymbol{h}}^2}{2\sigma_0^4}, \tag{2.88}$$

and, in the scalar coordinates (2.86),

$$G_{\theta_0}(\boldsymbol{h},\boldsymbol{h}) = \underbrace{\sum_{i=1}^{q}\frac{\left(2\boldsymbol{\Psi}_{ii} + b_{\boldsymbol{h}}\right)^2}{2\tau_i^2} + \frac{(p-q)b_{\boldsymbol{h}}^2}{2\sigma_0^4}}_{\text{coupled block}} + \underbrace{\sum_{1\leqslant i<j\leqslant q}\frac{4\boldsymbol{\Psi}_{ij}^2}{\tau_i\tau_j}}_{\boldsymbol{\Psi}\text{ off-diagonal}} + \underbrace{\sum_{i=1}^{q}\sum_{k=1}^{p-q}\left(\frac{1}{\sigma_0^2} - \frac{1}{\tau_i}\right)\boldsymbol{C}_{ki}^2}_{\boldsymbol{C}\text{-block}}. \tag{2.89}$$

In particular $\mathcal{G}$ is block diagonal with respect to the partition of (2.86) into the three groups $\{\boldsymbol{\Psi}_{ij}\}_{i<j}$, $\{\boldsymbol{C}_{ki}\}$ and $(\boldsymbol{\Psi}_{11},\ldots,\boldsymbol{\Psi}_{qq},b)$, with

$$\mathcal{G}_{\boldsymbol{\Psi}_{ij}\boldsymbol{\Psi}_{ij}} = \frac{4}{\tau_i\tau_j}\quad(i<j),\qquad \mathcal{G}_{\boldsymbol{C}_{ki}\boldsymbol{C}_{ki}} = \frac{1}{\sigma_0^2} - \frac{1}{\tau_i} = \frac{\sigma_i(\boldsymbol{W}_0)^2}{\sigma_0^2\tau_i},\qquad \mathcal{G}_{\mathrm{c}} = \begin{pmatrix}\boldsymbol{\Pi} & \boldsymbol{\nu}\\ \boldsymbol{\nu}^T & \eta\end{pmatrix}, \tag{2.90}$$

the last being the coupled block in the ordering $(\boldsymbol{\Psi}_{11},\ldots,\boldsymbol{\Psi}_{qq},b)$, where

$$\boldsymbol{\Pi} := \mathrm{diag}\left(2\tau_1^{-2},\ldots,2\tau_q^{-2}\right),\qquad \boldsymbol{\nu} := \left(\tau_1^{-2},\ldots,\tau_q^{-2}\right)^T \in \mathbb{R}^q,\qquad \eta := \frac{1}{2}\sum_{i=1}^{q}\frac{1}{\tau_i^2} + \frac{p-q}{2\sigma_0^4}. \tag{2.91}$$

Proof. Write $\boldsymbol{M} := 2\boldsymbol{\Psi} + b_{\boldsymbol{h}}\boldsymbol{I}_q$ and $b := b_{\boldsymbol{h}}$. Conjugating by $\boldsymbol{J}_0$ does not change traces, so by (1.24) with $\boldsymbol{h}_1 = \boldsymbol{h}_2 = \boldsymbol{h}$ it suffices to compute the trace of the square of $\boldsymbol{J}_0^T\boldsymbol{\Sigma}_0^{-1}\mathcal{D}_0(\boldsymbol{h})\boldsymbol{J}_0$. Multiplying the two matrices of Lemma 45,

$$\boldsymbol{J}_0^T\boldsymbol{\Sigma}_0^{-1}\mathcal{D}_0(\boldsymbol{h})\boldsymbol{J}_0 = \begin{pmatrix}\boldsymbol{T}_0^{-1}\boldsymbol{M} & \boldsymbol{T}_0^{-1}\boldsymbol{D}_0\boldsymbol{C}^T\\ \sigma_0^{-2}\boldsymbol{C}\boldsymbol{D}_0 & \sigma_0^{-2}b\,\boldsymbol{I}_{p-q}\end{pmatrix}. \tag{2.92}$$

The two diagonal blocks of the square of (2.92) are

$$\left(\boldsymbol{T}_0^{-1}\boldsymbol{M}\right)^2 + \sigma_0^{-2}\boldsymbol{T}_0^{-1}\boldsymbol{D}_0\boldsymbol{C}^T\boldsymbol{C}\boldsymbol{D}_0 \qquad\text{and}\qquad \sigma_0^{-2}\boldsymbol{C}\boldsymbol{D}_0\boldsymbol{T}_0^{-1}\boldsymbol{D}_0\boldsymbol{C}^T + \sigma_0^{-4}b^2\boldsymbol{I}_{p-q},$$

so, using the cyclic property of the trace and the fact that the diagonal matrices $\boldsymbol{T}_0$ and $\boldsymbol{D}_0$ commute,

$$\mathrm{tr}\left[\left(\boldsymbol{\Sigma}_0^{-1}\mathcal{D}_0(\boldsymbol{h})\right)^2\right] = \mathrm{tr}\left[\left(\boldsymbol{T}_0^{-1}\boldsymbol{M}\right)^2\right] + \frac{2}{\sigma_0^2}\,\mathrm{tr}\left(\boldsymbol{T}_0^{-1}\boldsymbol{D}_0^2\boldsymbol{C}^T\boldsymbol{C}\right) + \frac{(p-q)b^2}{\sigma_0^4}.$$

Halving gives (2.88).

For (2.89), both $\boldsymbol{T}_0^{-1}$ and $\boldsymbol{D}_0^2$ are diagonal, so

$$\mathrm{tr}\left(\boldsymbol{T}_0^{-1}\boldsymbol{D}_0^2\boldsymbol{C}^T\boldsymbol{C}\right) = \sum_{i=1}^{q}\frac{\sigma_i(\boldsymbol{W}_0)^2}{\tau_i}\left(\boldsymbol{C}^T\boldsymbol{C}\right)_{ii} = \sum_{i=1}^{q}\frac{\sigma_i(\boldsymbol{W}_0)^2}{\tau_i}\sum_{k=1}^{p-q}\boldsymbol{C}_{ki}^2,$$

and $\frac{\sigma_i(\boldsymbol{W}_0)^2}{\sigma_0^2\tau_i} = \frac{\tau_i - \sigma_0^2}{\sigma_0^2\tau_i} = \frac{1}{\sigma_0^2} - \frac{1}{\tau_i}$ by (2.82). For the first term, $\boldsymbol{M}$ is symmetric, so

$$\mathrm{tr}\left[\left(\boldsymbol{T}_0^{-1}\boldsymbol{M}\right)^2\right] = \sum_{i,j=1}^{q}\frac{\boldsymbol{M}_{ij}\boldsymbol{M}_{ji}}{\tau_i\tau_j} = \sum_{i=1}^{q}\frac{\boldsymbol{M}_{ii}^2}{\tau_i^2} + 2\sum_{i<j}\frac{\boldsymbol{M}_{ij}^2}{\tau_i\tau_j},$$

and $\boldsymbol{M}_{ii} = 2\boldsymbol{\Psi}_{ii} + b$, $\boldsymbol{M}_{ij} = 2\boldsymbol{\Psi}_{ij}$ for $i \neq j$; halving gives the first two groups of terms in (2.89).

Finally, (2.89) is a sum of three quadratic forms in *disjoint* sets of the scalar coordinates (2.86), which is exactly the assertion of block diagonality; reading off the coefficients gives the first two entries of (2.90). For the third, expanding the coupled part of (2.89),

$$\sum_{i=1}^{q} \frac{(2\boldsymbol{\Psi}_{ii}+b)^2}{2\tau_i^2} + \frac{(p-q)b^2}{2\sigma_0^4} = \sum_{i=1}^{q} \frac{2\boldsymbol{\Psi}_{ii}^2}{\tau_i^2} + 2b\sum_{i=1}^{q} \frac{\boldsymbol{\Psi}_{ii}}{\tau_i^2} + b^2\left(\frac{1}{2}\sum_{i=1}^{q}\frac{1}{\tau_i^2} + \frac{p-q}{2\sigma_0^4}\right),$$

whose matrix in the ordering $(\boldsymbol{\Psi}_{11}, \ldots, \boldsymbol{\Psi}_{qq}, b)$ is $\mathcal{G}_{\mathrm{c}}$ of (2.90): the coefficient of $\boldsymbol{\Psi}_{ii}^2$ is $2\tau_i^{-2} = \boldsymbol{\Pi}_{ii}$, that of $b\,\boldsymbol{\Psi}_{ii}$ is $2\tau_i^{-2} = 2\boldsymbol{\nu}_i$, and that of b^2 is η. □

Proposition 48 (Explicit inverse)**.** The matrix $\mathcal{G}$ of (2.87) is invertible, and $\mathcal{G}^{-1}$ is block diagonal with respect to the same partition, with

$$\left(\mathcal{G}^{-1}\right)_{\boldsymbol{\Psi}_{ij}\boldsymbol{\Psi}_{ij}} = \frac{\tau_i\tau_j}{4} \quad (i<j), \qquad \left(\mathcal{G}^{-1}\right)_{\boldsymbol{C}_{ki}\boldsymbol{C}_{ki}} = \frac{\sigma_0^2\tau_i}{\sigma_i(\boldsymbol{W}_0)^2} = \left(\frac{1}{\sigma_0^2} - \frac{1}{\tau_i}\right)^{-1}, \tag{2.93}$$

these two groups being diagonal, and, on the coupled block,

$$\left(\mathcal{G}^{-1}\right)_{bb} = \frac{2\sigma_0^4}{p-q}, \qquad \left(\mathcal{G}^{-1}\right)_{\boldsymbol{\Psi}_{ii}b} = -\frac{\sigma_0^4}{p-q}, \qquad \left(\mathcal{G}^{-1}\right)_{\boldsymbol{\Psi}_{ii}\boldsymbol{\Psi}_{jj}} = \delta_{ij}\frac{\tau_i^2}{2} + \frac{\sigma_0^4}{2(p-q)}. \tag{2.94}$$

Proof. $\mathcal{G}$ is positive definite because G_{θ_0} is, by Theorem 7, and κ is a bijection; so it is invertible, and the inverse of a block diagonal matrix is block diagonal with the blocks inverted. The first two groups are diagonal by (2.90), and their inverses are the reciprocals, which is (2.93).

For the coupled block, $\boldsymbol{\Pi}^{-1} = \operatorname{diag}\left(\frac{\tau_1^2}{2}, \ldots, \frac{\tau_q^2}{2}\right)$ and

$$\boldsymbol{\Pi}^{-1}\boldsymbol{\nu} = \left(\frac{\tau_i^2}{2}\cdot\frac{1}{\tau_i^2}\right)_{i=1}^{q} = \frac{1}{2}\mathbf{1}_q, \tag{2.95}$$

so the Schur complement of $\boldsymbol{\Pi}$ in $\mathcal{G}_{\mathrm{c}}$ is

$$\zeta := \eta - \boldsymbol{\nu}^T\boldsymbol{\Pi}^{-1}\boldsymbol{\nu} = \eta - \frac{1}{2}\sum_{i=1}^{q}\frac{1}{\tau_i^2} = \frac{p-q}{2\sigma_0^4} \tag{2.96}$$

by (2.91); it is strictly positive because $q<p$, which is where the standing assumption $q<p$ enters the inversion. The standard block inversion formula

$$\begin{pmatrix} \boldsymbol{\Pi} & \boldsymbol{\nu} \\ \boldsymbol{\nu}^T & \eta \end{pmatrix}^{-1} = \begin{pmatrix} \boldsymbol{\Pi}^{-1} + \zeta^{-1}\boldsymbol{\Pi}^{-1}\boldsymbol{\nu}\boldsymbol{\nu}^T\boldsymbol{\Pi}^{-1} & -\zeta^{-1}\boldsymbol{\Pi}^{-1}\boldsymbol{\nu} \\ -\zeta^{-1}\boldsymbol{\nu}^T\boldsymbol{\Pi}^{-1} & \zeta^{-1} \end{pmatrix}$$

together with (2.95) and (2.96) gives $\zeta^{-1} = \frac{2\sigma_0^4}{p-q}$, then $-\zeta^{-1}\left(\boldsymbol{\Pi}^{-1}\boldsymbol{\nu}\right)_i = -\frac{2\sigma_0^4}{p-q}\cdot\frac{1}{2} = -\frac{\sigma_0^4}{p-q}$, and finally

$$\left(\boldsymbol{\Pi}^{-1}\right)_{ij} + \zeta^{-1}\left(\boldsymbol{\Pi}^{-1}\boldsymbol{\nu}\right)_i\left(\boldsymbol{\Pi}^{-1}\boldsymbol{\nu}\right)_j = \delta_{ij}\frac{\tau_i^2}{2} + \frac{2\sigma_0^4}{p-q}\cdot\frac{1}{2}\cdot\frac{1}{2} = \delta_{ij}\frac{\tau_i^2}{2} + \frac{\sigma_0^4}{2(p-q)},$$

which is (2.94). □

Theorem 49 (Theorem 13 in block coordinates)**.** *Let $1 \leqslant q < p$ and $\theta_0 = [\boldsymbol{W}_0, \sigma_0^2] \in M$. Define, on the event $\mathcal{A}_n$ of (2.29), the aligned error blocks*

$$\begin{aligned} \boldsymbol{B}_n &:= \boldsymbol{U}_0^T\left(\hat{\boldsymbol{W}}_n\boldsymbol{Q}_n - \boldsymbol{W}_0\right)\boldsymbol{V}_0, \qquad \boldsymbol{C}_n := \left(\boldsymbol{U}_0^{\perp}\right)^T\left(\hat{\boldsymbol{W}}_n\boldsymbol{Q}_n - \boldsymbol{W}_0\right)\boldsymbol{V}_0, \\ \boldsymbol{\Psi}_n &:= \frac{1}{2}\left(\boldsymbol{B}_n\boldsymbol{D}_0 + \boldsymbol{D}_0\boldsymbol{B}_n^T\right), \qquad b_n := \hat{\sigma}_n^2 - \sigma_0^2, \end{aligned} \tag{2.97}$$

with $\boldsymbol{Q}_n$ as in (2.30), and set $(\boldsymbol{\Psi}_n, \boldsymbol{C}_n, b_n) := (\mathbf{0}, \mathbf{0}, 0)$ off $\mathcal{A}_n$. Then

$$\sqrt{n}\left(\boldsymbol{\Psi}_n,\ \boldsymbol{C}_n,\ b_n\right) \rightsquigarrow N\left(\mathbf{0},\ \mathcal{G}^{-1}\right) \qquad \textit{in } \mathit{Sym}(q)\times\mathbb{R}^{(p-q)\times q}\times\mathbb{R}, \tag{2.98}$$

with $\mathcal{G}^{-1}$ given by Proposition 48. Explicitly, writing $\boldsymbol{Z} = (\boldsymbol{Z}_{\boldsymbol{\Psi}}, \boldsymbol{Z}_{\boldsymbol{C}}, Z_b)$ for the limit:

1. *the $(p-q)q$ entries of $\boldsymbol{Z}_{\boldsymbol{C}}$ are mutually independent, independent of $(\boldsymbol{Z}_{\boldsymbol{\Psi}}, Z_b)$, and Var $((\boldsymbol{Z}_{\boldsymbol{C}})_{ki}) = \frac{\sigma_0^2\tau_i}{\sigma_i(\boldsymbol{W}_0)^2}$;*
2. *the $\frac{q(q-1)}{2}$ strictly upper triangular entries of $\boldsymbol{Z}_{\boldsymbol{\Psi}}$ are mutually independent, independent of everything else, and Var $((\boldsymbol{Z}_{\boldsymbol{\Psi}})_{ij}) = \frac{\tau_i\tau_j}{4}$ for $i<j$;*
3. *the remaining $q+1$ coordinates $((\boldsymbol{Z}_{\boldsymbol{\Psi}})_{11},\ldots,(\boldsymbol{Z}_{\boldsymbol{\Psi}})_{qq}, Z_b)$ are jointly Gaussian with the covariances (2.94); in particular*

$$\sqrt{n}\left(\hat{\sigma}_n^2-\sigma_0^2\right) \rightsquigarrow N\left(0,\ \frac{2\sigma_0^4}{p-q}\right). \tag{2.99}$$

Proof. By (2.30) we have $\iota(\hat{\boldsymbol{u}}_n) = \left(\hat{\boldsymbol{W}}_n\boldsymbol{Q}_n - \boldsymbol{W}_0,\ \hat{\sigma}_n^2-\sigma_0^2\right)$ on $\mathcal{A}_n$, so by (2.83) the triple $(\boldsymbol{\Psi}_n, \boldsymbol{C}_n, b_n)$ of (2.97) is exactly $\kappa(\hat{\boldsymbol{u}}_n)$ there; and both vanish off $\mathcal{A}_n$ because $\hat{\boldsymbol{u}}_n = \mathbf{0}$ there by (2.29). Hence $(\boldsymbol{\Psi}_n, \boldsymbol{C}_n, b_n) = \kappa(\hat{\boldsymbol{u}}_n)$ identically.

Let $\boldsymbol{K}$ denote the matrix of κ with respect to the g_{θ_0}-orthonormal basis $\boldsymbol{e}_1,\ldots,\boldsymbol{e}_d$ of (2.32) on the source and the scalar coordinates (2.86) on the target. Comparing (2.87) with $\boldsymbol{G}_{jk} = G_{\theta_0}(\boldsymbol{e}_j,\boldsymbol{e}_k)$ gives $\boldsymbol{G} = \boldsymbol{K}^T\mathcal{G}\boldsymbol{K}$, so

$$\boldsymbol{K}\boldsymbol{G}^{-1}\boldsymbol{K}^T = \boldsymbol{K}\left(\boldsymbol{K}^T\mathcal{G}\boldsymbol{K}\right)^{-1}\boldsymbol{K}^T = \boldsymbol{K}\boldsymbol{K}^{-1}\mathcal{G}^{-1}\boldsymbol{K}^{-T}\boldsymbol{K}^T = \mathcal{G}^{-1}. \tag{2.100}$$

By Proposition 13 we have $\sqrt{n}\hat{\boldsymbol{u}}_n \rightsquigarrow N(\mathbf{0},\boldsymbol{G}^{-1})$ in the coordinates of that basis, so the continuous mapping theorem applied to the linear map $\boldsymbol{K}$ and the identity (2.100) give (2.98).

The three assertions are then read off Proposition 48: a Gaussian vector whose covariance matrix is block diagonal has independent blocks, and within the first two groups the blocks are diagonal, so those coordinates are mutually independent. Statement (2.99) is the (b,b) entry of (2.94) together with $b_n = \hat{\sigma}_n^2-\sigma_0^2$. □

The score and the influence function in coordinates

The same block decomposition makes the score (1.22) and the influence operator (1.41) completely explicit. Throughout, for $\boldsymbol{x}\in\mathbb{R}^p$ we write

$$\boldsymbol{y} := \boldsymbol{U}_0^T\boldsymbol{x}\in\mathbb{R}^q, \qquad \boldsymbol{z} := \left(\boldsymbol{U}_0^{\perp}\right)^T\boldsymbol{x}\in\mathbb{R}^{p-q}, \tag{2.101}$$

so that, by part 1 of Lemma 45, when $\boldsymbol{X}\sim N_p(\mathbf{0},\Sigma_0)$ the vectors $\boldsymbol{U}_0^T\boldsymbol{X}$ and $(\boldsymbol{U}_0^{\perp})^T\boldsymbol{X}$ are independent, the first being $N_q(\mathbf{0},\boldsymbol{T}_0)$ and the second $N_{p-q}(\mathbf{0},\sigma_0^2\boldsymbol{I}_{p-q})$.

Proposition 50 (Score and influence function in coordinates)**.** Let $\tilde{\boldsymbol{s}}(\boldsymbol{x})\in\mathbb{R}^d$ be the coordinate vector of the score in the scalar coordinates (2.86), defined by $s(\boldsymbol{x};\theta_0)(\boldsymbol{h}) = \tilde{\boldsymbol{s}}(\boldsymbol{x})^T\kappa(\boldsymbol{h})$ for all $\boldsymbol{h}\in T_{\theta_0}M$; it is related to the vector $\boldsymbol{s}$ of (2.32) by $\boldsymbol{s} = \boldsymbol{K}^T\tilde{\boldsymbol{s}}$. Then

$$\tilde{\boldsymbol{s}}_{\boldsymbol{\Psi}_{ii}} = \frac{\boldsymbol{y}_i^2-\tau_i}{\tau_i^2}, \qquad \tilde{\boldsymbol{s}}_{\boldsymbol{\Psi}_{ij}} = \frac{2\boldsymbol{y}_i\boldsymbol{y}_j}{\tau_i\tau_j} \quad (i<j), \qquad \tilde{\boldsymbol{s}}_{\boldsymbol{C}_{ki}} = \frac{\sigma_i(\boldsymbol{W}_0)\,\boldsymbol{y}_i\boldsymbol{z}_k}{\sigma_0^2\tau_i}, \tag{2.102}$$

$$\tilde{\boldsymbol{s}}_b = \frac{1}{2}\left[\sum_{i=1}^q\frac{\boldsymbol{y}_i^2-\tau_i}{\tau_i^2} + \frac{\|\boldsymbol{z}\|^2-(p-q)\sigma_0^2}{\sigma_0^4}\right]. \tag{2.103}$$

Moreover $\mathbb{E}\left[\tilde{\boldsymbol{s}}(\boldsymbol{X})\right] = \mathbf{0}$ and $\mathbb{E}\left[\tilde{\boldsymbol{s}}(\boldsymbol{X})\tilde{\boldsymbol{s}}(\boldsymbol{X})^T\right] = \mathcal{G}$, the influence operator (1.41) has coordinate vector $\kappa\left(\mathrm{IF}(\boldsymbol{x})\right) = \mathcal{G}^{-1}\tilde{\boldsymbol{s}}(\boldsymbol{x})$, and explicitly

$$\mathrm{IF}_b(\boldsymbol{x}) = \frac{\boldsymbol{x}^T\left(\boldsymbol{I}_p-\boldsymbol{P}_0\right)\boldsymbol{x}-(p-q)\sigma_0^2}{p-q}, \qquad \mathrm{IF}_{\boldsymbol{C}_{ki}}(\boldsymbol{x}) = \frac{\boldsymbol{y}_i\boldsymbol{z}_k}{\sigma_i(\boldsymbol{W}_0)}, \tag{2.104}$$

$$\mathrm{IF}_{\boldsymbol{\Psi}_{ij}}(\boldsymbol{x}) = \frac{\boldsymbol{y}_i\boldsymbol{y}_j}{2} \quad (i<j), \qquad \mathrm{IF}_{\boldsymbol{\Psi}_{ii}}(\boldsymbol{x}) = \frac{1}{2}\left[\boldsymbol{y}_i^2 - \tau_i - \mathrm{IF}_b(\boldsymbol{x})\right]. \tag{2.105}$$

Proof. The relation $\boldsymbol{s} = \boldsymbol{K}^T\tilde{\boldsymbol{s}}$ is immediate from $\boldsymbol{s}(\boldsymbol{x})^T\boldsymbol{h} = s(\boldsymbol{x};\theta_0)(\boldsymbol{h}) = \tilde{\boldsymbol{s}}(\boldsymbol{x})^T\boldsymbol{K}\boldsymbol{h}$, valid for all $\boldsymbol{h}$.

Abbreviate $\boldsymbol{E} := \boldsymbol{\Sigma}_0^{-1}\left[\boldsymbol{x}\boldsymbol{x}^T - \boldsymbol{\Sigma}_0\right]\boldsymbol{\Sigma}_0^{-1}$, so that (1.22) reads $s(\boldsymbol{x};\theta_0)(\boldsymbol{h}) = \frac{1}{2}\operatorname{tr}\{\boldsymbol{E}\,\mathcal{D}_0(\boldsymbol{h})\}$. Since $\boldsymbol{J}_0^T\boldsymbol{x} = \left(\boldsymbol{y}^T, \boldsymbol{z}^T\right)^T$ by (2.101), part 1 of Lemma 45 gives

$$\boldsymbol{J}_0^T\boldsymbol{E}\boldsymbol{J}_0 = \begin{pmatrix} \boldsymbol{T}_0^{-1}\left(\boldsymbol{y}\boldsymbol{y}^T - \boldsymbol{T}_0\right)\boldsymbol{T}_0^{-1} & \sigma_0^{-2}\boldsymbol{T}_0^{-1}\boldsymbol{y}\boldsymbol{z}^T \\ \sigma_0^{-2}\boldsymbol{z}\boldsymbol{y}^T\boldsymbol{T}_0^{-1} & \sigma_0^{-4}\left(\boldsymbol{z}\boldsymbol{z}^T - \sigma_0^2\boldsymbol{I}_{p-q}\right) \end{pmatrix} =: \begin{pmatrix} \boldsymbol{E}_{11} & \boldsymbol{E}_{12} \\ \boldsymbol{E}_{12}^T & \boldsymbol{E}_{22} \end{pmatrix}. \tag{2.106}$$

Pairing (2.106) with (2.84), the two off-diagonal contributions to the trace are equal, because $\operatorname{tr}\left(\boldsymbol{E}_{12}^T\boldsymbol{D}_0\boldsymbol{C}^T\right) = \operatorname{tr}\left(\left(\boldsymbol{E}_{12}^T\boldsymbol{D}_0\boldsymbol{C}^T\right)^T\right) = \operatorname{tr}\left(\boldsymbol{C}\boldsymbol{D}_0\boldsymbol{E}_{12}\right) = \operatorname{tr}\left(\boldsymbol{E}_{12}\boldsymbol{C}\boldsymbol{D}_0\right)$, and therefore

$$2\,s(\boldsymbol{x};\theta_0)(\boldsymbol{h}) = \operatorname{tr}\left(\boldsymbol{E}_{11}\left[2\boldsymbol{\Psi} + b_{\boldsymbol{h}}\boldsymbol{I}_q\right]\right) + \frac{2}{\sigma_0^2}\,\boldsymbol{z}^T\boldsymbol{C}\boldsymbol{D}_0\boldsymbol{T}_0^{-1}\boldsymbol{y} + b_{\boldsymbol{h}}\operatorname{tr}\boldsymbol{E}_{22}. \tag{2.107}$$

Now read off the coefficient of each scalar coordinate. The matrix $\boldsymbol{\Psi}$ is symmetric, so $\operatorname{tr}\left(\boldsymbol{E}_{11}\boldsymbol{\Psi}\right) = \sum_i(\boldsymbol{E}_{11})_{ii}\boldsymbol{\Psi}_{ii} + 2\sum_{i<j}(\boldsymbol{E}_{11})_{ij}\boldsymbol{\Psi}_{ij}$, while $(\boldsymbol{E}_{11})_{ii} = \frac{\boldsymbol{y}_i^2-\tau_i}{\tau_i^2}$ and $(\boldsymbol{E}_{11})_{ij} = \frac{\boldsymbol{y}_i\boldsymbol{y}_j}{\tau_i\tau_j}$ for $i \neq j$; dividing the resulting $2\operatorname{tr}(\boldsymbol{E}_{11}\boldsymbol{\Psi})$ by the factor 2 in (2.107) gives the first two formulas of (2.102). The second term of (2.107) equals $\frac{2}{\sigma_0^2}\sum_{k,i}\boldsymbol{C}_{ki}\boldsymbol{z}_k\frac{\sigma_i(\boldsymbol{W}_0)\boldsymbol{y}_i}{\tau_i}$, which after division by 2 gives the third formula of (2.102). The coefficient of $b_{\boldsymbol{h}}$ is $\frac{1}{2}\left[\operatorname{tr}\boldsymbol{E}_{11} + \operatorname{tr}\boldsymbol{E}_{22}\right]$, which is (2.103).

The identities $\mathbb{E}\left[\tilde{\boldsymbol{s}}\right] = \mathbf{0}$ and $\mathbb{E}\left[\tilde{\boldsymbol{s}}\tilde{\boldsymbol{s}}^T\right] = \mathcal{G}$ are Theorem 7 read through (2.87), since for all $\boldsymbol{h}_1, \boldsymbol{h}_2 \in T_{\theta_0}M$

$$\mathbb{E}\left[s(\boldsymbol{X};\theta_0)(\boldsymbol{h}_1)\,s(\boldsymbol{X};\theta_0)(\boldsymbol{h}_2)\right] = G_{\theta_0}(\boldsymbol{h}_1,\boldsymbol{h}_2) = \kappa(\boldsymbol{h}_1)^T\mathcal{G}\kappa(\boldsymbol{h}_2)$$

and κ is onto. They can also be checked directly from (2.102): with $\boldsymbol{y}$ and $\boldsymbol{z}$ independent centred Gaussians, $\mathbb{E}\left[\tilde{\boldsymbol{s}}_{\boldsymbol{\Psi}_{ii}}^2\right] = \operatorname{Var}\left(\boldsymbol{y}_i^2\right)/\tau_i^4 = 2/\tau_i^2$, $\mathbb{E}\left[\tilde{\boldsymbol{s}}_{\boldsymbol{\Psi}_{ij}}^2\right] = 4/(\tau_i\tau_j)$, $\mathbb{E}\left[\tilde{\boldsymbol{s}}_{\boldsymbol{C}_{ki}}^2\right] = \sigma_i(\boldsymbol{W}_0)^2\tau_i\sigma_0^2/\left(\sigma_0^4\tau_i^2\right) = \sigma_i(\boldsymbol{W}_0)^2/\left(\sigma_0^2\tau_i\right)$, $\mathbb{E}\left[\tilde{\boldsymbol{s}}_{\boldsymbol{\Psi}_{ii}}\tilde{\boldsymbol{s}}_b\right] = 1/\tau_i^2$ and $\mathbb{E}\left[\tilde{\boldsymbol{s}}_b^2\right] = \eta$, every product containing an odd power of a coordinate of $\boldsymbol{z}$ having mean zero; these are exactly the entries (2.90).

By Definition 12 the vector $\mathrm{IF}(\boldsymbol{x})$ of (1.41) is characterized by $G_{\theta_0}\left(\mathrm{IF}(\boldsymbol{x}), \boldsymbol{h}\right) = s(\boldsymbol{x};\theta_0)(\boldsymbol{h})$ for all $\boldsymbol{h}$, that is, by (2.87), by $\kappa\left(\mathrm{IF}(\boldsymbol{x})\right)^T\mathcal{G}\kappa(\boldsymbol{h}) = \tilde{\boldsymbol{s}}(\boldsymbol{x})^T\kappa(\boldsymbol{h})$ for all $\boldsymbol{h}$; since κ is onto, $\mathcal{G}\kappa\left(\mathrm{IF}(\boldsymbol{x})\right) = \tilde{\boldsymbol{s}}(\boldsymbol{x})$.

It remains to compute $\mathcal{G}^{-1}\tilde{\boldsymbol{s}}$ block by block. For $i<j$, by (2.93), $\mathrm{IF}_{\boldsymbol{\Psi}_{ij}} = \frac{\tau_i\tau_j}{4}\cdot\frac{2\boldsymbol{y}_i\boldsymbol{y}_j}{\tau_i\tau_j} = \frac{\boldsymbol{y}_i\boldsymbol{y}_j}{2}$, and $\mathrm{IF}_{\boldsymbol{C}_{ki}} = \frac{\sigma_0^2\tau_i}{\sigma_i(\boldsymbol{W}_0)^2}\cdot\frac{\sigma_i(\boldsymbol{W}_0)\boldsymbol{y}_i\boldsymbol{z}_k}{\sigma_0^2\tau_i} = \frac{\boldsymbol{y}_i\boldsymbol{z}_k}{\sigma_i(\boldsymbol{W}_0)}$, which are the second formula of (2.104) and the first of (2.105). On the coupled block, (2.94) and (2.103) give

$$\begin{aligned}
\mathrm{IF}_b &= \frac{2\sigma_0^4}{p-q}\,\tilde{\boldsymbol{s}}_b - \frac{\sigma_0^4}{p-q}\sum_{i=1}^{q}\tilde{\boldsymbol{s}}_{\boldsymbol{\Psi}_{ii}} \\
&= \frac{\sigma_0^4}{p-q}\sum_{i=1}^{q}\frac{\boldsymbol{y}_i^2-\tau_i}{\tau_i^2} + \frac{\|\boldsymbol{z}\|^2-(p-q)\sigma_0^2}{p-q} - \frac{\sigma_0^4}{p-q}\sum_{i=1}^{q}\frac{\boldsymbol{y}_i^2-\tau_i}{\tau_i^2} = \frac{\|\boldsymbol{z}\|^2-(p-q)\sigma_0^2}{p-q},
\end{aligned}$$

the $\boldsymbol{y}$ terms cancelling identically; since $\|\boldsymbol{z}\|^2 = \boldsymbol{x}^T\boldsymbol{U}_0^{\perp}(\boldsymbol{U}_0^{\perp})^T\boldsymbol{x} = \boldsymbol{x}^T\left(\boldsymbol{I}_p - \boldsymbol{P}_0\right)\boldsymbol{x}$ by (2.81), this is the first formula of (2.104). Similarly

$$\begin{aligned}
\mathrm{IF}_{\boldsymbol{\Psi}_{ii}} &= \sum_{j=1}^{q}\left[\delta_{ij}\frac{\tau_i^2}{2} + \frac{\sigma_0^4}{2(p-q)}\right]\tilde{\boldsymbol{s}}_{\boldsymbol{\Psi}_{jj}} - \frac{\sigma_0^4}{p-q}\,\tilde{\boldsymbol{s}}_b \\
&= \frac{\boldsymbol{y}_i^2-\tau_i}{2} + \frac{\sigma_0^4}{2(p-q)}\sum_{j=1}^{q}\frac{\boldsymbol{y}_j^2-\tau_j}{\tau_j^2} - \frac{\sigma_0^4}{2(p-q)}\sum_{j=1}^{q}\frac{\boldsymbol{y}_j^2-\tau_j}{\tau_j^2} - \frac{\|\boldsymbol{z}\|^2-(p-q)\sigma_0^2}{2(p-q)},
\end{aligned}$$

which is the second formula of (2.105). □

Corollary 51 (The noise variance)**.** The maximum likelihood estimator of the noise variance satisfies (2.99), and it is asymptotically linear with influence function

$$\mathrm{IF}_{\sigma^2}(\boldsymbol{x}) = \frac{\boldsymbol{x}^T\left(\boldsymbol{I}_p - \boldsymbol{P}_0\right)\boldsymbol{x} - (p-q)\sigma_0^2}{p-q}, \qquad \mathbb{E}\left[\mathrm{IF}_{\sigma^2}(\boldsymbol{X})\right] = 0, \qquad \mathbb{E}\left[\mathrm{IF}_{\sigma^2}(\boldsymbol{X})^2\right] = \frac{2\sigma_0^4}{p-q}. \tag{2.108}$$

Proof. The weak limit is (2.99). The influence function is the first formula of (2.104), because $b_n = \hat{\sigma}_n^2 - \sigma_0^2$ is the b coordinate of $\kappa(\hat{\boldsymbol{u}}_n)$ by (2.97), so that the b coordinate of (1.42) reads $\sqrt{n}\left(\hat{\sigma}_n^2 - \sigma_0^2\right) = n^{-\frac{1}{2}}\sum_{i=1}^n \mathrm{IF}_b(\boldsymbol{X}_i) + o_{\mathbb{P}}(1)$. For the moments, $\left(\boldsymbol{U}_0^\perp\right)^T \boldsymbol{X} \sim N_{p-q}\left(\boldsymbol{0}, \sigma_0^2 \boldsymbol{I}_{p-q}\right)$ gives $\mathbb{E}\|\boldsymbol{z}\|^2 = (p-q)\sigma_0^2$ and $\mathrm{Var}\left(\|\boldsymbol{z}\|^2\right) = 2(p-q)\sigma_0^4$, whence $\mathbb{E}\left[\mathrm{IF}_{\sigma^2}^2\right] = \frac{2(p-q)\sigma_0^4}{(p-q)^2} = \frac{2\sigma_0^4}{p-q}$, consistent with (2.99) as it must be, since the asymptotic covariance of an asymptotically linear sequence is the covariance of its influence function. □

Remark 52 (Where the model helps, and where it does not)**.** Formulas (2.104)–(2.105) answer a question the asymptotic theory alone does not: in which directions is the constrained estimator $\hat{\boldsymbol{\Sigma}}_n$ actually better than the sample covariance matrix $\boldsymbol{S}_n$? The answer is: in exactly one block, and the reason is the computation of $\hat{\boldsymbol{\Sigma}}_n$ carried out in the proof of Theorem 9, which exhibits it as $\boldsymbol{S}_n$ with the trailing $p-q$ eigenvalues replaced by their average.

By (2.72) the influence function of $\hat{\boldsymbol{\Sigma}}_n$ is $\mathcal{D}_0 \circ \mathrm{IF}$, whose blocks are, by (2.84), $2\mathrm{IF}_{\boldsymbol{\Psi}} + \mathrm{IF}_b \boldsymbol{I}_q$, $\mathrm{IF}_{\boldsymbol{C}}\boldsymbol{D}_0$ and $\mathrm{IF}_b \boldsymbol{I}_{p-q}$. By (2.104)–(2.105) these are

$$\left(2\mathrm{IF}_{\boldsymbol{\Psi}} + \mathrm{IF}_b \boldsymbol{I}_q\right)_{ij} = \boldsymbol{y}_i\boldsymbol{y}_j - \delta_{ij}\tau_i, \qquad \left(\mathrm{IF}_{\boldsymbol{C}}\boldsymbol{D}_0\right)_{ki} = \sigma_i(\boldsymbol{W}_0)\frac{\boldsymbol{y}_i\boldsymbol{z}_k}{\sigma_i(\boldsymbol{W}_0)} = \boldsymbol{z}_k\boldsymbol{y}_i, \qquad \mathrm{IF}_b \boldsymbol{I}_{p-q}, \tag{2.109}$$

the diagonal case of the first identity being exactly the cancellation of IF_b in the second formula of (2.105). Now the influence function of $\boldsymbol{S}_n$, read in the same basis (2.81), is $\boldsymbol{J}_0^T\left(\boldsymbol{x}\boldsymbol{x}^T - \boldsymbol{\Sigma}_0\right)\boldsymbol{J}_0$, whose blocks are $\boldsymbol{y}\boldsymbol{y}^T - \boldsymbol{T}_0$, $\boldsymbol{z}\boldsymbol{y}^T$ and $\boldsymbol{z}\boldsymbol{z}^T - \sigma_0^2\boldsymbol{I}_{p-q}$. The first two coincide with (2.109): *to first order the maximum likelihood fit reproduces the sample covariance matrix in the signal block and in the signal–noise block, and gains nothing there.* Only the trailing block differs, where $\boldsymbol{z}\boldsymbol{z}^T - \sigma_0^2\boldsymbol{I}_{p-q}$ is replaced by its average

$$\frac{\mathrm{tr}\left(\boldsymbol{z}\boldsymbol{z}^T - \sigma_0^2\boldsymbol{I}_{p-q}\right)}{p-q}\boldsymbol{I}_{p-q} = \mathrm{IF}_b\,\boldsymbol{I}_{p-q}:$$

the isotropy constraint of the model averages the $p-q$ noise directions, dividing the variance by $p-q$. This is the infinitesimal shadow of the exact identity established in the proof of Theorem 9, which says the same thing non-asymptotically: $\hat{\boldsymbol{\Sigma}}_n$ differs from $\boldsymbol{S}_n$ only through the trailing eigenvalues, which it replaces by their mean. It also explains why the leading blocks cannot be improved: by (2.74) the model imposes no constraint there at all, the image of $\mathcal{D}_0$ being everything except for the isotropy of the trailing block.

Remark 53 (Cross-checks against classical principal component analysis)**.** Three consistency checks, each of which recovers a classical formula, and none of which is used elsewhere.

1. Pooled noise eigenvalues. The variance $\frac{2\sigma_0^4}{p-q}$ of (2.99) is the classical asymptotic variance of the average of the $p-q$ sample eigenvalues attached to a population eigenvalue of multiplicity $p-q$. Indeed $\hat{\sigma}_n^2 = \frac{1}{p-q}\sum_{j>q}\lambda_j(\boldsymbol{S}_n)$ exactly, by (1.39), and

$$\sum_{j>q}\lambda_j(\boldsymbol{S}_n) = \mathrm{tr}(\boldsymbol{S}_n) - \max\left\{\mathrm{tr}\left(\boldsymbol{P}\boldsymbol{S}_n\right) : \boldsymbol{P} \text{ an orthogonal projection of rank } q\right\}$$

by Ky Fan's variational principle. Write $\boldsymbol{P}_n$ for the maximizer, which is the projection onto the span of $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_q$. Since $\boldsymbol{P}_0$ is the maximizer at $\boldsymbol{\Sigma}_0$, by (2.80), we have on the one hand $\mathrm{tr}(\boldsymbol{P}_n\boldsymbol{S}_n) \geqslant \mathrm{tr}(\boldsymbol{P}_0\boldsymbol{S}_n)$ and on the other

$$\mathrm{tr}\left(\boldsymbol{P}_n\boldsymbol{S}_n\right) - \mathrm{tr}\left(\boldsymbol{P}_0\boldsymbol{S}_n\right) \leqslant \mathrm{tr}\left(\left(\boldsymbol{P}_n - \boldsymbol{P}_0\right)\left(\boldsymbol{S}_n - \boldsymbol{\Sigma}_0\right)\right) \leqslant \|\boldsymbol{P}_n - \boldsymbol{P}_0\|_F\|\boldsymbol{S}_n - \boldsymbol{\Sigma}_0\|_F,$$

because $\operatorname{tr}(\boldsymbol{P}_n\boldsymbol{\Sigma}_0) \leqslant \operatorname{tr}(\boldsymbol{P}_0\boldsymbol{\Sigma}_0)$. The multivariate central limit theorem applied to the independent, identically distributed and square integrable matrices $\boldsymbol{X}_i\boldsymbol{X}_i^T$ gives $\|\boldsymbol{S}_n - \boldsymbol{\Sigma}_0\|_F = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right)$, and the Davis–Kahan $\sin\Theta$ theorem, with the eigengap (2.80), gives $\|\boldsymbol{P}_n - \boldsymbol{P}_0\|_F = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right)$ as well (see for example Chen et al., 2021, Chapter 2, or Kato, 1995, Chapter II). Hence $\boldsymbol{P}_n$ may be replaced by $\boldsymbol{P}_0$ at a cost of $O_{\mathbb{P}}(n^{-1})$, and the statistic that remains is $\frac{1}{p-q}\operatorname{tr}\left((\boldsymbol{I}_p - \boldsymbol{P}_0)\,\boldsymbol{S}_n\right)$, whose centred and $\sqrt{n}$-scaled version is exactly the average of the influence functions (2.108).

2. The leading block is not improved. By (2.94), and writing $\text{Var}\left(2(\boldsymbol{Z}_{\boldsymbol{\Psi}})_{ii} + Z_b\right) = 4\text{Var}\left((\boldsymbol{Z}_{\boldsymbol{\Psi}})_{ii}\right) + 4\text{Cov}\left((\boldsymbol{Z}_{\boldsymbol{\Psi}})_{ii}, Z_b\right) + \text{Var}(Z_b)$,

$$\text{Var}\left(2(\boldsymbol{Z}_{\boldsymbol{\Psi}})_{ii} + Z_b\right) = 4\left(\frac{\tau_i^2}{2} + \frac{\sigma_0^4}{2(p-q)}\right) + 4\left(-\frac{\sigma_0^4}{p-q}\right) + \frac{2\sigma_0^4}{p-q} = 2\tau_i^2 = \text{Var}\left(\left(\boldsymbol{U}_0^T\boldsymbol{X}\right)_i^2\right),$$

and likewise $\text{Var}\left(2(\boldsymbol{Z}_{\boldsymbol{\Psi}})_{ij}\right) = \tau_i\tau_j = \text{Var}\left(\boldsymbol{y}_i\boldsymbol{y}_j\right)$ for $i < j$ — the sample covariance variances for those coordinates, as Remark 52 requires.

3. Anderson's eigenvector variances. Suppose the $\sigma_i(\boldsymbol{W}_0)$ are simple. By part 2 of Lemma 45, the component of the aligned loading error orthogonal to the column space of $\boldsymbol{W}_0$ is $\boldsymbol{U}_0^{\perp}\boldsymbol{C}_n\boldsymbol{V}_0^T$; since the i-th column of $\boldsymbol{W}_0\boldsymbol{V}_0$ is $\sigma_i(\boldsymbol{W}_0)$ times the i-th column of $\boldsymbol{U}_0$, the tilt of the estimated i-th direction out of the signal subspace is $\boldsymbol{U}_0^{\perp}$ applied to the i-th column of $\boldsymbol{C}_n$, divided by $\sigma_i(\boldsymbol{W}_0)$, with limiting per-coordinate variance

$$\frac{1}{\sigma_i(\boldsymbol{W}_0)^2} \cdot \frac{\sigma_0^2\tau_i}{\sigma_i(\boldsymbol{W}_0)^2} = \frac{\sigma_0^2\tau_i}{\left(\tau_i - \sigma_0^2\right)^2} = \frac{\lambda_i(\boldsymbol{\Sigma}_0)\lambda_j(\boldsymbol{\Sigma}_0)}{\left(\lambda_i(\boldsymbol{\Sigma}_0) - \lambda_j(\boldsymbol{\Sigma}_0)\right)^2} \qquad \text{for any } j > q,$$

which is Anderson's formula for the noise directions. When the $\sigma_i(\boldsymbol{W}_0)$ are repeated this eigenvector-by-eigenvector reading breaks down — individual eigenvectors are then not asymptotically normal — but nothing in Propositions 47 and 48 is affected: no formula in this section divides by a difference of singular values, and the only inverted quantities are τ_i, σ_0^2, $\sigma_i(\boldsymbol{W}_0)$ and $p - q$, all strictly positive under the standing assumptions $\operatorname{rank}(\boldsymbol{W}_0) = q$, $\sigma_0^2 > 0$ and $q < p$. This is the promised quantitative form of the third comment of Remark 22.

References

Yuxin Chen, Yuejie Chi, Jianqing Fan, and Cong Ma. *Spectral Methods for Data Science: A Statistical Perspective*, volume 14 of *Foundations and Trends in Machine Learning*. Now Publishers, Boston, 2021. doi: 10.1561/2200000079.

Arghya Datta and Sayak Chakrabarty. On the consistency of maximum likelihood estimation of probabilistic principal component analysis. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

A. K. Gupta and D. K. Nagar. *Matrix Variate Distributions*, volume 104 of *Chapman & Hall/CRC Monographs and Surveys in Pure and Applied Mathematics*. Chapman & Hall/CRC, Boca Raton, 2000. ISBN 1-58488-046-5.

Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, Cambridge, second edition, 2013. ISBN 978-0-521-83940-2.

Tosio Kato. *Perturbation Theory for Linear Operators*. Classics in Mathematics. Springer-Verlag, Berlin, 1995. ISBN 3-540-58661-X. Reprint of the 1980 corrected printing of the second edition; originally Volume 132 of the Grundlehren der mathematischen Wissenschaften.

John M. Lee. *Introduction to Smooth Manifolds*, volume 218 of *Graduate Texts in Mathematics*. Springer New York, second edition, 2013. ISBN 978-1-4419-9981-8. doi: 10.1007/978-1-4419-9982-5.

John M. Lee. *Introduction to Riemannian Manifolds*, volume 176 of *Graduate Texts in Mathematics*. Springer International Publishing, second edition, 2018. ISBN 978-3-319-91754-2. doi: 10.1007/978-3-319-91755-9.

Barrett O'Neill. Submersions and geodesics. *Duke Math. J.*, 34:363–373, 1967.

Richard Redner. Note on the consistency of the maximum likelihood estimate for nonidentifiable distributions. *The Annals of Statistics*, 9(1):225–228, 1981.

Lvfang Sun, Zhenhua Lin, and Lin Liu. Towards an asymptotic efficiency theory on regular parameter manifolds. arXiv:2510.13703v2 [math.ST], July 2026. Version 2, 12 July 2026.

Michael E. Tipping and Christopher M. Bishop. Probabilistic principal component analysis. *Journal of the Royal Statistical Society, Series B*, 61(3):611–622, 1999.

Fabricio Valencia. Riemannian submersions. Unpublished notes, Instituto de Matemática e Estatística, Universidade de São Paulo, July 2021.

A. W. van der Vaart. *Asymptotic Statistics*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, Cambridge, 1998. ISBN 0-521-49603-9. First published 1998; first paperback edition 2000, ISBN 0-521-78450-6.